Directed Cyclic Networks
Research Contract and Proof Compendium

P11 spectral draft: 3 October 2026

Current mathematical status: P01–P04 are VERIFIED / AUDITED; P05 v1.1 is VERIFIED / AUDITED AFTER MINOR REVISION; P08 v1.1 is VERIFIED / AUDITED AFTER MINOR REVISION; P07 v1.1 is VERIFIED / AUDITED; P09 v1.1 is VERIFIED / AUDITED; P10 v1.1 is VERIFIED / AUDITED; P11 v1.0 is DRAFT; P06 and P12–P18 are TODO. The independent audit `audits/A01_wellposedness_and_absorbing_box.tex` records CORE VERIFIED and EXTENDED MINOR REVISION. Its sole required correction, E01, is applied: the archived radius is described as the larger real root and as nonnegative. The extended verdict is now VERIFIED AFTER MINOR REVISION. No formula, assumption, inequality, conclusion, or proof is changed. All remaining P01 body text, including its historical DRAFT-status sentence, is preserved; the current P01 header and project manifest govern status.

P02 v1.1 is VERIFIED / AUDITED after A02's sole required correction E01: the boundary-continuity paragraph explicitly names the ordered pair (s_-, s_+). A02's original verdicts are CORE MINOR REVISION and EXTENDED VERIFIED; closure is CORE VERIFIED AFTER MINOR REVISION, EXTENDED VERIFIED. The registered audit source is `audits/A02_synchronous_subspace_and_equilibria.tex`. Its original source has now been located in the companion audit package and restored unchanged, with its matching PDF reading copy under the same basename. No theorem, proof, displayed formula, table, count, or assumption changed. Historical DRAFT wording in the P02 body is preserved; its header and the manifest govern current status. Compilation is not mathematical verification.

P03 v1.1 is VERIFIED / AUDITED after A03, which verifies the exact P03 v1.0 with no required mathematical or wording correction. The audit source is `audits/A03_origin_and_linearization.tex`. Closure changes only comment metadata; every byte of the mathematical body is preserved. Historical DRAFT/audit-pending wording in that body is retained; the current header and manifest govern status. The manifest records the original audited hash and the current verified hash. A03 and its supplied support records are included unchanged. Compilation is not mathematical verification.

P04 v1.1 is VERIFIED / AUDITED after A04, whose mathematical verdict on exact P04 v1.0 is VERIFIED with no required mathematical or wording correction. The audit source is `audits/A04_arbitrary_nq_fourier_spectrum.tex`. Closure changes comment metadata only; the entire P04 body is byte-identical. Its historical DRAFT/audit-pending prose is preserved; the current header and manifest govern status and record both version hashes. A04 and its support records are included unchanged. The original P04 validation companion was unavailable to the independent auditor, so its historical numerical-check statistics were not independently reproduced (A04 U01). This reproducibility note is nonblocking; A04's mathematical verdict remains VERIFIED. P04 remains supporting known machinery.

P05 v1.1 is VERIFIED / AUDITED AFTER MINOR REVISION. The immutable audit `audits/A05_q2_mode_selection.tex` records the original verdicts CORE MINOR REVISION, REALIZABILITY VERIFIED and OVERALL MINOR REVISION for exact P05 v1.0. Its sole required correction E01 is applied verbatim: mode zero belongs to the candidate Fourier labels over which the maximum is taken, not necessarily to the selected set. All other P05 body bytes, including historical DRAFT/audit-pending prose, are preserved; its current header and manifest govern status and record the original and corrected hashes. No formula, theorem or proof is revised. A05 and its supplied support records are unchanged. A05 U01 records that the original validation companion was unavailable to the auditor; this nonblocking historical provenance

note remains, and no checker is rerun for closure. At that closure P08 was cleared to begin. A05 supplies P05's independent audit; compilation is not mathematical verification.

P08 v1.1 is VERIFIED / AUDITED AFTER MINOR REVISION. The immutable audit `audits/A08_matched_bilinear_analysis.tex` verifies the MB foundation, MB spectrum, and parity/obstruction results for exact P08 v1.0. Its same-tuple comparison and overall verdicts are MINOR REVISION. The sole required issue E01 is closed verbatim: the universal witness selects MB representative 1 for $n = 5, 6, 7$ and representative 2 for $n = 8$, corresponding to labels $\{2, 6\}$. All theorem statements, proofs, witness tuples, equilibrium amplitudes, dispersion formulas and all-n conclusions are unchanged. Other P08 body bytes, including historical DRAFT/audit-pending prose, are retained; the current header and manifest govern status and record both hashes. A08 and all its supplied support files are preserved unchanged.

A08 located, authenticated and exactly reproduced the original P08 validation summaries and results. No validation checker is rerun for this closure. Earlier A04/A05 provenance notes are unchanged. At the P08 closure, P07 was cleared to begin. A08 supplies P08's independent audit; its historical closure records remain unchanged.

P07 v1.1 is VERIFIED / AUDITED. A07 verifies exact P07 v1.0 in all five layers: Fourier and trigonometry, ordering and selection, frequency and coincidence, stability and critical modes, and scope and dependencies. Its overall verdict is VERIFIED, with no required mathematical or wording correction. Closure changes header metadata only; every mathematical-body byte, including historical DRAFT/audit-pending wording, is preserved. The current header and manifest govern status and record both full-file hashes and the unchanged mathematical-body hash. The immutable source is `audits/A07_five_node_specialization.tex`; it and all supplied support files are retained unchanged.

A07 authenticated and reproduced the original P07 validation: 161 exact checks, 108 numerical cases, and byte-identical summary and result files. No historical validation suite is rerun for this closure. Validation and compilation are corroboration, not proof; A07 supplies the independent mathematical verification. The direct edge remains P05 to P07, with P03/P04 and P01/P02 inherited; no P08 theorem dependency is introduced. Frozen scope and novelty documents are unchanged. At the P07 closure, P09 was cleared to begin.

P09 v1.1 is VERIFIED / AUDITED. The audit `audits/A09_transcritical_bifurcation.tex` verifies exact P09 v1.0 in all six layers: critical spectrum and reduction, generic transcritical bifurcation, stability and physical branches, generic critical dynamics, degenerate critical stability, and codimension and scope. Its overall verdict is VERIFIED, with no mathematical or wording correction. Closure changes header metadata only; every byte of the audited mathematical body is retained, including historical DRAFT/audit-pending prose. The current header and manifest govern status and record both full-file hashes and the unchanged mathematical-body hash. A09 and all its support are unchanged.

A09 authenticated and reproduced the original P09 validation: 1,260 exact checks, 2,700 matrix cases and 12 slope checks, with byte-identical result and summary files. These saved records are preserved; no historical checker is rerun for closure. A09 independently reconstructed the mathematics. Validation and compilation remain corroboration, not proof. The theorem inputs remain P02–P04 with P01 inherited; no P05, P07 or P08 theorem dependency is introduced. Prior proofs, audits and frozen documents are unchanged. At that closure P10 was cleared to begin.

P10 v1.1 is VERIFIED / AUDITED. A10 verifies exact P10 v1.0 in all six layers and overall, without corrections. Metadata-only closure preserves the entire mathematical body; its current header and manifest govern status. A10 and its support are immutable. Basin conclusions remain local except on the synchronous ray. A10 inventoried the developer validation but did not rerun it; its own computations corroborate its analytical verification. P10 inputs remain P02–P04/P09, with P01 inherited.

P11 v1.0 is DRAFT; independent A11 audit pending. It supplies an exact spectral candidate, regular upper branch and fixed path for P12. Transversality, nonlinear nondegeneracy and periodic orbits remain unproved. P06 and P12–P18 remain TODO; P12 is not begun. Earlier contract and planning fragments retain historical status wording. The literature phase is frozen. The immutable blueprint is archived separately and is not included in this document.

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1 Canonical model contract

The explicit initialization prompt is authoritative. This fragment specifies the main models; formulas in the archived blueprint remain research candidates.

1.1 Main demand-modulated model

For $i \in \mathbb{Z}_n$, define the vector field F^{DM} by

$$\dot{x}_i = ax_i - kx_i^2 + \frac{cx_{i-1}}{1+x_i} - \frac{gx_i}{1+x_{i+q}} - hx_ix_{i-q}. \tag{$\mathrm{DM}_{n,q}$}$$

The variables are dimensionless normalized activities. The preferred term is *demand-modulated model*. The positive input is called demand-modulated predecessor input; no standard Holling type-II classification is adopted. The intended initial conditions are $x(0) \in \mathbb{R}^n_{\geq 0}$. For local statements and boundary bifurcations, use the specified open extension domain $\mathcal{U} = (-1,\infty)^n$. This domain choice asserts no invariance theorem.

1.2 Primary matched bilinear comparator

Define F^{MB} by

$$\dot{y}_i = ay_i - ky_i^2 + cy_iy_{i-1} - gy_iy_{i+q} - hy_iy_{i-q}, \qquad i \in \mathbb{Z}_n. \tag{$\mathrm{MB}_{n,q}$}$$

This is the topology- and parameter-count-matched comparator in the main program. Its intended initial data are nonnegative; the polynomial vector field is specified on $\mathbb{R}^n$. No existence, invariance, boundedness or stability conclusion is included in this contract. Matching parameter names does not settle a quantitative calibration between the two laws.

1.3 Parameter and indexing contract

$$n \in \mathbb{Z}, \quad n \geq 5, \qquad q \in \{1,\dots,n-1\}, \qquad \boldsymbol{\pi} = (a,k,c,g,h) \in (0,\infty)^5.$$

All indices are modulo n, with representatives $0,\dots,n-1$. Arbitrary q is allowed. The interpretation as one connected step-q cycle requires $\gcd(n,q) = 1$; this condition is not imposed on the general analytical program. In particular, even n is not excluded when $q = 2$. Coincident offsets do not remove terms from either defining equation. The intended exact wave-number-selection program focuses on $q = 2$. The five-node Wu-Xing-inspired specialization is $n = 5, q = 2$.

1.4 Archived secondary six-term comparator

The following is a transcription of the blueprint's proposed arbitrary-n,q extension, not a verified attribution to the original Lin–Han publication:

$$\dot{y}_i = ay_i - by_i^2 + cy_iy_{i-1} - dy_iy_{i-q} - g_{\mathrm{c}}y_iy_{i+1} - hy_iy_{i+q}. \tag{$\mathrm{LB}_{n,q}$ archive}$$

Its archived five-node equation is obtained by specifying $n = 5, q = 2$ in this transcription. The six coefficient names are archival and are not a parameter identification with DM or MB. The original source, coefficient domain, index orientation, and attribution remain to be checked before analysis or use as a literature application.

1.5 Scope boundary

The main program covers homogeneous DM, primary comparator MB, their synchronous equilibria and spectra, the planned bifurcations, and the planned small-heterogeneity response theory. All requested P01–P18 tasks remain in the project; final main-text versus supplement placement is open. The heterogeneous coefficient assignment is not frozen by this contract. The six-term model is archived for a possible secondary literature application, never a replacement for MB. Source-saturating replacements, local motif theory, pathology extensions, adaptive set points, interventions, noise, delays, data calibration, and prior Lyapunov-spectrum reanalysis are postponed to other work. The master remains unchanged.

2 Notation and assumptions

2.1 Standing assumptions

A1. $n \in \mathbb{Z}$, $n \geq 5$, and $q \in \{1, \ldots, n-1\}$.

A2. Homogeneous parameters $\boldsymbol{\pi} = (a, k, c, g, h) \in (0, \infty)^5$.

A3. Dimensionless real activities; initial states in the nonnegative orthant when forward biological-admissibility statements are sought. For DM the local extension domain is $\mathcal{U} = (-1, \infty)^n$.

A4, conditional. Impose $\gcd(n, q) = 1$ only for the connected step-q cycle interpretation; specify $q = 2$ or $(n, q) = (5, 2)$ when required by a module.

A5, conditional. Positivity of an equilibrium, invertibility, hyperbolicity, stability, Hopf nondegeneracy, and perturbation regularity are additional hypotheses to be stated and checked in the appropriate future proof. None is a standing established result.

2.2 Indices and cyclic shifts

Define $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$, using representatives $0, \ldots, n-1$. For $z \in \mathbb{C}^n$, define the predecessor and successor shift matrices by

$$(Pz)_j = z_{j-1}, \qquad (P^{-1}z)_j = z_{j+1}, \qquad P_{j\ell} = \begin{cases} 1, & \ell \equiv j-1 \pmod{n}, \\ 0, & \text{otherwise.} \end{cases}$$

Thus the symbols P^q and P^{-q} denote repeated predecessor and successor shifts, respectively. The group-action convention is $\rho(r)z = P^r z$. For transcription of the original five-node labels, the original $x_1, \ldots, x_5$ are relabeled as $x_0, \ldots, x_4$ in the same cyclic order. This is a label translation only; no historical or physiological statement is certified.

2.3 Fourier vectors, transform and pairing

Use i for the imaginary unit and j for a node index. Define

$$\theta_m = \frac{2\pi m}{n}, \qquad v_j^{(m)} = \frac{1}{\sqrt{n}} \exp(\mathrm{i}\theta_m j), \qquad \mathsf{Q}_{jm} = v_j^{(m)}, \quad 0 \leq j, m < n.$$

The canonical transform convention is

$$\widehat{z}_m = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} z_j \exp(-\mathrm{i}\theta_m j), \qquad z_j = \frac{1}{\sqrt{n}} \sum_{m=0}^{n-1} \widehat{z}_m \exp(\mathrm{i}\theta_m j).$$

The normalization is the unitary DFT convention. The blueprint's displayed unscaled Fourier vectors are not copied as normalized eigenvectors. Use A^* for conjugate transpose, A^{T} for transpose, and

$$\langle u, v\rangle = u^* v = \sum_j \overline{u_j} v_j, \qquad \|z\|_2 = (z^* z)^{1/2}.$$

The pairing is conjugate-linear in its first argument. Matrix 2-norms are induced Euclidean norms. The adjoint eigenvector equation, its phase/scale, Taylor tensors and first-Lyapunov-coefficient convention must be reconciled in P13 before any blueprint normal-form formula is imported.

2.4 State sets and equilibrium notation

$$\mathbb{R}^n_{\geq 0} = \{x \in \mathbb{R}^n : x_j \geq 0 \text{ for all } j\}, \qquad \mathbb{R}^n_{>0} = \{x \in \mathbb{R}^n : x_j > 0 \text{ for all } j\},$$

$$\mathbf{1} = (1, \ldots, 1)^{\mathsf{T}}, \qquad \mathcal{S} = \{s\mathbf{1} : s \in \mathbb{R}\}.$$

The phrase positive orthant means the strict set $\mathbb{R}^n_{>0}$; nonnegative orthant means $\mathbb{R}^n_{\geq 0}$. A synchronous state is a member of $\mathcal{S}$; a positive synchronous equilibrium, if established, is denoted $x^*\mathbf{1}$ with $x^* > 0$. The admissible synchronous domain for DM is $\mathcal{S} \cap \mathcal{U}$. No invariance or equilibrium classification is asserted by these definitions.

2.5 Spectrum, hyperbolicity and mode terminology

For a square matrix J, define

$$\operatorname{sa}(J) = \max\{\operatorname{Re}\lambda : \lambda \in \operatorname{spec}(J)\}.$$

An equilibrium of a differentiable vector field is hyperbolic when its Jacobian has no eigenvalue with zero real part. A matrix is Hurwitz if all its eigenvalues have negative real part, and normal if $JJ^* = J^*J$. These definitions assert neither stability nor normality for a model. Mode $m = 0$ is called the uniform or synchronous mode; modes $m \neq 0$ are called nonuniform or symmetry-breaking modes. The mode-pair notation is $\{m, (-m) \bmod n\}$, usually written $\{m, n-m\}$. A pair representative is chosen in $0 \leq m \leq \lfloor n/2 \rfloor$. Keep the uniform mode and, for even n, $m = n/2$ as singleton labels. Any eigenvalue-conjugacy or simplicity claim belongs to P04 or P11. When a spectrum has been derived, the selected-mode set will mean $\arg\max_m \operatorname{Re}\lambda_m$, including every tie. A first unstable mode additionally requires a specified equilibrium branch and oriented parameter path.

2.6 Reserved scalar notation

For continuity with the archive, reserve $A = k + h$ and $\mu = a + c - g$ as abbreviations, without asserting an equilibrium equation or bifurcation. Use ν for a future scalar continuation parameter and ε for a heterogeneity amplitude. Reserve $\omega > 0$ for a chosen positive Hopf frequency. Use $\operatorname{sa}(J)$ for spectral abscissa and $\delta = -\operatorname{sa}(J) > 0$ only under an explicitly stated Hurwitz hypothesis. This avoids reusing α as both a stability margin and a parameter perturbation. The spectral coefficient called T_0 in the blueprint will be named τ if needed, keeping T_{scale} available for a physical time scale. Heterogeneity profiles will be denoted collectively by $\boldsymbol{\zeta}$ if adopted; no coefficient-assignment convention is implied.

3 Theorem dependency map and completion gates

Arrows mean a prerequisite proof module, not a theorem already proved. Every module also depends on the canonical contract and notation. The following displayed adjacency graph is the full direct-dependency graph; transitive prerequisites are inherited. Conditional edges are listed separately.

$$
\begin{aligned}
\text{Contract} &\longrightarrow \text{P01}\\
\text{P01} &\longrightarrow \text{P02}\\
\text{P01, P02} &\longrightarrow \text{P03}\\
\text{P03} &\longrightarrow \text{P04}\\
\text{P04} &\longrightarrow \text{P05}\\
\text{P05} &\longrightarrow \text{P06}\\
\text{P05} &\longrightarrow \text{P07}\\
\text{P01, P02, P04, P05} &\longrightarrow \text{P08}\\
\text{P02, P03, P04} &\longrightarrow \text{P09}\\
\text{P02, P03, P04, P09} &\longrightarrow \text{P10}\\
\text{P02, P04, P05} &\longrightarrow \text{P11}\\
\text{P11} &\longrightarrow \text{P12}\\
\text{P11, P12} &\longrightarrow \text{P13}\\
\text{P11, P12, P13} &\longrightarrow \text{P14}\\
\text{P02, P04} &\longrightarrow \text{P15}\\
\text{P04, P15} &\longrightarrow \text{P16}\\
\text{P04, P16} &\longrightarrow \text{P17}\\
\text{P04, P12, P13, P15, P16} &\longrightarrow \text{P18}
\end{aligned}
$$

The graph is acyclic. P07 is an additional prerequisite when a five-node specialization is used in P11–P14. An optional periodic-orbit persistence extension in P15 would additionally require P14; it is not part of the core equilibrium-persistence gate. Source verification is a conditional input to any six-term extension in P08. An unresolved optional extension does not block the core module if its statement is explicitly excluded from the completed scope.

3.1 One dependency-respecting completion order

1. P01: Well-posedness and attracting bounded region.
2. P02: Synchrony and complete synchronous-equilibrium classification.
3. P03: Origin and homogeneous linearization.
4. P04: Arbitrary-(n,q) Fourier spectrum and normality.
5. P05: Exact step-two mode selection.
6. P06: Large-n selected-wave-number limit.
7. P07: Five-node golden-ratio specialization.
8. P08: Matched bilinear analysis and exact comparison.
9. P09: Transcritical bifurcation at the origin.
10. P10: Saddle-node bifurcation and true bistable region.

11. P11: Hopf spectral conditions and spectral simplicity.
12. P12: Hopf transversality.
13. P13: First Lyapunov coefficient and nondegenerate Hopf theorem.
14. P14: Equivariant rotating waves.
15. P15: Persistence under parameter heterogeneity.
16. P16: Static sensitivity and localized Green function.
17. P17: Transient and harmonic response.
18. P18: Stability radius and Hopf-threshold sensitivity.

After P05, P06, P07 and the later independent branches need not be worked in strict numerical order. This list is one valid order for the compendium and for eventual completion. P11 collects spectral conditions, P12 establishes transversality, and P13 is the home of the assembled nondegenerate Hopf theorem. P14 owns the additional equivariant rotating-wave conclusion.

3.2 Numerical dependency gates

Task	Required analytical inputs before the claim is validated
N01 Jacobian and boundary checks	P01–P03; P08 for MB.
N01 equilibrium and spectrum checks	P02–P04; P05–P07 for selection, limits and specialization.
N01/N02 bifurcation validation	P09–P13; complete-spectrum, transversality and independent l1 checks.
N02 periodic branch and wave validation	P13–P14; use Floquet and phase checks, not a time-series image alone.
N01/N02 heterogeneity validation	P15–P16 for equilibrium/remainder; P18 for threshold shifts.
N01/N03 response and robustness	P16–P18; specify norms, forcing and direct versus equilibrium-induced perturbations.
N03 final figures	Each figure's listed proof owners, passed N01 checks and reproducible N02 output where relevant.

Exploratory future checks may help find errors in DRAFT derivations, but cannot promote a theorem or a numerical claim to verified evidence on their own. No computational checks are performed in this initialization.

3.3 Manuscript claims withheld until verification

Withheld claim	Minimum proof gate
Local well-posedness, positivity, global existence and bounded attraction	P01 with precise attraction/absorption language.
Complete synchronous-equilibrium classification and origin stability	P02–P03; nonlinear boundary cases handled separately.
Arbitrary-(n,q) spectrum and normality	P04.
Exact q=2 selection, large-n limit and five-node constants	P05, P06 and P07, respectively.
Exact matched-bilinear comparison	P08 plus its prerequisites and an explicit calibration for quantitative comparisons.
Transcritical, fold and true bistability claims	P09–P10 with all-mode stability and explicit basin scope.
Nondegenerate symmetry-breaking Hopf	P11–P13, including rigorous nonzero frequency, simplicity, transversality and l1 nonvanishing.
Stable rotating-wave branch	P14 and applicable transverse stability from P11–P13.
Persistence, sensitivity and localized response	P15–P16; positivity, invertibility and stability assumptions distinguished.
Transient/harmonic response and robustness	P17–P18, with explicit norms and remainder bounds.
Numerical values, periods, figures or observed regimes	Applicable verified proofs plus N01/N02 evidence and recorded reproduction.
Novelty, priority or literature comparisons	Completed source-specific literature audits; none confirmed here.

Before submission, mathematical claims must also have a completed independent audit for the exact version cited. Unsupported candidate claims remain in the archive and registry. No final-paper narrative is supplied by this map.

3.4 Phase 1C frozen scope and current attempt order

The adjacency graph above remains in force. The numerical ordering above and in the compendium is a valid reading order. The current research attempt order, superseding earlier scheduling recommendations, is

$$\begin{gathered}\mathrm{P01} \to \mathrm{P02} \to \mathrm{P03} \to \mathrm{P04} \to \mathrm{P05} \to \mathrm{P08} \to \mathrm{P07} \to \mathrm{P09} \to \mathrm{P10},\\ \mathrm{P11} \to \mathrm{P12} \to \mathrm{P13} \to \mathrm{P14} \to \mathrm{P06},\\ \mathrm{P15} \to \mathrm{P16} \to \mathrm{P17} \to \mathrm{P18} \quad \text{(postponed extension queue)}.\end{gathered}$$

Arrows in this schedule indicate order of attempts, not additional mathematical dependencies. P06 is attempted only if it adds a useful large-n consequence. P07 precedes any five-node use in P11–P14. P15–P18 are outside the frozen paper.

Intended claim	Assembly gate	Frozen role
Realizable $q = 2$ selection and switching	P05	Main theorem target; quadratic maximization alone is insufficient.

Intended claim	Assembly gate	Frozen role
Matched DM–MB contrast	P08; P13 if comparing criticality	Main theorem target on a common admissible branch domain.
First symmetry-breaking Hopf	P11, P12, P13	Assembled main theorem; P11 alone is spectral support.
Nontrivial l_1 sign region	P13	Main theorem target; currently unresolved.
Coefficient reduction	P13	Proposition/lemma; detailed algebra may be supplementary.
Exact rotating-wave branch	P14	Corollary of verified Hopf and symmetry hypotheses.
Five-node motivating application	P07 plus applicable P11–P14	Corollary within the main paper.
General spectrum; branch classification and local branch bifurcations	P01–P04, P09–P10	Supporting propositions/lemmas.
Useful large-n consequence	P06	Conditional corollary.
Heterogeneity, response and robustness	P15–P18	Postpone.

The primary comparison uses identical positive coefficient tuples with each model's own named positive synchronous branch; branch/path choices must be stated before their use. A criticality comparison is assembled after P13 and does not introduce a circular prerequisite for P08's spectral analysis.

The numerical gates above remain valid. Checks and figures for P15–P18 are dormant in this paper. No numerical task is executed in Phase 1C. All proof statuses remain TODO. Exact-model priority is not closed because the named 2021–2024 main-text equation/theorem inventories remain incomplete; this is a separate publication-claim gate, not a theorem dependency. See `frozen_theorem_package.md` and `final_novelty_verdict.md`.

4 Literature audit template

Copy this form for each source in a future audit. Keep an unfilled field unresolved; do not infer absence of overlap from absence of evidence.

Record ID, reviewer, date and version:

Source:

Full authors, title, year, venue, DOI or stable URL; primary source file and provenance; verification of bibliographic metadata.

Exact model:

Transcribed equations, parameter domain, normalization, node and edge conventions, boundary assumptions; source transcription check.

Theorem or numerical result:
Exact statement and evidence type; separate a proved theorem from a simulation or interpretation.

Assumptions:
All stated and implicit hypotheses, exclusions, nondegeneracy conditions, parameter restrictions and computational settings.

Page/equation/theorem reference:
Exact location in the consulted source, including version and supplement where relevant.

Relation to our work:
Corresponding registry IDs and proof modules; explicit dictionary between equations, parameters and indexing conventions.

Possible overlap:
Matching conclusions, proof methods or model classes; whether a specialization or direct corollary would cover our proposed result.

Confirmed novelty:
Leave empty until comparison is complete; attach evidence and reviewer sign-off for a narrowly stated novelty claim.

Unresolved concern:
Missing hypotheses, ambiguous attribution, unavailable source, inconsistent formula or unreproduced numerical claim.

Disposition:
Unresolved / source verified / overlap confirmed / qualified distinction documented. Record downstream claim and scope changes.

4.1 Future search clusters, not reviewed here

Cyclic feedback and circulant spectra; coupled-cell synchrony; equivariant Hopf and rotating waves; perturbation theory, heterogeneity and Green functions; exact Wu-Xing models and primary sources for interpretive language. No bibliography entry in the blueprint has been certified by this template.

5 P01: Well-posedness and bounded absorbing sets

5.1 Domain, hypotheses, and set terminology

Use exactly the demand-modulated vector field in $(\mathrm{DM}_{n,q})$, denoted here by $F = (F_i)_{i\in\mathbb{Z}_n}$. The canonical assumptions imply the weaker hypotheses

$$n \geq 1, \quad q \in \mathbb{Z} \pmod{n}, \quad a \in \mathbb{R}, \quad k > 0, \quad c, g, h \geq 0. \tag{P01-W}$$

We prove the main results under (P01-W), without changing the canonical contract for later modules. Coincident offsets retain every term of the defining equation. Write

$$\mathcal{U} = (-1, \infty)^n, \qquad K_r = [0, r]^n \ (r \geq 0), \qquad X(t) = \max_{i\in\mathbb{Z}_n} x_i(t), \qquad s_+ = \max\{s, 0\}.$$

For a continuous real function v, define

$$D^+ v(t) = \limsup_{\tau\downarrow 0} \frac{v(t+\tau) - v(t)}{\tau}.$$

Distances use the maximum norm; for $x \in \mathbb{R}^n_{\geq 0}$,

$$\mathrm{dist}_\infty(x, K_r) = (\max_i x_i - r)_+. \tag{1}$$

A set is *positively invariant* if solutions starting there remain there for all forward times for which they exist. It *absorbs bounded subsets of* $\mathbb{R}^n_{\geq 0}$ if every bounded $E \subset \mathbb{R}^n_{\geq 0}$ has a finite T_E such that all solutions starting in E lie in the set for every $t \geq T_E$. Attraction means convergence of the distance to zero; uniform attraction of bounded sets means convergence uniform over each such E. We will also establish stronger absorption bounds with a single time valid for the entire orthant.

5.2 Finite maxima, local solutions, and coordinate boundaries

Lemma 5.1 (Finite maxima, including ties)**.** *For finitely many continuously differentiable functions y_j, set $v(t) = \max_j y_j(t)$ and $I(t) = \{j : y_j(t) = v(t)\}$. Then v is locally Lipschitz, its right derivative exists at every time with a right neighborhood in the interval, and*

$$D^+ v(t) = \max_{j\in I(t)} \dot{y}_j(t). \tag{2}$$

Proof. Bounded derivatives on compact time intervals make the finite maximum Lipschitz. At a fixed t, every nonmaximizing index has a positive gap from $v(t)$. Finiteness and continuity ensure that none of these indices maximizes at $t + \tau$ for sufficiently small $\tau > 0$. Uniformly over $I(t)$, $y_j(t + \tau) = v(t) + \tau\dot{y}_j(t) + o(\tau)$. Taking the maximum and dividing by τ proves the assertion. □

Proposition 5.2 (Local well-posedness and nonnegative invariance)**.** *For any finite real coefficients, F is C^∞ on $\mathcal{U}$. Every $x^0 \in \mathcal{U}$ has a unique maximal classical forward solution $x : [0, T_{\max}) \to \mathcal{U}$, with continuous dependence on initial data. If $c \geq 0$ and $x^0 \in \mathbb{R}^n_{\geq 0}$, then $x(t) \in \mathbb{R}^n_{\geq 0}$ for $t < T_{\max}$, regardless of the signs of the other coefficients.*

Proof. The denominators $1 + x_i$ and $1 + x_{i+q}$ are strictly positive on $\mathcal{U}$. Each component of F is a sum of polynomials and rational functions with nonvanishing denominators, hence is smooth and locally Lipschitz. Standard local ODE existence, uniqueness, and maximal continuation theory on an open domain applies.

On each coordinate face,

$$x_i = 0, \quad x \in \mathbb{R}^n_{\geq 0} \quad \Longrightarrow \quad F_i(x) = cx_{i-1} \geq 0. \tag{3}$$

To treat tangent faces rigorously, fix $T < T_{\max}$ and let $x^+(t)$ be the coordinatewise positive part of $x(t)$. The curves x, x^+ and their connecting line segments lie in a compact rectangle in $\mathcal{U}$. Choose a Lipschitz constant L for F on this rectangle, in the maximum norm, and put

$$V(t) = \max\{0, -x_0(t), \ldots, -x_{n-1}(t)\} = \left\|x(t) - x^+(t)\right\|_\infty .$$

If $V > 0$ and $-x_i = V$, then $x_i^+ = 0$, so $F_i(x^+) \geq 0$ and $-F_i(x) \leq |F_i(x) - F_i(x^+)| \leq LV$. If $V = 0$, any active coordinate has $x_i = 0$ and $-F_i(x) \leq 0$; the constant zero is also active. Lemma 5.1 gives $D^+V \leq LV$. Since V is Lipschitz, this inequality holds for its ordinary derivative almost everywhere. Integration of $(e^{-Lt}V)' \leq 0$, with $V(0) = 0$, gives $V = 0$ on $[0, T]$. The choice of T was arbitrary. □

Proposition 5.3 (Persistence and propagation of positivity)**.** *For $c \geq 0$ and nonnegative initial data, every initially positive coordinate remains strictly positive on $[0, T_{\max})$. In particular, strictly positive initial data stay strictly positive. If $c > 0$ and $x^0 \neq 0$, all coordinates are strictly positive at every $0 < t < T_{\max}$. If $c = 0$, initially zero coordinates remain zero. The zero initial state remains identically zero.*

Proof. Along a nonnegative solution write

$$\dot{x}_i = b_i(t)x_i + d_i(t), \qquad b_i = a - kx_i - \frac{g}{1 + x_{i+q}} - hx_{i-q}, \qquad d_i = \frac{cx_{i-1}}{1 + x_i} \geq 0.$$

These coefficients are continuous on every compact time interval in the maximal interval. The integrating-factor formula gives

$$x_i(t) = x_i^0 e^{\int_0^t b_i(s)\, ds} + \int_0^t e^{\int_s^t b_i(r)\, dr} d_i(s)\, ds. \tag{4}$$

This proves persistence of an initially positive coordinate. If $c > 0$, a predecessor positive at every $s > 0$ supplies a strictly positive integrand to its successor at every $s > 0$, making that successor positive at every $t > 0$. At most $n - 1$ iterations along the step-one cycle reach all nodes, independently of q. For $c = 0$ the integral vanishes, proving preservation of zero coordinates. Finally $F(0) = 0$ and uniqueness proves the zero-data assertion. □

5.3 Maximum comparison and the exact comparison radius

Under (P01-W), define

$$b(s) = a - ks + \frac{c}{1 + s}, \qquad H(s) = sb(s) \quad (s \geq 0),$$

and set

$$X_{\max} = R := \max\left\{0, \frac{a - k + \sqrt{(a + k)^2 + 4kc}}{2k}\right\}. \tag{5}$$

This radius is defined by a scalar upper comparison; it is not asserted to be the smallest invariant cube of the full vector field.

Lemma 5.4 (Dini inequality and scalar root)**.** *Every nonnegative solution under* (P01-W) *satisfies*

$$D^+X = \max_{i:x_i=X} F_i(x) \le H(X) - \frac{gX}{1+X} \le H(X). \tag{6}$$

If $a+c>0$, then $R>0$ is the unique positive zero of b, and

$$kR^2 + (k-a)R - (a+c) = 0. \tag{7}$$

If $a+c\le 0$, then $R=0$ and $H(s)\le -ks^2$ for $s\ge 0$. For $R>0$ and $s\ge R$,

$$H(s) = s\left[-k(s-R) + c\left(\frac{1}{1+s} - \frac{1}{1+R}\right)\right] \le -ks(s-R). \tag{8}$$

Proof. Lemma 5.1 includes every tied maximizing coordinate. At $x_i = X$, nonnegative invariance gives $0 \le x_{i-1}, x_{i+q}, x_{i-q} \le X$. Consequently

$$\frac{cx_{i-1}}{1+X} \le \frac{cX}{1+X}, \qquad -\frac{gX}{1+x_{i+q}} \le -\frac{gX}{1+X}, \qquad -hXx_{i-q} \le 0.$$

The middle inequality uses the negative prefactor and $g \ge 0$. Adding proves (6); no independence of neighbor coordinates is required.

Now $b'(s) = -k - c/(1+s)^2 < 0$, $b(0) = a+c$, and $b(s) \to -\infty$. If $a+c>0$, multiply $b(R)=0$ by $1+R$ to obtain (7). Its constant term is negative and its discriminant is $(a+k)^2 + 4kc$, so its positive root is (5). If $a+c\le 0$, then $k-a>0$ and

$$(a+k)^2 + 4kc - (k-a)^2 = 4k(a+c) \le 0.$$

Thus the raw root in (5) is nonpositive, $R=0$, and $b(s) \le a+c-ks \le -ks$. Subtracting $b(R)=0$ when $R>0$ gives (8); its inequality follows from $c\ge 0$ and $s\ge R$. □

Theorem 5.5 (Global continuation and invariant attracting cubes)**.** *Under* (P01-W)*, every $x^0 \in \mathbb{R}^n_{\ge 0}$ has a unique classical solution for all $t\ge 0$. The scalar solution*

$$\dot{u} = H(u), \qquad u(0) = X(0) \tag{9}$$

is global and nonnegative, and

$$0 \le X(t) \le u(t) \le \max\{X(0), R\}. \tag{10}$$

Every K_r with $r\ge R$ is positively invariant. Furthermore,

$$\operatorname{dist}_\infty(x(t), K_R) \le e^{-kRt}(X(0)-R)_+ \qquad (R>0), \tag{11}$$

$$X(t) \le \frac{X(0)}{1+kX(0)t} \qquad (R=0). \tag{12}$$

There are also initial-state-independent bounds, for $t>0$:

$$X(t) \le \frac{R}{1-e^{-kRt}} \qquad (R>0), \tag{13}$$

$$X(t) \le \frac{1}{kt} \qquad (R=0). \tag{14}$$

Thus K_R attracts all nonnegative solutions uniformly on bounded initial sets, and indeed uniformly over the entire orthant as $t\to\infty$. Every larger cube K_r, $r>R$, absorbs the entire orthant by the common time

$$T_r = \begin{cases} \dfrac{1}{kR}\log\dfrac{r}{r-R}, & R>0, \\ \dfrac{1}{kr}, & R=0. \end{cases} \tag{15}$$

Proof. The scalar field H is smooth on $(-1,\infty)$, $H(0)=0$, and $H(s)\le 0$ for $s\ge R$. Its solution starting nonnegative stays in $[0,M]$, where $M=\max\{u(0),R\}$: uniqueness prevents crossing zero, and the upper endpoint has nonpositive field. The latter barrier also follows by the positive-part argument below applied to $(u-M)_+$, using $H(M)\le 0$. Compact containment gives global scalar continuation.

We justify the vector comparison without assuming differentiability of X everywhere. On a compact subinterval of the vector maximal interval, let L be a Lipschitz constant for H on an interval containing X and u. The Lipschitz function

$$W=(X-u)_+=\max\{0,x_i-u : i\in\mathbb{Z}_n\}$$

satisfies $D^+W\le H(X)-H(u)\le LW$ where $W>0$. At $W=0$, any active coordinate has $x_i=u=X$, unless the constant zero is the only active function; Lemma 5.1 and (6) give $D^+W\le 0$. Integrating the almost-everywhere inequality, as in Proposition 5.2, yields $W=0$. This proves (10) on the maximal interval.

If $T_{\max}<\infty$, the vector trajectory stays in K_M, a compact subset of $\mathcal{U}$ at distance at least one from every singular hyperplane. The field is bounded on a compact neighborhood in $\mathcal{U}$, hence $\dot{x}$ is bounded. Thus $x(t)$ has a limit in K_M as $t\uparrow T_{\max}$. Local existence at that limit and uniqueness extend the solution past $T_{\max}$, a contradiction. This proves global continuation and absence of finite-time blow-up. If $X(0)\le r$ and $r\ge R$, (10) proves invariance.

For $R>0$, (8) gives $\dot{u}\le -ku(u-R)\le -kR(u-R)$ where $u\ge R$. It follows that $(u(t)-R)_+\le e^{-kRt}(u(0)-R)_+$, including solutions initially at or below R, and hence (11). If $u(0)>R$, set $w=1-R/u$ while $u>R$. Then

$$\dot{w}=\frac{R\dot{u}}{u^2}\le -kRw,\qquad 0<w(0)<1.$$

Solving this inequality gives (13) for u and therefore for X. If u reaches R, invariance supplies the same bound afterwards; if $u(0)\le R$, it is immediate. For $R=0$, integrating $(1/u)'\ge k$ where $u>0$, using $\dot{u}\le -ku^2$, proves (12) and (14); zero initial data follow by uniqueness.

Attraction follows from these estimates and (1). For $R>0$ the distance is at most $R/(e^{kRt}-1)$ independently of initial data; for $R=0$ use $1/(kt)$. Substitution of T_r into (13) or (14) gives $X(T_r)\le r$, and invariance supplies all subsequent times. □

5.4 Finite-time absorption at the comparison radius

Theorem 5.6 (Exact absorption criterion for K_R)**.** *Assume* (P01-W). *If $R=0$, no initially nonzero nonnegative solution enters $K_R=\{0\}$ at finite time. This cube is invariant and globally attracting but not absorbing.*

If $R>0$, the following are equivalent:

(i) K_R absorbs every nonnegative solution in finite time.

(ii) K_R absorbs bounded subsets of $\mathbb{R}^n_{\ge 0}$.

(iii) One finite absorption time is valid for the entire orthant.

(iv) The coefficients and offset satisfy

$$g>0 \quad or \quad [\,h>0 \text{ and } (c>0 \text{ or } q\equiv 0 \pmod n)\,]. \tag{16}$$

Canonically $R > 0$, $g > 0$, and the entire orthant enters K_R by

$$T_* = \frac{\log 2}{kR} + \frac{1+R}{g}. \tag{17}$$

If $g > 0$ and $X(0) \le M$, an alternative entry-time bound is

$$T_M = \frac{(M-R)_+(1+R)}{gR}. \tag{18}$$

Proof. When $R = 0$, Proposition 5.3 ensures that some coordinate of any initially nonzero solution stays positive at each finite time.

Now suppose $R > 0$ and set $t_0 = (\log 2)/(kR)$. By (13), every trajectory is in the invariant cube K_{2R} by t_0. If $g > 0$, then whenever $X \ge R$,

$$D^+X \le -\frac{gX}{1+X} \le -\frac{gR}{1+R} =: -\delta < 0, \tag{19}$$

using $H(X) \le 0$. Absolute continuity permits integration before the first entry into $X \le R$. If $X(0) \le M$, a longer time outside than $(M-R)_+/\delta$ would force $X \le R$, a contradiction. Invariance gives all later times. Applying this after t_0 with $M = 2R$ proves (17).

If $g = 0$, $h > 0$, and $q \equiv 0 \pmod n$, a maximizing coordinate has $x_{i-q} = X$. Thus $D^+X \le H(X) - hX^2 \le -hR^2$ when $X \ge R$, giving common absorption by $t_0 + 1/(hR)$.

For the remaining sufficient case $g = 0$, $h > 0$, $c > 0$, we give a coordinatewise quantitative argument. Put $M = 2R$ and fix any $\tau > 0$. On K_M,

$$\dot{x}_i \ge -\beta x_i + \eta x_{i-1}, \qquad \beta = (-a)_+ + (k+h)M + g, \quad \eta = \frac{c}{1+M} > 0. \tag{20}$$

Indeed $a \ge -(-a)_+$, $kx_i \le kM$, $g/(1+x_{i+q}) \le g$, $hx_{i-q} \le hM$, and $c/(1+x_i) \ge c/(1+M)$. On $[s, s+\tau]$ in this cube, the transformed variables $z_i(r) = e^{\beta(r-s)}x_i(r)$ satisfy

$$z_i(r) \ge x_i(s) + \eta \int_s^r z_{i-1}(v)\, dv.$$

Repeated substitution, keeping the remaining nested integral nonnegative, yields

$$x_i(s+\tau) \ge e^{-\beta\tau} \sum_{j=0}^{n-1} \frac{(\eta\tau)^j}{j!} x_{i-j}(s). \tag{21}$$

A j-fold integral of a constant over the time simplex equals $\tau^j/j!$, which gives the displayed coefficients.

For a trajectory that has not entered K_R by time $t \ge t_0 + \tau$, we have $X(t-\tau) > R$. The indices $i-j$, $0 \le j \le n-1$, cover the cycle once. Thus (21) gives, for every i,

$$x_i(t) \ge d := Re^{-\beta\tau} \min_{0 \le j \le n-1} \frac{(\eta\tau)^j}{j!} > 0. \tag{22}$$

At such times, every maximizing coordinate satisfies $F_i(x) \le H(X) - hXx_{i-q} \le -hRd$. Starting from $X \le 2R$ at $t_0 + \tau$, integration proves entry by $t_0 + \tau + 1/(hd)$, or earlier. We have proved (iv)⇒(iii).

The implications (iii)⇒(ii)⇒(i) are immediate. If (iv) fails and $g = h = 0$, choose all initial coordinates equal to $s_0 > R$. Direct substitution verifies the solution $x_i(t) = s(t)$ with $\dot{s} = H(s)$. It

stays above R at each finite time: a first meeting with the constant scalar solution R would violate uniqueness, also backwards near that meeting. Hence it never enters K_R. The other failure case is $g = 0$, $h > 0$, $c = 0$, $q \not\equiv 0 \pmod{n}$. Here $R = a/k > 0$. Choose one initial coordinate $x_j(0) > R$ and all others zero. The zero coordinates stay zero and $j - q \neq j$, so $\dot{x}_j = ax_j - kx_j^2$. It likewise stays strictly above a/k at every finite time. These cases exhaust failure of (iv) and contradict (i), completing the equivalence. □

5.5 Canonical conclusions and weaker-assumption limits

Corollary 5.7 (Canonical foundation)**.** *For canonical $n \geq 5$, $1 \leq q \leq n-1$ and $a, k, c, g, h > 0$, every nonnegative initial state has a unique global nonnegative solution. Every initially positive coordinate stays positive, and every nonzero initial state becomes strictly positive in every coordinate at each positive time. The cube $[0, X_{\max}]^n$, with*

$$X_{\max} = \frac{a - k + \sqrt{(a+k)^2 + 4kc}}{2k} > 0,$$

is positively invariant, globally attracting, and finite-time absorbing for the entire orthant with the common bound (17)*. All larger cubes are also invariant and absorbing. No condition on* $\gcd(n, q)$ *is required.*

Proof. The canonical assumptions imply (P01-W), $a + c > 0$, $c > 0$ and $g > 0$. Apply the preceding results. □

Remark 5.8 (Archived radius and the meaning of stability). The blueprint's candidate radius

$$B = \frac{a + \sqrt{a^2 + 4kc}}{2k}$$

is the larger real root of $as - ks^2 + c = 0$, and is nonnegative. If $c > 0$, then $B > 0$ and $H(B) = -c/(1 + B) < 0$, so $B > R$. Its cube is therefore also invariant and absorbing by Theorem 5.5, even when $g = h = 0$. The smaller radius R retains $cX/(1+X)$ instead of replacing it by c. If $c = 0$, then $B = R$ and the exact-radius criterion applies. Neither radius is claimed optimal after all g, h, q information is used.

Invariance concerns remaining inside; absorption concerns finite-time entry; attraction alone permits approaching from outside forever. The distance estimates also show Lyapunov stability of K_R as a set, relative to $\mathbb{R}^n_{\geq 0}$, because distance never exceeds its initial value. Under the convention that asymptotic stability of a set means Lyapunov stability plus attraction, K_R is globally asymptotically stable as a set. This is not convergence to a particular point, nor a claim that the cube is a minimal global attractor or is invariant in both time directions.

Proposition 5.9 (Global existence when $k = 0$ is allowed)**.** *Global nonnegative existence and the positivity conclusions still hold for $k \geq 0$, $a \in \mathbb{R}$, $c, g, h \geq 0$, any $n \geq 1$, and any residue q. For $k = 0$, no bounded absorbing cube is guaranteed over this parameter class.*

Proof. Local existence and nonnegative invariance did not require $k > 0$. Discarding nonpositive losses at a maximizing coordinate gives

$$D^+ X \leq aX + \frac{cX}{1+X} \leq (a_+ + c)X.$$

The Lipschitz/Dini argument yields $X(t) \leq X(0)e^{(a_+ + c)t}$. If the maximal time were finite, this would confine the trajectory to a compact subset of $\mathcal{U}$, and the earlier continuation argument would extend it. Positivity follows from Proposition 5.3. For the last assertion take $a > 0$, $k = c = g = h = 0$: an initially positive coordinate equals $x_i(0)e^{at}$ and is unbounded. □

5.6 Checks and possible failure points

1. **Boundary tangency.** The negative-part maximum proves invariance even when the derivative on a face vanishes. A bare first-contact derivative assertion is insufficient. If $c < 0$ and $n \geq 2$, a face state with $x_i = 0$, $x_{i-1} > 0$ has $\dot{x}_i < 0$, so invariance can fail. No forward assertion concerns initial data outside $\mathbb{R}^n_{\geq 0}$.

2. **Tied maxima and loss signs.** All active indices occur in (2). The negative prefactor in the g term reverses the reciprocal inequality. Coincident offsets are harmless because the estimates do not optimize neighbors independently.

3. **Root degeneracy.** Equation (7) characterizes the positive comparison root. If $a + c \leq 0$, the selected radius is zero and need not be a root of that quadratic: the factor s in $H(s)$ supplies the zero. No expression containing $1/R$ is used when $R = 0$.

4. **Absorption mechanism.** The equality $H(R) = 0$ alone does not imply finite-time entry. Canonical absorption uses the retained g loss; when it vanishes, the exact criterion supplies another strict mechanism or a nonabsorbing trajectory. Positivity prevents finite-time entry into K_0 from nonzero data.

5. **Dimension and connectivity.** Nothing requires $n \geq 5$, coprimality, or distinct offsets. Positivity propagates through the step-one input when $c > 0$. The $q \equiv 0$ absorption case belongs only to the weaker extension of the canonical range. Strict positivity at each finite time is not a uniform positive lower bound as $t \to \infty$.

6. **Coefficient edges.** Strict $a, c, g, h > 0$ is unnecessary for the bounded-region theorem when $k > 0$; even $a \in \mathbb{R}$ is allowed. The condition $k \geq 0$ suffices for global existence. If $k < 0$ and $a = c = g = h = 0$, then $\dot{x}_i = |k| x_i^2$ blows up in finite time for a positive coordinate. If $h < 0$, the dissipative bounds also fail in general: take $k > 0$, $a = c = g = 0$, $h < -k$ and equal positive initial coordinates. Direct substitution gives $\dot{s} = -(k + h)s^2$ and finite-time blow-up. These examples are outside the proved domain, not necessity claims for every signed parameter choice.

7. **Continuation and status.** Boundedness is used together with compact containment in the smooth open domain, excluding denominator singularities. All maximum, comparison, and coordinate integral inequalities are proved here. Only standard local ODE existence/uniqueness/continuation theory is invoked. This module remains DRAFT pending a separate independent audit; no numerical experiment or later proof-module result is used.

6 P02: Synchrony and synchronous equilibria

Status: DRAFT; independent audit pending. The standing assumptions are the canonical A1–A3. In particular, $n \geq 5$, $1 \leq q \leq n-1$, $a, k, c, g, h > 0$, and the open extension domain is $\mathcal{U} = (-1, \infty)^n$. The nonnegative model state space is $\mathbb{R}^n_{\geq 0}$. P01 v1.1 is VERIFIED / AUDITED: its required wording correction E01 is already incorporated. Only its local uniqueness and, for nonnegative initial data, global continuation are needed here. The calculations below start from $(\mathrm{DM}_{n,q})$. No assertion about nonsynchronous equilibria, equilibrium stability, or bifurcation type is made.

6.1 Invariant synchronous subspace

Proposition 6.1 (Invariance of the synchronous subspace)**.** *The set*

$$\mathcal{S} \cap \mathcal{U} = \{s\mathbf{1} : s \in (-1, \infty)\}$$

is invariant for the canonical vector field for the entire maximal solution interval in $\mathcal{U}$. Its scalar restriction is

$$\dot{s} = f(s) := as - ks^2 + \frac{cs}{1+s} - \frac{gs}{1+s} - hs^2, \qquad s \in (-1, \infty). \tag{23}$$

This conclusion needs no coprimality or distinct-offset assumption. Indeed, the substitution and local invariance remain valid for any integer $n \geq 1$, any residue q modulo n, and any finite real coefficients. The canonical restrictions are retained for the equilibrium classification below.

Proof. Put $x = s\mathbf{1}$ with $s > -1$. For every $i \in \mathbb{Z}_n$, each of $x_i, x_{i-1}, x_{i+q}, x_{i-q}$ equals s. Direct substitution gives

$$F_i(s\mathbf{1}) = as - ks^2 + \frac{cs}{1+s} - \frac{gs}{1+s} - hs^2 = f(s).$$

Every component is identical, so $F(s\mathbf{1}) = f(s)\mathbf{1}$. Coincident offsets retain all the displayed terms; no distinctness of neighbors is used. Nor is the size or connectedness of a step-q cycle used.

The scalar field is smooth on $(-1, \infty)$. A scalar solution $s(t)$ lifts to the vector solution $s(t)\mathbf{1}$. Uniqueness from Proposition 5.2 makes this lift agree with the vector solution having initial value $s(0)\mathbf{1}$ wherever both exist. If the scalar maximal interval ended earlier while the vector solution continued in $\mathcal{U}$, the latter would supply a limit $s_*\mathbf{1} \in \mathcal{U}$ at that time. Local scalar continuation at $s_* > -1$ would extend the lift, a contradiction. Conversely, a longer scalar interval would extend the vector solution. Thus the maximal intervals coincide and synchrony persists throughout. The same local uniqueness argument applies on either side of an initial time. The scalar domain inherited from $\mathcal{U}$ is exactly $(-1, \infty)$. □

6.2 Scalar synchronous dynamics

Use the reserved abbreviations and one local auxiliary abbreviation

$$A = k + h > 0, \qquad \mu = a + c - g, \qquad b = a - A. \tag{24}$$

Here b is used only within P02; it is not the archived comparator's coefficient. In particular $b > -A$ because $a > 0$. The exact restriction is

$$f(s) = s\left(a - As + \frac{c-g}{1+s}\right) = \frac{s}{1+s}\big(\mu + bs - As^2\big), \qquad s > -1. \tag{25}$$

For $s(0) \geq 0$, P01 and Proposition 6.1 give a global nonnegative scalar solution. For $-1 < s(0) < 0$, only the maximal interval within this open domain is asserted. Even when $c = g$ cancels the two rational terms along synchrony, this does not enlarge the canonical domain or admit $s = -1$.

6.3 The origin and the nonzero equilibrium equation

Proposition 6.2 (Equilibrium equation with domain restrictions)**.** *The origin* $0\mathbf{1}$ *is an equilibrium for every canonical parameter tuple. Every other synchronous equilibrium in* $\mathcal{U}$ *is exactly* $s\mathbf{1}$*, where* $s \neq 0$*,* $s > -1$*, and*

$$\begin{aligned} 0 = Q(s) &:= As^2 - bs - \mu \\ &= (k+h)s^2 + (k+h-a)s - (a+c-g). \end{aligned} \tag{26}$$

Define the discriminant

$$\begin{aligned} D &:= b^2 + 4A\mu \\ &= (a-A)^2 + 4A(a+c-g) = (a+A)^2 + 4A(c-g). \end{aligned} \tag{27}$$

The complete list of real polynomial roots, when $D \geq 0$*, is*

$$s_\pm = \frac{b \pm \sqrt{D}}{2A} = \frac{a-(k+h) \pm \sqrt{(a+k+h)^2 + 4(k+h)(c-g)}}{2(k+h)}. \tag{28}$$

They are distinct for $D > 0$ *and coincide for* $D = 0$*. When* $D < 0$ *there are no real roots of* Q*.*

Proof. Directly in (23), all five terms vanish at $s = 0$. Thus the origin is always an equilibrium, before any division is performed. Now take $s \neq 0$ with $s > -1$. Dividing (25) by s and multiplying by the strictly positive number $1 + s$ gives

$$(a - As)(1+s) + c - g = \mu + (a-A)s - As^2 = 0.$$

Both operations are reversible on this stated set, giving (26). Completing the square yields

$$Q(s) = A\left(s - \frac{b}{2A}\right)^2 - \frac{D}{4A}. \tag{29}$$

Expansion of $b^2 + 4A\mu$ proves both discriminant forms and the quadratic formula gives (28). Finally,

$$Q(-1) = A + b - \mu = g - c. \tag{30}$$

Thus $s = -1$ is a polynomial root precisely when $g = c$, but it is never a point of the canonical domain. Multiplication by $1 + s$ does not authorize retaining that root. A root $s = 0$ of Q, when present, is the already counted origin rather than another equilibrium. □

6.4 Complete classification in the canonical parameter regime

In this subsection, a *mathematical root* means a real root of Q, including a zero or an inadmissible root. A *DM-domain root* additionally satisfies $s > -1$. A *nonnegative model equilibrium* has $s \geq 0$, and a *positive model equilibrium* has $s > 0$. The ever-present origin is added once to the admissible nonzero roots. All counts below count distinct values, not algebraic multiplicity.

Theorem 6.3 (Complete synchronous-equilibrium classification)**.** *Under the canonical assumptions, the synchronous equilibrium set is*

$$\{0\mathbf{1}\} \ \cup \ \{s\mathbf{1} : Q(s) = 0,\ s > -1,\ s \neq 0\}. \tag{31}$$

The following eleven cases are exhaustive. In the table, r *counts real nonzero polynomial roots,* d *counts those with* $s > -1$*,* N_0 *counts all nonnegative synchronous equilibria (including the origin), and* N_+ *counts the strictly positive ones. The total number of synchronous equilibria in* $\mathcal{U}$ *is* $1 + d$*.*

Case	*Parameter conditions*	*Real roots of Q and location*	r	d	N_0	N_+
I	$\mu > 0,\ g < c$	$s_- < -1 < 0 < s_+$	*2*	*1*	*2*	*1*
II	$\mu > 0,\ g = c$	$s_- = -1,\ s_+ = a/A > 0$	*2*	*1*	*2*	*1*
III	$\mu > 0,\ g > c$	$-1 < s_- < 0 < s_+$	*2*	*2*	*2*	*1*
IV	$\mu = 0,\ b > 0$	$s_- = 0,\ s_+ = b/A > 0$	*1*	*1*	*2*	*1*
V	$\mu = 0,\ b = 0$	$s_- = s_+ = 0$ *(double)*	*0*	*0*	*1*	*0*
VI	$\mu = 0,\ b < 0$	$-1 < s_- = b/A < 0 = s_+$	*1*	*1*	*1*	*0*
VII	$\mu < 0,\ D < 0$	*No real roots; any sign of b*	*0*	*0*	*1*	*0*
VIII	$\mu < 0,\ D > 0,\ b > 0$	$0 < s_- < s_+$	*2*	*2*	*3*	*2*
IX	$\mu < 0,\ D = 0,\ b > 0$	$s_- = s_+ = b/(2A) > 0$	*1*	*1*	*2*	*1*
X	$\mu < 0,\ D > 0,\ b < 0$	$-1 < s_- < s_+ < 0$	*2*	*2*	*1*	*0*
XI	$\mu < 0,\ D = 0,\ b < 0$	$-\frac{1}{2} < s_- = s_+ = b/(2A) < 0$	*1*	*1*	*1*	*0*

When $\mu < 0$ and $b = 0$, necessarily $D = 4A\mu < 0$, so case VII applies. Each appearance of $s_\pm$ refers to (28)*. The only repeated polynomial roots are in cases V, IX and XI.*

As zeros of the original smooth scalar field f on $(-1, \infty)$, all nonzero equilibria are simple except the double roots in IX and XI. The origin is simple if $\mu \neq 0$, double if $\mu = 0$, $b \neq 0$, and triple if $\mu = b = 0$. The algebraic continuation of the nonzero equilibrium curve meets the origin exactly at $\mu = 0$: through s_- in case IV, through s_+ in case VI, and through both coincident root labels in case V. No negative root represents a state in $\mathbb{R}^n_{\geq 0}$.

Proof. Proposition 6.2 proves completeness of (31). The Vieta relations are

$$s_- + s_+ = \frac{b}{A}, \qquad s_- s_+ = -\frac{\mu}{A}. \tag{32}$$

We first establish the domain test for every real pair of roots. Because $a > 0$ and $A > 0$,

$$s_+ \geq \frac{b}{2A} = \frac{a}{2A} - \frac{1}{2} > -\frac{1}{2} > -1.$$

For the smaller root,

$$s_- > -1 \iff \sqrt{D} < a + A \iff D < (a + A)^2 \iff g > c. \tag{33}$$

The middle equivalence uses $a + A > 0$ and $D \geq 0$. Equality gives $s_- = -1$ exactly when $g = c$; the reverse inequality gives $s_- < -1$ exactly when $g < c$. These are polynomial-root statements on the real-discriminant set, not assertions that such roots are admissible.

If $\mu > 0$, then $D > b^2$, so $s_- < 0 < s_+$ and both roots are nonzero and simple. The domain test gives I–III. When $g = c$, $D = (a + A)^2$ and the stated roots $-1, a/A$ follow directly.

If $\mu = 0$, then $Q(s) = s(As - b)$. If $b \neq 0$, its distinct roots are 0 and b/A. Since $b/A = a/A - 1 > -1$, the latter is always domain-admissible and has the sign of b. Their ordering gives IV and VI. If $b = 0$, $Q(s) = As^2$ gives V. The origin is counted only once in the equilibrium set in all three cases.

If $\mu < 0$, real roots exist precisely when $D \geq 0$. For $D < 0$ we obtain VII. For $D \geq 0$, the product in (32) is strictly positive and $b \neq 0$ (otherwise $D < 0$). Thus both roots are nonzero with the sign of their sum, namely the sign of b. They are distinct for $D > 0$ and double for $D = 0$. Also $g > a + c > c$, so both are admissible by (33). This gives VIII–XI, including the sharper negative repeated-root bound from $b > -A$. These arguments yield every count in the table.

On $s > -1$, the identity $f(s) = -sQ(s)/(1 + s)$ shows that the nonzero zeros have the same multiplicity as their polynomial roots. At the origin, use (25) without dividing: the numerator is

$\mu s + bs^2 - As^3$ and $1 + s$ does not vanish. This gives the stated orders one, two and three. Finally $Q(0) = -\mu$, so the nonzero equation's algebraic continuation contains $s = 0$ exactly when $\mu = 0$. The formulas at that locus give its root labels, and the local branch description below confirms the approach by nonzero roots. The counting and multiplicity assertions are purely algebraic and carry no stability conclusion. □

6.5 Positive synchronous equilibria: necessary and sufficient conditions

Theorem 6.4 (Exact positive-equilibrium alternatives)**.** *In the canonical parameter regime the following four alternatives partition parameter space:*

(i) There is no positive synchronous equilibrium if and only if

$$[\,\mu = 0,\ b \le 0\,] \quad \text{or} \quad [\,\mu < 0,\ (b \le 0 \text{ or } D < 0)\,]. \tag{34}$$

(ii) There is exactly one positive synchronous equilibrium and it is simple if and only if $\mu > 0$, or $\mu = 0$ and $b > 0$. It is $s_+\mathbf{1}$; in the second case $s_+ = b/A$.

(iii) There are exactly two distinct positive synchronous equilibria if and only if $\mu < 0$, $b > 0$, and $D > 0$. They are $s_-\mathbf{1}, s_+\mathbf{1}$, both simple.

(iv) There is exactly one positive synchronous equilibrium and it is repeated if and only if $\mu < 0$, $b > 0$, and $D = 0$. It is a double root, $s_-\mathbf{1} = s_+\mathbf{1} = (b/(2A))\mathbf{1}$.

Consequently, exactly one distinct positive equilibrium (without a simplicity qualification) occurs if and only if (ii) or (iv) holds. There are never more than two positive synchronous equilibria.

Proof. The cases of Theorem 6.3 prove every implication, including the necessity and all equalities. One can also reconstruct the positive-root count directly from $H(s) = \mu + bs - As^2$ for $s > 0$. If $b \le 0$, then $H'(s) = b - 2As < 0$ on this interval: it has one positive zero exactly when $\mu > 0$. If $b > 0$, its maximum on $s \ge 0$ is at $s = b/(2A) > 0$ and has value $D/(4A)$. For $\mu < 0$, that maximum is positive, zero, or negative precisely in alternatives (iii), (iv), or (i). For $\mu = 0$, the zero at the endpoint is excluded from the positive count and the other root is $b/A > 0$. For $\mu > 0$, the opposite-sign Vieta product leaves one positive root. This gives an independent sign-and-vertex justification of the list. □

6.6 Algebraic branch loci and degeneracies

Proposition 6.5 (Branch-intersection and repeated-root loci)**.** *For fixed positive a, k, c, h, define*

$$g_f := a + c + \frac{(a-A)^2}{4A} = c + \frac{(a+A)^2}{4A}, \qquad s_f := \frac{a-A}{2A}. \tag{35}$$

The following statements concern only the scalar equilibrium equation.

(i) The branch-intersection locus *is $\mu = 0$, or $g = a + c$. It is the set where the algebraic continuation of the nonzero equilibrium equation meets the origin branch.*

(ii) The repeated-root locus *of Q is $D = 0$, or $g = g_f$, and its root is s_f. Indeed,*

$$D = 4A(g_f - g), \qquad g_f - (a + c) = \frac{(a-A)^2}{4A} \ge 0. \tag{36}$$

The root is positive for $a > A$, negative for $0 < a < A$, and zero for $a = A$. It always lies in $(-1/2, \infty)$ and hence in the DM domain. It is a nonzero double equilibrium only when $a \ne A$.

(iii) The two parameter loci meet exactly at the algebraic degeneracy $a = A$, $g = a + c$. *There* $Q(s) = As^2$ *and*

$$f(s) = -\frac{As^3}{1+s}. \tag{37}$$

The origin is the sole synchronous equilibrium in the domain at that parameter tuple; it is a triple zero of f *and a double root of* Q*, not an additional nonzero equilibrium.*

Proof. The nonzero equation can be written, without approximation, as

$$\mu = As^2 - bs, \qquad g = a + c + bs - As^2 = g_f - A(s - s_f)^2. \tag{38}$$

This is its algebraic continuation, to be filtered by $s > -1$, $g > 0$, and $s \neq 0$ when counting nonzero model equilibria. At $s = 0$ it gives exactly $g = a + c$. Completing the square or using (29) gives $D = 0$, $g = g_f$, and $s = s_f$ as the unique repeated-root condition. The difference of the two parameter values in (36) vanishes exactly at $a = A$. Since $a > 0$, $s_f = a/(2A) - 1/2 > -1/2$. At the common locus, substitution into (25) proves (37) and all the remaining assertions. □

For additional local algebra at the origin, fix a, A, c and vary g (equivalently μ). If $b \neq 0$, the root that tends to zero is

$$s_0(\mu) = \frac{b - \operatorname{sgn}(b)\sqrt{b^2 + 4A\mu}}{2A} = -\frac{\mu}{b} + \frac{A\mu^2}{b^3} + O(\mu^3). \tag{39}$$

This follows by expansion of the explicit square root near $\mu = 0$; $g = a + c - \mu$ remains positive in a sufficiently small neighborhood. Thus nonzero domain-admissible roots approach the origin from the neighboring parameter regions. At the intersection itself,

$$f(s)\big|_{\mu=0} = \frac{s^2(b - As)}{1+s} = bs^2 - as^3 + O(s^4). \tag{40}$$

The coefficient of the usual quadratic scalar term is $b = a - A$. It vanishes when $a = A$; this is precisely the degeneracy above.

More explicitly, for $a = A$ the nonzero polynomial equation is $As^2 = \mu$. For $\mu > 0$ its roots are

$$s_\pm = \pm\sqrt{\mu/A}. \tag{41}$$

The negative one is admissible exactly when $0 < \mu < A$ (equivalently $c < g < a + c$); it is -1 when $\mu = A$, and below -1 when $\mu > A$. At $\mu = 0$ both labels give the origin; for $\mu < 0$ no real nonzero roots exist. These are algebraic branch descriptions only. Any bifurcation theorem requires the separate work assigned to P09 and P10; none is asserted here.

6.7 Canonical parameter corollary

Corollary 6.6 (Positive branches in the original canonical parameters)**.** *Let* $a, k, c, g, h > 0$, $A = k + h$*, and let* g_f *be as in* (35)*. If* $a \leq k + h$*, there is one simple positive synchronous equilibrium exactly for* $0 < g < a + c$*, and none for* $g \geq a + c$*. If* $a > k + h$*, the exact alternatives are*

Parameter range	*Positive synchronous equilibria*
$0 < g \leq a + c$	*One simple equilibrium,* $s_+\mathbf{1}$
$a + c < g < g_f$	*Two distinct simple equilibria,* $s_-\mathbf{1}, s_+\mathbf{1}$
$g = g_f$	*One double equilibrium,* $s_f\mathbf{1}$
$g > g_f$	*None*

In particular, $a+c<g$ does not by itself preclude positive synchrony: two distinct positive equilibria exist if and only if

$$a > k+h, \qquad a+c < g < a+c+\frac{(a-k-h)^2}{4(k+h)}. \tag{42}$$

The upper equality gives one repeated positive equilibrium when $a > k+h$; the lower equality gives one simple positive equilibrium and the origin. The origin exists in every row, independently of the displayed positive count. There is no dependence on n, q, $\gcd(n,q)$, or distinctness of the offsets.

Proof. The sign of b is the sign of $a-(k+h)$; the sign of μ is the sign of $a+c-g$; and $D = 4A(g_f-g)$ with $A>0$. Substitute these identities into Theorem 6.4. □

6.8 Weaker assumptions and limits of the classification

The synchronous substitution and the scalar identity require only homogeneity, finite real coefficients, and nonvanishing denominators. They therefore apply under P01's weaker assumptions as well. For $A>0$, the polynomial formula, discriminant, Vieta relations, and the positive-root alternatives of Theorem 6.4 remain valid for real a, c, g: their positive-root proofs use only $A>0$. This observation is not a generalization of the canonical negative-root table. In particular, the domain test (33) uses $a+A>0$, and the bound $s_f > -1/2$ uses $a>0$. P01 allows real a and does not supply these inequalities. The complete domain-filtered classification in this module is intentionally restricted to the canonical positive parameter regime. No classification for $A=0$ or $A<0$ is claimed.

6.9 Checks and possible failure points

Root substitution and discriminant reconstruction. For $D \geq 0$, put $d=\sqrt{D}$ and $r = (b+\varepsilon d)/(2A)$, $\varepsilon \in \{-1,1\}$. Direct expansion gives

$$\mu + br - Ar^2 = \mu + \frac{2b(b+\varepsilon d)-(b+\varepsilon d)^2}{4A} = \mu + \frac{b^2-d^2}{4A} = 0.$$

For $r \neq -1$, substitution in the original scalar expression gives

$$ar - kr^2 + \frac{cr}{1+r} - \frac{gr}{1+r} - hr^2 = \frac{r}{1+r}(\mu + br - Ar^2) = 0.$$

Only $r > -1$ is retained as a DM equilibrium. If $r=-1$, the two original rational terms are undefined even when $c=g$, so no substitution or canceled extension admits that point. At $\mu=0$, the specialized root $r=b/A$ also gives $br-Ar^2=0$. At $D=0$, $r=s_f=b/(2A)$ gives $\mu+br-Ar^2 = \mu+b^2/(4A)=0$. At $g=c$, the retained root a/A gives $ar-Ar^2=0$. At $b=0$, the square-root branches give $\mu-Ar^2=0$. Finally, direct substitution at $r=0$ gives zero for every parameter tuple. These checks include every exact root formula displayed above, with s_0 in (39) being one of $s_\pm$; its truncated expansion is not used as a root.

Independently expanding the polynomial discriminant gives

$$(A-a)^2 - 4A[-(a+c-g)] = a^2+2aA+A^2+4A(c-g) = (a+A)^2+4A(c-g).$$

Also the sum of the two explicit roots is b/A, and their product is $(b^2-D)/(4A^2) = -\mu/A$, confirming both Vieta relations.

Exact representative cells and equality boundaries. The following rational tuples realize every row of the classification. Set $k = h = 1/2$, hence $A = 1$, and $c = 1$ throughout. All original coefficients are strictly positive. Roots are obtained by the formulas already proved; the counts refer to the preceding table.

Case	a	g	μ	D	(r, d, N_0, N_+)
I	2	1/2	5/2	11	$(2, 1, 2, 1)$
II	2	1	2	9	$(2, 1, 2, 1)$
III	2	2	1	5	$(2, 2, 2, 1)$
IV	2	3	0	1	$(1, 1, 2, 1)$
V	1	2	0	0	$(0, 0, 1, 0)$
VI	1/2	3/2	0	1/4	$(1, 1, 1, 0)$
VII	1	3	-1	-4	$(0, 0, 1, 0)$
VIII	2	25/8	$-1/8$	1/2	$(2, 2, 3, 2)$
IX	2	13/4	$-1/4$	0	$(1, 1, 2, 1)$
X	1/2	49/32	$-1/32$	1/8	$(2, 2, 1, 0)$
XI	1/2	25/16	$-1/16$	0	$(1, 1, 1, 0)$

Cases II, IV–VI, IX and XI test all relevant equality surfaces, including the simultaneous equality $a = A$, $g = a + c$. For $b < 0, 0, > 0$ inside the $\mu > 0$ cases, one may respectively take $a = 1/2, 1, 2$, and $g = 1/2, 1, 1 + a/2$ for I, II, III. For all three signs of b in VII, take the same three a values and $g = g_f + 1$. These choices also check that omitted sign subcases do not alter the counts. Internal exact-arithmetic checks supplement these displayed examples; they are not numerical dependencies or independent verification of the module.

For reproducibility of that finite check, take $V = \{1/4, 1/2, 1, 2, 3, 4\}$ and all $(a, A, c, g) \in V^4$. For each $(a, A, c) \in V^3$, also take $g \in \{c, a + c, g_f\} + \{-\eta, 0, \eta\}$, where $\eta = \min\{a, A, c, 1\}/64$. The union contains 2,868 distinct positive tuples. Set $k = A/3$, $h = 2A/3$ for this grid; use the displayed $k = h = 1/2$ for the eleven table examples. Exact arithmetic in $u + v\sqrt{D}$, with rational u, v, D, checks root residuals, Vieta relations, domain comparisons, and every positive-count equivalence and its converse. Including the eleven table examples, 4,438 original scalar residuals vanish exactly and 217 occurrences of $s = -1$ are explicitly excluded. All eleven cells and all equality surfaces are represented. These finite checks found no counterexample; they supplement rather than establish the exhaustive proof.

Boundary continuity. The ordered formulas $s_\pm$ are continuous on the parameter set $D \geq 0$, since $A > 0$. At $D \downarrow 0$ they both tend to s_f; for $D < 0$ their absence as real roots is recorded, not bridged by a real continuation. At $\mu \to 0$ with fixed $b > 0$, $(s_-, s_+) \to (0, b/A)$; with fixed $b < 0$, $(s_-, s_+) \to (b/A, 0)$. With $b = 0$ and $\mu \downarrow 0$, both labels tend to zero as in (41). At $g \to c$, $(s_-, s_+) \to (-1, a/A)$; the first root is excluded from the DM domain at the boundary. At $b \to 0$ all limits within $D \geq 0$ are again given by the same continuous formulas. The sign-specific table is therefore compatible with every boundary, although admissible root counts can change there.

Logical completeness and attempted counterexamples. The origin was retained before division, and every other equilibrium is equivalent to a nonzero root of Q in $s > -1$. This proves that none was lost or introduced. In particular, the excluded root -1 at $g = c$ and the zero root at $\mu = 0$ cannot be counted as additional nonzero equilibria. Positivity, discriminant sign, and root order partition all parameter tuples as in the theorem; each implication in the necessary-and-sufficient statements is checked in every cell. The independent positive-half-line argument supplies a second check. Case VIII is an explicit canonical counterexample to the rejected claim that $a + c \leq g$ rules out positive synchrony. Case IV is a counterexample even at equality. Case X shows that two admissible nonzero roots need not be positive; case XI shows that a repeated domain root need not be positive. Case V prevents counting a repeated zero as a positive equilibrium, and case II prevents admitting a canceled denominator root. All these distinctions are essential.

Priority checks for the independent P02 auditor. Independently reconstruct the following steps.

(1) Componentwise substitution with coincident offsets and uniqueness on the maximal interval in $\mathcal{U}$.

(2) The origin before division and equivalence to $Q = 0$ only for $s \neq 0$, $s > -1$.

(3) All discriminant identities, root formulas, and Vieta relations.

(4) The smaller-root domain test at $g = c$, including exclusion of $s = -1$ even when the synchronous rational terms cancel.

(5) Every cell, count, and multiplicity in Theorem 6.3, including $\mu < 0$ and negative roots.

(6) Both directions of every alternative in Theorem 6.4 and its original-parameter corollary.

(7) The branch graph, g_f, s_f, and the simultaneous degeneracy, distinguishing multiplicity of Q from that of f.

(8) All boundary limits and the quadratic coefficient in (40).

(9) The exact representative tuples and original-equation residuals, including the excluded-root and double-counting tests.

(10) The scope of weaker assumptions and the absence of any stability or bifurcation conclusion.

Internal checking and compilation do not change the DRAFT status.

7 P03: Origin and homogeneous linearization

Status: DRAFT; independent audit pending. Use the canonical demand-modulated field ($\mathrm{DM}_{n,q}$), with $n \geq 5$, $1 \leq q \leq n-1$, $a, k, c, g, h > 0$, and $\mathcal{U} = (-1, \infty)^n$. All indices and equalities of indices are modulo n. Write $\mu = a + c - g$ and $A = k + h$ as in the project notation. P01 supplies smoothness on $\mathcal{U}$; P02 supplies the synchronous restriction and, only where explicitly invoked below, its equilibrium relation. No formula is imported from the archived blueprint. The stability equivalences below concern the *linearized system*. Nonlinear behavior at a zero spectral abscissa and bifurcation type are reserved for P09. The positive-state Fourier spectrum and the homogeneous normality theorem are reserved for P04.

7.1 General Jacobian

Let $\delta^{(n)}_{j\ell}$ be one if $j \equiv \ell \pmod{n}$ and zero otherwise. These are modular Kronecker symbols, not mutually exclusive case labels.

Lemma 7.1 (Jacobian entries)**.** *For every $x \in \mathcal{U}$ and $i, j \in \mathbb{Z}_n$, $J(x) = DF(x)$ has entries*

$$\begin{aligned} J_{ij}(x) = &\left(a - 2kx_i - \frac{cx_{i-1}}{(1+x_i)^2} - \frac{g}{1+x_{i+q}} - hx_{i-q}\right)\delta^{(n)}_{ji} \\ &+ \frac{c}{1+x_i}\delta^{(n)}_{j,i-1} + \frac{gx_i}{(1+x_{i+q})^2}\delta^{(n)}_{j,i+q} - hx_i\delta^{(n)}_{j,i-q}. \end{aligned} \tag{43}$$

All contributions are added when indices coincide; all other entries are zero. The three off-diagonal indices differ from i, but need not differ from each other.

Proof. The denominators are positive on $\mathcal{U}$. The product and chain rules give, for the two rational terms separately,

$$\begin{aligned} \frac{\partial}{\partial x_j}\frac{cx_{i-1}}{1+x_i} &= \frac{c}{1+x_i}\delta^{(n)}_{j,i-1} - \frac{cx_{i-1}}{(1+x_i)^2}\delta^{(n)}_{ji}, \\ \frac{\partial}{\partial x_j}\frac{-gx_i}{1+x_{i+q}} &= -\frac{g}{1+x_{i+q}}\delta^{(n)}_{ji} + \frac{gx_i}{(1+x_{i+q})^2}\delta^{(n)}_{j,i+q}. \end{aligned}$$

The derivative of the negative rational term with respect to its denominator coordinate appears with an algebraic plus sign; its value has the sign of x_i. The other derivatives are

$$(a - 2kx_i)\delta^{(n)}_{ji}, \qquad -hx_{i-q}\delta^{(n)}_{ji} - hx_i\delta^{(n)}_{j,i-q}.$$

Adding proves (43). Differentiating with respect to one coordinate differentiates every occurrence of it, including when two neighbor indices coincide. The sum of Kronecker terms implements exactly that rule. □

For a second formulation, put

$$D_i = a - 2kx_i - \frac{cx_{i-1}}{(1+x_i)^2} - \frac{g}{1+x_{i+q}} - hx_{i-q}, \quad C_i = \frac{c}{1+x_i}, \quad G_i = \frac{gx_i}{(1+x_{i+q})^2}, \quad H_i = hx_i.$$

The directional derivative is

$$(DF(x)z)_i = D_iz_i + C_iz_{i-1} + G_iz_{i+q} - H_iz_{i-q}. \tag{44}$$

Since $(P^r z)_i = z_{i-r}$ for every integer r, this gives

$$J(x) = \mathrm{diag}(D_i) + \mathrm{diag}(C_i)P + \mathrm{diag}(G_i)P^{-q} - \mathrm{diag}(H_i)P^q. \tag{45}$$

The diagonal factors are on the left because their coefficients belong to receiving row i. This general matrix need not be circulant.

7.2 Linearization at the origin

Direct substitution gives $F(0) = 0$. In (43), $D_i(0) = a - g$, $C_i(0) = c$, and $G_i(0) = H_i(0) = 0$. Both derivatives of the bilinear h term contain a state coordinate and vanish. The denominator-coordinate (target) derivative of the input contains x_{i-1} and vanishes; that of the rational loss contains x_i and vanishes as well. The numerator derivative of the loss survives as $-g$ on the diagonal. Thus, including every collision case,

$$\boxed{J(0) = (a - g)I + cP.} \tag{46}$$

Proposition 7.2 (Origin spectrum)**.** *With the project's DFT convention $\theta_m = 2\pi m/n$ and $v_j^{(m)} = n^{-1/2} e^{\mathrm{i}\theta_m j}$, the complete spectrum is*

$$\boxed{\lambda_m(0) = a - g + ce^{-\mathrm{i}\theta_m}, \qquad 0 \le m < n.} \tag{47}$$

Every eigenvalue is algebraically and geometrically simple over $\mathbb{C}$. Its real and imaginary parts are

$$\operatorname{Re}\lambda_m(0) = \mu - c(1 - \cos\theta_m), \qquad \operatorname{Im}\lambda_m(0) = -c\sin\theta_m. \tag{48}$$

The unique mode attaining the largest real part is $m = 0$, and

$$\boxed{\operatorname{sa}(J(0)) = \lambda_0(0) = \mu = a + c - g.} \tag{49}$$

Proof. The direction of the shift is checked componentwise:

$$(Pv^{(m)})_j = v_{j-1}^{(m)} = e^{-\mathrm{i}\theta_m} v_j^{(m)}.$$

This also holds at wraparound since $e^{\mathrm{i}n\theta_m} = 1$. For $m \neq \ell$, the geometric sum $\sum_{j=0}^{n-1} e^{\mathrm{i}(\theta_m - \theta_\ell)j} = 0$; for $m = \ell$ it equals n. Thus these n vectors form an orthonormal basis. Applying (46) gives (47) with no missing eigenvalues. The n roots $e^{-\mathrm{i}\theta_m}$ are distinct, and multiplication by $c > 0$ followed by translation preserves distinctness. This proves both multiplicity assertions. Independently, their product gives

$$\det(\lambda I - J(0)) = (\lambda - (a - g))^n - c^n, \tag{50}$$

whose derivative at any root is nonzero because $\lambda - (a - g)$ has modulus $c > 0$.

Taking real and imaginary parts proves (48). Since $\cos\theta_m \le 1$, every real part is at most μ. Equality means $\theta_m \in 2\pi\mathbb{Z}$, hence $m = 0$ among the chosen representatives. Positivity of c makes the inequality strict for every other mode, proving the assertion. □

7.3 Origin stability and the nonhyperbolic boundary

Proposition 7.3 (Exact linear stability and hyperbolicity conditions)**.** *For $\dot{z} = J(0)z$, zero is asymptotically (indeed exponentially) stable if and only if $\mu < 0$, and unstable if and only if $\mu > 0$. At $\mu = 0$ it is Lyapunov stable but not asymptotically stable. The origin of the nonlinear field is nonhyperbolic if and only if*

$$g = a + c\cos\frac{2\pi m}{n} \quad \text{for some } m \in \{0, \dots, n-1\}, \tag{51}$$

with $g > 0$ as always. Nonhyperbolicity is therefore not restricted to the stability boundary $\mu = 0$: every locus in (51) with $m \neq 0$ has $\mu > 0$.

Proof. In the orthonormal basis of Proposition 7.2, each coefficient solves $\dot{\widehat{z}}_m = \lambda_m(0)\widehat{z}_m$. For $\mu < 0$, all coefficients decay exponentially, with $\|z(t)\|_2 \le e^{\mu t}\|z(0)\|_2$ for $t \ge 0$. For $\mu > 0$, the real synchronous solution $z(t) = \varepsilon e^{\mu t}\mathbf{1}$ proves instability for arbitrarily small $\varepsilon > 0$. For $\mu = 0$, every nonsynchronous coefficient decays and the synchronous coefficient is constant. Therefore $\|z(t)\|_2 \le \|z(0)\|_2$, while nonzero constant synchronous solutions prevent attraction. This also proves both converses without a nonlinear stability theorem. The definition of hyperbolicity and (48) give (51). For $m \neq 0$ its value of μ is $c(1-\cos\theta_m) > 0$. □

The spectral-abscissa sign and the linear stability threshold depend on neither n, q, nor $\gcd(n,q)$. Individual eigenvalues and the *full set* of nonhyperbolic loci depend on n, but not on q or a coprimality condition. For example, $n = 6$, $a = 2$, $c = 1$, $g = 5/2$ gives $\mu = 1/2 > 0$ and purely imaginary eigenvalues in modes $1, 5$; mode zero is already unstable. This is not a Hopf claim. More generally, using $0 \le m \le \lfloor n/2 \rfloor$ as pair representatives, each interior representative in (51) contributes exactly the two simple nonzero imaginary eigenvalues in modes $m, n-m$. Strict decrease of the cosine on $[0,\pi]$ excludes any other simultaneous zero-real-part mode. The endpoint cases are treated below.

Proposition 7.4 (Simple zero at the stability boundary)**.** *At $g = a + c$ the only zero eigenvalue is $\lambda_0(0) = 0$. Its algebraic and geometric multiplicities are both one, with kernel* $\operatorname{span}\{\mathbf{1}\}$*. Every nonzero mode has strictly negative real part; none can simultaneously have zero real part.*

Proof. Here $J(0) = c(P - I)$ and

$$\lambda_m(0) = c(e^{-\mathrm{i}\theta_m} - 1), \qquad \operatorname{Re}\lambda_m(0) = -c(1-\cos\theta_m).$$

The latter vanishes only for $m = 0$. The root-of-unity condition also shows that only λ_0 is zero. The equation $Pz = z$ means $z_j = z_{j-1}$ at every index, giving the stated kernel. The characteristic polynomial is now $(\lambda + c)^n - c^n$, whose derivative at zero is $nc^{n-1} \neq 0$, verifying algebraic simplicity separately. For even n, mode $m = n/2$ has $\lambda_{n/2}(0) = -2c < 0$, so it is no exception. This covers every canonical dimension, including $n = 5$ and $n = 6$. □

Remark 7.5 (A zero eigenvalue away from the stability boundary). Over *all* canonical parameter values, a nonzero-mode zero is possible. Its imaginary part requires $\sin\theta_m = 0$; the only nonzero possibility is even n and $m = n/2$. The real part then vanishes exactly at $g = a - c$, which is canonical precisely when $a > c$. That zero is also simple, but $\mu = 2c > 0$, so the origin is already linearly unstable. For example, $a = 2$, $c = g = 1$ works in every even canonical dimension. Thus $c > 0$ excludes nonzero-mode zeros at $g = a + c$, not at every parameter value. For odd n there is no nonzero-mode zero.

These are spectral and linear-system conclusions. "Nonhyperbolic" and "linearly unstable" are not disjoint categories. The Hurwitz criterion is not an assertion that nonlinear asymptotic stability holds if and only if $\mu < 0$. Linearization alone does not decide nonlinear behavior at $\mu = 0$, either in $\mathcal{U}$ or relative to $\mathbb{R}^n_{\ge 0}$. The simple zero does not establish a transcritical bifurcation or a nonlinear nondegeneracy condition.

7.4 Jacobian at a synchronous state

Let $x = x^*\mathbf{1}$ be any synchronous state with $x^* > -1$ and set $s = 1 + x^* > 0$. Here s denotes the positive denominator, not the scalar state variable denoted by s in P02.

Proposition 7.6 (Homogeneous Jacobian)**.** *At every such state, whether or not it is an equilibrium, define*

$$d = a - 2kx^* - \frac{cx^*}{s^2} - \frac{g}{s} - hx^*, \tag{52}$$

$$C = \frac{c}{s}, \qquad G = \frac{gx^*}{s^2}, \qquad H = hx^*. \tag{53}$$

Then

$$\boxed{J(x^*\mathbf{1}) = dI + CP + GP^{-q} - HP^q.} \tag{54}$$

Its entries are $d\delta^{(n)}_{ji} + C\delta^{(n)}_{j,i-1} + G\delta^{(n)}_{j,i+q} - H\delta^{(n)}_{j,i-q}$*, with sums retained at coincident indices.*

Proof. Substitute $x_i = x_{i-1} = x_{i+q} = x_{i-q} = x^*$ in each separately derived coefficient of (43). This gives (52)–(53) without an equilibrium relation. Acting on an arbitrary vector z gives

$$(Jz)_i = dz_i + Cz_{i-1} + Gz_{i+q} - Hz_{i-q}.$$

The $i+q$ term is $(P^{-q}z)_i$, and the $i-q$ term is $(P^q z)_i$. This proves both signs and orientations from indices. □

Always $C > 0$. If $x^* > 0$, then $G, H > 0$, and the separate neighbor contributions have signs $+C, +G, -H$. If $x^* = 0$, then $G = H = 0$; if $-1 < x^* < 0$, then $G, H < 0$. No fixed sign for d at an arbitrary synchronous state is assumed. Signs at colliding offsets must be assessed after addition.

Optional equilibrium simplification using P02. Only now suppose $x^* \neq 0$ and $x^*\mathbf{1}$ is an equilibrium. P02's (25), divided by this nonzero state, gives

$$a - (k+h)x^* + \frac{c-g}{1+x^*} = 0. \tag{55}$$

Substitution of $a = (k+h)x^* - (c-g)/s$ in the unsimplified diagonal coefficient yields

$$d = -kx^* - \frac{c}{s} - \frac{cx^*}{s^2} = -kx^* - \frac{c(1+2x^*)}{s^2}. \tag{56}$$

This applies in particular at a positive synchronous equilibrium. It is not available at an arbitrary state or at the origin merely because the origin is an equilibrium. No stability conclusion is drawn.

7.5 Circulant representation and offset collisions

Since $P^n = I$, (54) is the ordinary polynomial $dI + CP + GP^{n-q} - HP^q$. Each power of P has entries depending only on the cyclic difference $i - j$; their linear combination has rows obtained by cyclic shifts of the preceding row. This proves that the homogeneous Jacobian is circulant without requiring four distinct cyclic diagonals. The standard consequence is Fourier diagonalizability; the actual homogeneous spectrum and normality theorem are left to P04.

The diagonal cannot collide with a neighbor since $1 \not\equiv 0$ and $q \not\equiv 0$ modulo n. Neighbor collisions are exhausted by

$$\begin{aligned} i - 1 \equiv i - q &\iff q = 1, \\ i - 1 \equiv i + q &\iff q = n - 1, \\ i + q \equiv i - q &\iff 2q \equiv 0 \pmod{n}. \end{aligned}$$

The last case means even n and $q = n/2$. For $n \geq 5$, these three cases are mutually exclusive; no triple collision occurs. A nontrivial $\gcd(n, q)$ creates no further coincident offsets. The resulting matrices are

$$q = 1: \quad J = dI + (C - H)P + GP^{-1}, \tag{57}$$

$$q = n - 1: \quad J = dI + (C + G)P - HP^{-1}, \tag{58}$$

$$n \text{ even}, \ q = n/2: \quad J = dI + CP + (G - H)P^{n/2}. \tag{59}$$

For other canonical q the three neighbor offsets are distinct. Even then a coefficient may vanish; the representation lists contributions, not a claim that every entry is nonzero.

At nonsynchronous states the same rule gives, for $q = 1$, the entry $c/(1 + x_i) - hx_i$ at $j = i - 1$; for $q = n - 1$, the entry $c/(1 + x_i) + gx_i/(1 + x_{i-1})^2$ at $j = i - 1$; and for even n, $q = n/2$, the entry $gx_i/(1 + x_{i+q})^2 - hx_i$ at $j = i + q = i - q$. The actual repeated coordinate is identified before combining contributions. At homogeneous states $C - H$ or $G - H$ can vanish through cancellation. The summed polynomial remains valid.

7.6 Checks and possible failure points

Independent analytic consistency checks. Expand $F_i(x + tz)$ to first order using $(u + tv)^{-1} = u^{-1} - tv/u^2 + O(t^2)$ for $u > 0$. The first-order rational contributions are

$$c\left(\frac{z_{i-1}}{1 + x_i} - \frac{x_{i-1}z_i}{(1 + x_i)^2}\right), \qquad -\frac{gz_i}{1 + x_{i+q}} + \frac{gx_iz_{i+q}}{(1 + x_{i+q})^2},$$

and the bilinear contribution is $-h(x_{i-q}z_i + x_iz_{i-q})$. They reproduce (44), including collisions. Putting $x^* = 0$ in the homogeneous coefficients gives $d = a - g$, $C = c$, $G = H = 0$, agreeing with the direct origin calculation.

An independent check along synchrony uses P02's original scalar field, written with a fresh variable r: $f(r) = ar - (k + h)r^2 + (c - g)r/(1 + r)$. Its derivative is

$$f'(r) = a - 2(k + h)r + \frac{c - g}{(1 + r)^2} = d + C + G - H \quad (x^* = r). \tag{60}$$

Combining denominators in the unsimplified coefficients proves the last equality without an equilibrium relation. This is a row-sum check, not the positive-equilibrium spectrum assigned to P04.

The characteristic polynomial and direct boundary-kernel calculation check origin multiplicities separately. The exact difference $\lambda_0(0) - \operatorname{Re}\lambda_m(0) = c(1 - \cos\theta_m) > 0$ for $m \neq 0$ checks spectral ordering. Equality of conjugate real parts does not mean repeated eigenvalues; the even singleton $m = n/2$ is distinct. None of these arguments relies on a numerical sample.

Computational corroboration. Exact sparse-polynomial differentiation, with modular indices already identified, checks all 2,570 Jacobian entries for the 39 pairs $5 \leq n \leq 10$, $1 \leq q < n$, as well as both rational derivatives, origin reductions, generic synchronous states, shift orientations, and the three collision forms. Independent finite determinant and rank checks corroborate origin simplicity. Secondary central differences cover 312 states: origins, positive and negative synchronous states, positive synchronous equilibria, and seeded random nonsynchronous states, including every required collision. The maximum scaled entry error is 2.70×10^{-9} (rounded up), below the 10^{-7} tolerance. The manifest records the exact method, seed, software, counts, and reproducibility details. These checks detect transcription errors; they are neither proof premises nor an independent audit. No nonlinear trajectory, bifurcation, or positive-state Fourier spectrum is computed here.

Priority checks for the independent P03 auditor.

(1) Recompute every partial derivative from the canonical field, especially both rational denominator derivatives and their opposite algebraic signs.

(2) Identify coincident indices before combining their contributions; check addition and cancellation rather than overwriting.

(3) Verify $(P^r z)_i = z_{i-r}$ and the negative Fourier exponent for P; do not reverse the G and H offsets.

(4) Derive the origin matrix independently; check vanishing terms, the complete spectrum, and its unique real-part maximizer.

(5) Separate linear stability from nonlinear boundary behavior; test all nonhyperbolic loci, including overlap with the unstable region.

(6) Check both multiplicities of the boundary zero, strict decay of other modes, and the even-n zero away from the boundary.

(7) Reconstruct unsimplified homogeneous coefficients for every $x^* > -1$ before using P02's nonzero-equilibrium relation.

(8) Check the exhaustive collision list, all summed matrix forms, the origin reduction, and the scalar row-sum identity.

(9) Confirm exact versions and assumptions, and that P04's spectrum/normality and P09's nonlinear conclusions remain deferred.

Internal checking and compilation do not promote this module beyond DRAFT.

8 P04: Arbitrary-(n, q) Fourier spectrum and normality

Status: DRAFT; independent audit pending. The standing assumptions are $n \geq 5$, $1 \leq q < n$ and $a, k, c, g, h > 0$. Use the canonical domain $\mathcal{U} = (-1, \infty)^n$, indices in $\mathbb{Z}_n$, predecessor shift $(Pz)_j = z_{j-1}$ and the unitarily normalized, positive-exponent Fourier vectors from Section 2. No coprimality condition is imposed. This is supporting, known Fourier machinery specialized to the verified homogeneous Jacobian, not a novelty claim.

For any synchronous state $x^*\mathbf{1} \in \mathcal{U}$, set $s = 1 + x^* > 0$. Here s is the denominator, whereas the scalar state variable below is r. Import Proposition 7.6, without re-deriving its Jacobian:

$$J := DF(x^*\mathbf{1}) = dI + CP + GP^{-q} - HP^q, \tag{61}$$

$$d = a - 2kx^* - \frac{cx^*}{s^2} - \frac{g}{s} - hx^*, \qquad C = \frac{c}{s}, \quad G = \frac{gx^*}{s^2}, \quad H = hx^*. \tag{62}$$

All coefficients are real; $C > 0$, while G, H have the sign of x^* (and vanish at zero). The spectral statements through the synchronous-mode identity apply even when the state is not an equilibrium. At a nonstationary state they describe its Jacobian, not an autonomous linearization about an equilibrium.

8.1 Fourier diagonalization

Theorem 8.1 (Complete homogeneous Fourier spectrum)**.** *Let $\theta_m = 2\pi m/n$, $v_j^{(m)} = n^{-1/2} e^{\mathrm{i}\theta_m j}$, and $Q := \mathsf{Q} = [v^{(0)} \ \cdots \ v^{(n-1)}]$. Then $Q^*Q = QQ^* = I$, and the full spectrum, listed with algebraic multiplicity, is*

$$\boxed{\lambda_m = d + Ce^{-\mathrm{i}\theta_m} + Ge^{\mathrm{i}q\theta_m} - He^{-\mathrm{i}q\theta_m}, \qquad m = 0, \ldots, n-1.} \tag{63}$$

In particular,

$$Jv^{(m)} = \lambda_m v^{(m)}, \qquad Q^*JQ = \Lambda := \operatorname{diag}(\lambda_0, \ldots, \lambda_{n-1}), \qquad J = Q\Lambda Q^*. \tag{64}$$

These statements do not require distinct shift offsets, distinct numerical eigenvalues, nonzero combined coefficients, or $\gcd(n, q) = 1$.

Proof. Since $e^{\mathrm{i}n\theta_m} = 1$, every wraparound of an index changes its exponential by a factor one. Thus for every integer ℓ,

$$(P^\ell v^{(m)})_j = v_{j-\ell}^{(m)} = e^{-\mathrm{i}\ell\theta_m} v_j^{(m)}.$$

In particular, the three orientations are

$$Pv^{(m)} = e^{-\mathrm{i}\theta_m} v^{(m)}, \quad P^{-q}v^{(m)} = e^{\mathrm{i}q\theta_m} v^{(m)}, \quad P^q v^{(m)} = e^{-\mathrm{i}q\theta_m} v^{(m)}. \tag{65}$$

Apply each term of (61) to $v^{(m)}$ and add. The successor term has coefficient $+G$ and positive phase; the predecessor-q term has coefficient $-H$ and negative phase. This reconstructs (63) directly from P03.

For $0 \leq m, \ell < n$, the exact normalization gives

$$(Q^*Q)_{\ell m} = \frac{1}{n} \sum_{j=0}^{n-1} e^{\mathrm{i}(\theta_m - \theta_\ell)j} = \begin{cases} 1, & m = \ell, \\ 0, & m \neq \ell. \end{cases}$$

In the second case the ratio $w = e^{2\pi \mathrm{i}(m-\ell)/n}$ satisfies $w \neq 1$ and $w^n = 1$, so the sum is $(1-w^n)/(1-w) = 0$. Hence $Q^*Q = I$. Being square, Q is invertible with inverse Q^*, and $QQ^* = I$. The column identities give $JQ = Q\Lambda$; multiplication by Q^* yields (64). Consequently

$$\det(\zeta I - J) = \prod_{m=0}^{n-1} (\zeta - \lambda_m), \tag{66}$$

so all eigenvalues and their algebraic multiplicities are included. Only the stated powers of P and the geometric sum were used. □

8.2 Real and imaginary dispersion relations

Writing $\lambda_m = r_m + \mathrm{i}\omega_m$, Euler's identities give

$$\boxed{r_m = d + C\cos\theta_m + (G-H)\cos(q\theta_m),} \tag{67}$$

$$\boxed{\omega_m = -C\sin\theta_m + (G+H)\sin(q\theta_m).} \tag{68}$$

Indeed, the imaginary contributions of the three shift terms are $-C\sin\theta_m$, $+G\sin(q\theta_m)$ and $+H\sin(q\theta_m)$: the last plus sign is the product of $-H$ and the negative imaginary part of $e^{-\mathrm{i}q\theta_m}$. Their real contributions are $C\cos\theta_m$, $G\cos(q\theta_m)$ and $-H\cos(q\theta_m)$. Here ω_m is a signed imaginary part, possibly zero; it is not the reserved positive frequency ω of a future Hopf theorem.

8.3 Conjugacy and multiplicities

For a label m, write $\bar{m} = (-m) \bmod n$ in $\{0, \dots, n-1\}$; in particular $\bar{0} = 0$, not an additional label n. The phases at $\bar{m}$ are conjugates of those at m, so

$$v^{(\bar{m})} = \overline{v^{(m)}}, \qquad \lambda_{\bar{m}} = \overline{\lambda_m}, \qquad r_{\bar{m}} = r_m, \quad \omega_{\bar{m}} = -\omega_m. \tag{69}$$

This verifies conjugacy explicitly from the formula and real coefficients, consistently with J being real.

A label is self-conjugate precisely when $2m = 0 \pmod n$. For odd n there is the singleton 0 and the $(n-1)/2$ pairs $\{m, n-m\}$ with $1 \le m \le (n-1)/2$. For even n the singletons are $0, n/2$, with $n/2 - 1$ pairs $1 \le m < n/2$. Each singleton necessarily has a real eigenvalue. The converse is false: a non-singleton pair can have $\omega_m = 0$, and then its two labels give the same real eigenvalue.

Proposition 8.2 (Numerical eigenvalue multiplicities)**.** *For a numerical eigenvalue η, define its label set $M_\eta = \{m : 0 \le m < n,\ \lambda_m = \eta\}$. Its algebraic and geometric multiplicities over $\mathbb{C}$ both equal $|M_\eta|$, and*

$$\ker(J - \eta I) = \mathrm{span}_{\mathbb{C}}\{v^{(m)} : m \in M_\eta\}.$$

Every eigenvalue is semisimple. Each Fourier label contributes one one-dimensional Fourier coordinate, but this does not imply that its numerical eigenvalue is simple.

Proof. Equation (66) counts the algebraic multiplicity. Write $z = Qw$; then $(J - \eta I)z = 0$ if and only if $(\lambda_m - \eta)w_m = 0$ for every m. Exactly the coordinates in M_η are free, proving the eigenspace formula and dimension. Diagonalization also excludes nontrivial Jordan blocks. □

Thus "Fourier-mode multiplicity" means the number of labels producing a given value, if that term is used at all; it is not a second source of multiplicity. A nonreal conjugate pair consists of two different numerical eigenvalues, each of which can also coincide with eigenvalues at other labels. A tie $r_m = r_\ell$ alone says nothing about equality of λ_m, λ_ℓ. Singleton labels can likewise share their numerical value with other labels; singleton refers to label conjugacy, not simplicity.

For a concrete canonical example, take $q = 1$, $x^* = 1$ and $(a, k, c, g, h) = (2, 1, 4, 4, 1)$. P02's scalar equation vanishes, and P03 gives $d = -4, C = 2, G = H = 1$. Hence $J = -4I + P + P^{-1}$ and $\lambda_m = -4 + 2\cos\theta_m$; each non-singleton pair has a repeated real eigenvalue. At the origin, by contrast, $G = H = 0$, $C = c > 0$ and $d = a - g$, so (63) reduces to P03's $a - g + ce^{-\mathrm{i}\theta_m}$. The distinct roots of unity and $c > 0$ make all those origin eigenvalues simple, as established in Proposition 7.2.

8.4 Offset collisions and self-conjugate modes

P03's exhaustive collision list requires no replacement spectrum. Direct substitution in the master formula gives the following checks, with phases understood modulo 2π.

(i) If $q = 1$, then

$$J = dI + (C - H)P + GP^{-1}, \qquad \lambda_m = d + (C - H)e^{-\mathrm{i}\theta_m} + Ge^{\mathrm{i}\theta_m}.$$

The same expression follows by combining the C and $-H$ terms in (63). Here $r_m = d + (C + G - H)\cos\theta_m$ and $\omega_m = (-C + G + H)\sin\theta_m$.

(ii) If $q = n - 1$, then $P^{-q} = P$ and $P^q = P^{-1}$. Since $e^{\mathrm{i}(n-1)\theta_m} = e^{-\mathrm{i}\theta_m}$ and $e^{-\mathrm{i}(n-1)\theta_m} = e^{\mathrm{i}\theta_m}$,

$$J = dI + (C + G)P - HP^{-1}, \qquad \lambda_m = d + (C + G)e^{-\mathrm{i}\theta_m} - He^{\mathrm{i}\theta_m}.$$

Thus $r_m = d + (C + G - H)\cos\theta_m$ and $\omega_m = -(C + G + H)\sin\theta_m$, also obtained from (67)–(68).

(iii) If n is even and $q = n/2$, then $e^{\pm\mathrm{i}q\theta_m} = e^{\pm\mathrm{i}\pi m} = (-1)^m$ and $P^{-q} = P^q = P^{n/2}$. Therefore

$$J = dI + CP + (G - H)P^{n/2}, \qquad \lambda_m = d + Ce^{-\mathrm{i}\theta_m} + (G - H)(-1)^m.$$

Here $r_m = d + C\cos\theta_m + (G - H)(-1)^m$ and $\omega_m = -C\sin\theta_m$.

These equalities include cancellations such as $C = H$ in the first case or $G = H$ in the third. Adding contributions before or after Fourier transformation gives identical results; coincident entries must never be overwritten.

The singleton formulas, for every admissible q, are

$$\lambda_0 = d + C + G - H \in \mathbb{R}, \tag{70}$$

$$\lambda_{n/2} = d - C + (G - H)(-1)^q \in \mathbb{R} \quad (n \text{ even}). \tag{71}$$

For the second formula use $e^{-\mathrm{i}\pi} = -1$ and $e^{\pm\mathrm{i}q\pi} = (-1)^q$; all sine terms vanish. It equals $d - C + G - H$ for even q and $d - C - G + H$ for odd q. When also $q = n/2$, the factor is $(-1)^{n/2}$, agreeing with the half-offset formula at $m = n/2$. Thus no real singleton critical mode is omitted from subsequent all-mode tests.

8.5 Normality

Proposition 8.3 (Normality)**.** *Every homogeneous Jacobian* (61) *is normal over* $\mathbb{C}^n$*:* $JJ^* = J^*J$*.*

Proof. Theorem 8.1 gives $J^* = Q\Lambda^*Q^*$ and

$$JJ^* = Q\Lambda\Lambda^*Q^* = Q\,\mathrm{diag}(|\lambda_m|^2)Q^* = Q\Lambda^*\Lambda Q^* = J^*J.$$

No eigenvalue distinctness was used. □

Alternatively, P is a permutation matrix with $P^* = P^{-1}$. Both J and $J^* = dI + CP^{-1} + GP^q - HP^{-q}$ are finite linear combinations of integer powers of the same P. Every such power commutes with every other, proving normality directly. Both proofs apply with repeated eigenvalues, colliding offsets, cancelled coefficients, or nontrivial $\gcd(n, q)$. They apply at negative synchronous states as well. They assert no normality at general nonsynchronous or heterogeneous states.

8.6 Synchronous mode

P02's scalar field, using a fresh state variable r, is

$$f(r) = ar - (k+h)r^2 + \frac{(c-g)r}{1+r}, \qquad r > -1.$$

Since every shift fixes $\mathbf{1}$, the row sum of J is $d + C + G - H$. Combining the unsimplified coefficients gives

$$\begin{aligned} d + C + G - H &= a - 2(k+h)x^* + \left(\frac{c}{s} - \frac{cx^*}{s^2}\right) + \left(-\frac{g}{s} + \frac{gx^*}{s^2}\right) \\ &= a - 2(k+h)x^* + \frac{c-g}{s^2}. \end{aligned}$$

Here $s - x^* = 1$ establishes both rational cancellations. Hence

$$\boxed{\lambda_0 = d + C + G - H = f'(x^*).} \tag{72}$$

This is P03's row-sum identity (60), now identified with the Fourier eigenvalue, without assuming equilibrium. Independently, differentiating P02's identity $F(r\mathbf{1}) = f(r)\mathbf{1}$ gives $DF(r\mathbf{1})\mathbf{1} = f'(r)\mathbf{1}$. The vector $v^{(0)} = \mathbf{1}/\sqrt{n}$ spans the tangent direction to the invariant synchronous subspace. This explains the equality mathematically and also covers a nonstationary synchronous trajectory, whose scalar tangential variational coefficient is $f'(r(t))$.

Positive synchronous equilibrium specialization. Now assume additionally $x^* > 0$ and $F(x^*\mathbf{1}) = 0$. Then $C, G, H > 0$. Put $A = k + h$ as in the canonical notation. P02 and P03 give

$$a - Ax^* + \frac{c-g}{s} = 0, \qquad d = -kx^* - \frac{c}{s} - \frac{cx^*}{s^2} = -kx^* - \frac{c(1+2x^*)}{s^2}.$$

This is an optional equilibrium-only replacement for d in (63); the unsimplified expression (62) remains valid at all synchronous states. Writing $f(r) = rB(r)$ with $B(r) = a - Ar + (c-g)/(1+r)$ and $B(x^*) = 0$ yields

$$\boxed{\lambda_0 = -x^*\left(A + \frac{c-g}{s^2}\right) = \frac{x^*}{s}\big(a - A - 2Ax^*\big).} \tag{73}$$

The first equality uses $f'(x^*) = x^*B'(x^*)$; for the second, substitute $c - g = (Ax^* - a)s$. The equilibrium identities require a nonzero equilibrium; they also hold at a negative nonzero equilibrium in $\mathcal{U}$, but $G, H > 0$ is then false. At the origin use (62), which gives $\lambda_0 = a + c - g$. No comparison with other r_m or selection of a dominant mode is made here.

8.7 Linear stability and hyperbolicity

In this subsection $x^*\mathbf{1}$ is a synchronous *equilibrium* in $\mathcal{U}$. Adopt the project's spectral-abscissa notation

$$\operatorname{sa}(J) = \max_{0 \le m < n} r_m. \tag{74}$$

For $\dot{z} = Jz$, exponential asymptotic stability means there exist $M \ge 1$ and $\beta > 0$ such that $\|z(t)\|_2 \le M e^{-\beta t} \|z(0)\|_2$ for all $t \ge 0$.

Corollary 8.4 (Exact linear stability criterion)**.** *The linearized system is exponentially asymptotically stable if and only if*

$$\boxed{\operatorname{sa}(J) < 0 \quad \Longleftrightarrow \quad d + C\cos\theta_m + (G - H)\cos(q\theta_m) < 0 \quad \text{for every } m.} \tag{75}$$

For all $t \ge 0$, its exact induced Euclidean propagator norm is

$$\boxed{\left\|e^{tJ}\right\|_2 = e^{t\,\operatorname{sa}(J)}.} \tag{76}$$

If $\operatorname{sa}(J) = 0$, the linear system is Lyapunov stable but not asymptotically stable. If $\operatorname{sa}(J) > 0$, it is unstable.

Proof. The exponential power series and (64) give $e^{tJ} = Q\,\mathrm{diag}(e^{t\lambda_m})Q^*$. Unitary matrices preserve Euclidean norms. The induced norm of a diagonal matrix is the maximum absolute value of a diagonal entry, since the upper bound from the sum of squared coordinates is attained at a coordinate vector. Consequently, for $t \ge 0$,

$$\left\|e^{tJ}\right\|_2 = \max_m |e^{t\lambda_m}| = \max_m e^{tr_m} = e^{t\,\operatorname{sa}(J)}.$$

This proves exponential stability when $\operatorname{sa}(J) < 0$, with $M = 1$. Conversely, an exponential estimate would require $e^{t\,\operatorname{sa}(J)} \le M e^{-\beta t}$ for all $t \ge 0$; taking logarithms, dividing by t and letting $t \to \infty$ gives $\operatorname{sa}(J) \le -\beta < 0$.

These norm and stability claims also hold over the real state space. For a real matrix T, writing $z = u + \mathrm{i}v$ shows $\|Tz\|_2^2 = \|Tu\|_2^2 + \|Tv\|_2^2$, so the induced real and complex norms coincide. More explicitly, each singleton has a real eigenvector. On a non-singleton pair the real orthonormal vectors $\sqrt{2}\,\mathrm{Re}\,v^{(m)}, \sqrt{2}\,\mathrm{Im}\,v^{(m)}$ give the block

$$\begin{pmatrix} r_m & \omega_m \\ -\omega_m & r_m \end{pmatrix}.$$

Its exponential is e^{tr_m} times an orthogonal rotation. Orthonormality follows by expressing the two vectors as combinations of the orthonormal $v^{(m)}, v^{(\bar{m})}$; applying J gives the stated block. A maximizing block with $r_m = 0$ preserves nonzero norm, preventing attraction while the full propagator norm is one. A block with $r_m > 0$ has real solutions with exponentially increasing norm, proving Lyapunov instability from arbitrarily small data. This also shows linear asymptotic stability is equivalent here to linear exponential stability. □

Proposition 8.5 (Hyperbolicity)**.** *The equilibrium is hyperbolic if and only if $r_m \ne 0$ for every Fourier label. It is nonhyperbolic whenever any $r_m = 0$, including when $\omega_m \ne 0$. Invertibility requires instead $\lambda_m \ne 0$ for every label, and is strictly weaker than hyperbolicity.*

Proof. The complete spectrum is (63); substitute it in the definitions of hyperbolicity and invertibility. A nonzero purely imaginary eigenvalue meets the invertibility test but fails the hyperbolicity test. For an instance within the canonical family, P03's origin example $n = 6$, $a = 2, c = 1, g = 5/2$ (any positive k, h, any canonical q) has $r_1 = r_5 = 0$ and nonzero imaginary parts, whereas no eigenvalue is zero: $ce^{-\mathrm{i}\theta_m} = 1/2$ is impossible by modulus. Thus the weaker criterion is strictly weaker here as well. □

Hyperbolicity alone is not stability: hyperbolic equilibria may have positive real parts. Nonhyperbolicity and instability may coexist when one mode has zero real part and another has positive real part.

Nonlinear implications and their limits. The following conclusions concern the nonlinear equilibrium in the open domain $\mathcal{U}$, using P01's C^∞ smoothness there:

(i) If $\mathrm{sa}(J) < 0$, the equilibrium is locally asymptotically stable; in fact it is locally exponentially stable.

(ii) If $\mathrm{sa}(J) > 0$, it is unstable, even if some other eigenvalues have zero real part.

(iii) If $\mathrm{sa}(J) = 0$, linearization alone is generally inconclusive about nonlinear stability or attraction. Normality only settles the linear system's boundedness in this case.

For the first assertion, translate to $y = x - x^*\mathbf{1}$ and write $\dot{y} = Jy + R(y)$, where $R(y) = o(\|y\|_2)$. Let $\delta = -\mathrm{sa}(J) > 0$, and choose a closed ball of radius ρ inside the translated domain on which $\|R(y)\|_2 \le (\delta/2)\,\|y\|_2$. Variation of constants, (76), and Gronwall's inequality give, while the solution stays in this ball,

$$\|y(t)\|_2 \le e^{-\delta t}\,\|y(0)\|_2 + \frac{\delta}{2}\int_0^t e^{-\delta(t-u)}\,\|y(u)\|_2\,du \quad\Longrightarrow\quad \|y(t)\|_2 \le e^{-\delta t/2}\,\|y(0)\|_2\,.$$

If $\|y(0)\|_2 < \rho$, this precludes a first exit; compact local continuation from P01 then supplies all forward times. This proves the stated local exponential conclusion.

For the second assertion we use the standard finite-dimensional linearized-instability theorem: a C^1 vector field on an open neighborhood of an equilibrium is Lyapunov unstable there if its Jacobian has at least one eigenvalue with positive real part. The theorem does not require the other eigenvalues to avoid the imaginary axis. Here the equilibrium is interior to $\mathcal{U}$, P01 supplies C^1 regularity, and (63) supplies exactly the required positive eigenvalue real part, checking every hypothesis.

For why no universal boundary conclusion follows, the smooth scalar fields $\dot{y} = -y^3$ and $\dot{y} = y^3$ both have the normal linearization $J = 0$. In the first, $|y(t)| = |y(0)|/\sqrt{1 + 2y(0)^2 t}$ decreases to zero, with Lyapunov stability; in the second, any small positive initial state increases and leaves a fixed small neighborhood. These are examples of the limitation of linearization theory, not claims about a boundary parameter of the DM model.

All stability implications above are local. They establish neither global attraction nor a global basin, even given P01's separate boundedness result. At a positive equilibrium one may choose the local ball inside $\mathbb{R}^n_{>0}$, so the local assertions apply there as well. At the origin the conclusions above are stated in $\mathcal{U}$; instability relative to a restricted state set is a separate question unless its admissible perturbations are checked. A Hurwitz Jacobian is sufficient for nonlinear local asymptotic stability, not a necessary condition in general. Thus the rejected blanket nonlinear equivalence C040 and global assertion C153 are not reinstated.

Dominant modes and first loss of stability. At a fixed parameter point and specified synchronous equilibrium, the dominant-label set is $\arg\max_{0\le m<n} r_m$, retaining all ties. This is the fixed-point "selected-mode set" reserved in Section 2; it need not contain an unstable mode. A *first unstable mode along a path* additionally needs a specified equilibrium branch $x^*(\nu)\mathbf{1}$, a specified oriented parameter path $\boldsymbol{\pi}(\nu)$, a prior interval of stability, and a loss-of-stability crossing event. For a continuous branch/path a finite boundary event from $\mathrm{sa}(J) < 0$ must have $\mathrm{sa}(J) = 0$; calling it an instability crossing also requires positive real part on the unstable side, not merely tangency at zero. Several labels or pairs can participate, so uniqueness and transverse crossing cannot be presumed. No such path, crossing, or $q = 2$ selection analysis is supplied by this module.

8.8 Arbitrary offset and the step-q subgraph

Let $\kappa = \gcd(n, q)$, $n = \kappa n'$ and $q = \kappa q'$, where $\gcd(n', q') = 1$. Repeated addition of q returns an index to itself after the least positive L for which $n \mid Lq$. This is equivalent to $n' \mid Lq'$, hence to $n' \mid L$, so $L = n/\kappa$. The step-q permutation therefore has exactly κ disjoint cycles, each of length n/κ; they are the residue classes modulo κ. Reversing the step gives the same partition. Thus coprimality is required only to call this step-q subgraph a single cycle. The full canonical coupling also includes the step-one predecessor cycle, and the step-q partition is not an assertion of full-network disconnection or of a block decomposition of J by those residue classes.

The Fourier sampling grid and label count depend on n; parity of n determines whether the $n/2$ singleton exists. The general phases and possible eigenvalue coincidences depend on q modulo n as well as n and the coefficients. The value of λ_0 is independent of n, q at fixed coefficients and state; the half-mode value depends on the parity of q. No proof of diagonalization, conjugacy, multiplicity, normality, or the all-mode stability and hyperbolicity criteria requires coprimality. The actual spectral abscissa generally depends on the grid and phases; P03's n, q-independent origin threshold is a special case.

8.9 Checks and possible failure points

Analytic checks. Equations (65), (67) and (68) check every phase and sign independently. The geometric sum checks normalization and completeness. Equation (69) checks modular conjugacy; the singleton and collision substitutions check all requested special cases. The characteristic polynomial and explicit eigenspace calculation check both multiplicities without assuming simplicity. The independent scalar derivative checks the row sum at all states. Both normality proofs and the exact propagator norm check the spectral-abscissa criterion. None of these arguments is numerical.

Computational corroboration. A separate portable checker compares the formula against direct complex eigendecomposition of real matrices assembled by adding P03's contributions at modular column indices. This includes $(n, q) = (5, 2), (6, 2), (6, 3), (7, 1), (8, 4)$, all last offsets $q = n - 1$ in the tested dimensions, noncoprime pairs, positive synchronous states, and negative synchronous states in $\mathcal{U}$. It also tests positive equilibria, origin controls, cancelled coefficients, and repeated numerical eigenvalues. All 39 pairs with $5 \le n \le 10$, $1 \le q < n$ were tested: 522 matrices and 4,144 eigenvalue occurrences, matched one-to-one including repeats. The largest scaled eigenvalue discrepancy was below 5.95×10^{-15}; 2,088 independent matrix-exponential checks had norm errors below 1.69×10^{-13} relative to $e^{t\,\mathrm{sa}(J)}$. Five exact cleared-denominator polynomial checks also passed. The manifest records the methods, tolerances, inputs and reproducibility details. These tests corroborate the written proof and supply no theorem premise or audit verdict.

Priority checks for the independent P04 auditor.

(1) Authenticate the current P01–P03/audit versions and retain P04's DRAFT status and supporting role.

(2) Reconstruct the phases of P, P^{-q} and P^q from node indices; inspect especially $+G$ versus $-H$ and $G + H$ in ω_m.

(3) Check the $n^{-1/2}$ normalization, completeness, and both unitary-diagonalization identities, including repeated values.

(4) Distinguish singleton labels, real eigenvalues, conjugate pairs, and algebraic/geometric multiplicity; test the real repeated-pair example.

(5) Check all three collision reductions, cancellations, $m = 0$, and the even singleton's dependence on the parity of q.

(6) Verify $\lambda_0 = f'(x^*)$ without equilibrium, then check the two nonzero-equilibrium simplifications only under their hypotheses.

(7) Check normality, the real and complex norm equality for $t \geq 0$, and both directions of the exponential-stability equivalence.

(8) Keep zero spectral abscissa, zero eigenvalue, nonhyperbolicity, nonlinear local stability and global attraction distinct; check the hypotheses of the standard nonlinear instability theorem.

(9) Check the step-q cycle count without imposing coprimality on spectral results; retain all ties and require branch/path/crossing data before first-instability language.

(10) Treat finite computations and compilation as corroboration only; confirm that no P05 selection or P11–P13 bifurcation analysis was begun.

9 P05: Exact step-two mode selection

Status: DRAFT; independent audit pending. Fix the canonical demand-modulated model with $q = 2$, $n \geq 5$ and $a, k, c, g, h > 0$. Let $x^*\mathbf{1}$ be a specified positive synchronous equilibrium and put $s = 1 + x^* > 1$. Import only the audited P04 coefficients and real dispersion relation:

$$\begin{aligned} C = \frac{c}{s} > 0, \qquad G = \frac{gx^*}{s^2} > 0, \qquad H = hx^* > 0, \\ r_m = d + C\cos\theta_m + (G - H)\cos(2\theta_m), \qquad \theta_m = \frac{2\pi m}{n}. \end{aligned} \tag{77}$$

Here $r_m = \operatorname{Re}\lambda_m$ and d is P04's diagonal coefficient. All selection conclusions below are derived from (77). The maximization is over all Fourier labels, including mode zero, and selection means the complete set maximizing the real part at this equilibrium, retaining all ties. No coprimality condition is used. P01 supplies smoothness, P02 supplies the equilibrium relation, and P03–P04 supply the homogeneous coefficients. No later proof module is used.

Notation and novelty scope. Use $T = G - H$ and the dimensionless scalar

$$\chi = \frac{4(H - G)}{C} = -\frac{4T}{C}. \tag{78}$$

The letter D remains available for P02's discriminant, and ρ retains the group-action meaning in Section 2. Here j denotes a Fourier representative, rather than a node index. The frozen Phase 1B/1C verdict treats quadratic maximization as routine. The intended main-theorem content is the complete finite-grid classification, its realization by positive canonical parameters, and local switch geometry. No priority claim or new literature assessment is made.

9.1 Quadratic dispersion relation

Set $u_m = \cos\theta_m$. The identity $\cos(2\theta) = 2\cos^2\theta - 1$ gives

$$r_m = R(u_m), \qquad \boxed{R(u) = d - T + Cu + 2Tu^2.} \tag{79}$$

The additive constant is $d - T$; the part controlling selection is $Cu + 2Tu^2$. In particular, differences of growth rates and the complete maximizer set depend only on C, T, not on d. This is an algebraic selection statement, not permission to vary d independently in the model.

Put $N = \lfloor n/2 \rfloor$ and $u_j = \cos(2\pi j/n)$ for $0 \leq j \leq N$. Since cosine is strictly decreasing on $[0, \pi]$,

$$1 = u_0 > u_1 > \cdots > u_N, \qquad u_N = \begin{cases} -1, & n \text{ even}, \\ -\cos(\pi/n), & n \text{ odd}. \end{cases} \tag{80}$$

Equation (77) gives $r_{n-m} = r_m$ for $1 \leq m < n$. Define the distinct label sets

$$L_j = \begin{cases} \{0\}, & j = 0, \\ \{n/2\}, & n \text{ even and } j = N, \\ \{j, n - j\}, & \text{otherwise}. \end{cases} \tag{81}$$

For a maximizing representative set S, the full dominant-label set is $\bigcup_{j \in S} L_j$, a disjoint union.

9.2 Continuous maximizer

If $T \geq 0$, the polynomial is convex (linear when $T = 0$). Explicitly,

$$R(1) - R(-1) = 2C > 0, \qquad R(1) - R(u) = (1-u)\big[C + 2T(1+u)\big] > 0 \quad (-1 \leq u < 1).$$

Thus $j = 0$ is the unique dominant representative, including $T = 0$ and odd n, when -1 itself is absent from the grid.

If $T < 0$, then $H - G > 0$, $R'' = -4(H - G) < 0$, and

$$R'(u) = C - 4(H-G)u, \qquad u_* := \frac{C}{4(H-G)} = \frac{1}{\chi} > 0. \tag{82}$$

If $u_* \geq 1$, the derivative is positive on $[-1, 1)$, hence $j = 0$ is still uniquely selected. The boundary $u_* = 1$ is exactly $H - G = C/4$, or $\chi = 1$; the zero derivative at $u = 1$ does not give a second grid maximizer. If $0 < u_* < 1$, the continuous maximizer on $[-1, 1]$ is the interior vertex. A discrete change need not occur when this vertex first leaves the endpoint.

9.3 Exact discrete selection rule

For $T < 0$, direct completion of the square gives

$$R(u) = d + (H - G) + \frac{C^2}{8(H-G)} - 2(H-G)(u - u_*)^2. \tag{83}$$

Consequently maximizing the growth rate is exactly minimizing $|u_j - u_*|$. This holds for every $u_* > 0$, including $u_* \geq 1$. Distances are in cosine space; the grid in that space is nonuniform. No angular rounding rule is invoked.

Define the strictly decreasing midpoint sequence

$$\beta_j = \frac{u_j + u_{j+1}}{2} = \cos\frac{\pi}{n}\cos\frac{(2j+1)\pi}{n}, \qquad 0 \leq j < N. \tag{84}$$

Strict decrease follows from $\beta_j - \beta_{j+1} = (u_j - u_{j+2})/2 > 0$.

Theorem 9.1 (A: Exact selection, including endpoints and all ties)**.** *Under the stated canonical equilibrium assumptions, if $T \geq 0$ then $S = \{0\}$. If $T < 0$, the selected representatives are exactly those whose closed cells below contain u_*:*

$$\begin{aligned} V_0 &= [\beta_0, \infty), \\ V_j &= [\beta_j, \beta_{j-1}] \quad (1 \leq j < N), \\ V_N &= (-\infty, \beta_{N-1}]. \end{aligned} \tag{85}$$

A representative is the unique maximizer precisely when u_* lies in the corresponding open cell. At $u_* = \beta_j$ the selected set is exactly $\{j, j+1\}$. No nonadjacent or triple representative tie can be maximal. Only positive midpoints can be attained when $T < 0$. The full Fourier-label set is given by (81)*.*

Proof. The preceding endpoint comparison proves the $T \geq 0$ assertion. For $T < 0$ use (83). For any adjacent grid points,

$$(u_j - u_*)^2 - (u_{j+1} - u_*)^2 = 2(u_j - u_{j+1})(\beta_j - u_*).$$

Since $u_j > u_{j+1}$, representative j is closer exactly when $u_* > \beta_j$, with equality exactly at the midpoint. For $1 \le j < N$ the two neighboring comparisons require $\beta_j \le u_* \le \beta_{j-1}$. These inequalities imply $u_{j+1} < u_* < u_{j-1}$; every grid point beyond either neighbor is even farther in that direction. Thus the neighboring comparisons are also sufficient. For $j = 0$, the single inequality $u_* \ge \beta_0$ makes every u_ℓ, $\ell > 1$, farther than u_1; the analogous argument at N uses $u_* \le \beta_{N-1}$. Strict inequalities give uniqueness, and equality gives exactly the stated adjacent pair.

For completeness, if two nonadjacent representatives had equal maximal value, a grid point strictly between their cosines would have strictly larger value by strict concavity. This is impossible. Three maximizers would include a nonadjacent pair. Equality of two nonadjacent values can occur away from the maximum; the theorem excludes only maximal ties. Finally $u_* > 0$ excludes all nonpositive midpoint equalities. □

9.4 Mode ties and selection cells

The real-part comparison also has the useful exact form

$$R(u_j) - R(u_{j+1}) = C(u_j - u_{j+1})(1 - \chi\beta_j). \tag{86}$$

For $T < 0$ an adjacent *maximal* tie therefore occurs if and only if $u_* = \beta_j > 0$, equivalently $\chi = 1/\beta_j$. The word maximal is essential: adjacent equalities outside the concave regime need not be ties at the spectral abscissa.

Let $w_j = |L_j|$. A strict cell has w_j selected Fourier labels; a representative tie has $w_j + w_{j+1}$. Thus two nonsingleton representatives give four labels, a tie involving 0 gives three, and an even-n endpoint would contribute only one label. Below we prove that the $N = n/2$ endpoint is never selected here. In the canonical reachable cells every nonzero selected representative gives exactly two labels, and every reachable nonzero/nonzero switch gives four. These counts are counts of labels, not of distinct numerical eigenvalues.

9.5 Reachable Fourier representatives

Theorem 9.2 (B: Reachable cells and parity)**.** *The full list of representatives whose nearest-cosine cell intersects* $(0, 1)$*, and also the full list attainable at some canonical positive synchronous equilibrium, is*

$$\boxed{0, 1, \ldots, J, \qquad J = j_{\max}(n) = \left\lfloor \frac{n+1}{4} \right\rfloor.} \tag{87}$$

Each is attainable as the unique representative. Precisely the midpoints $\beta_0, \ldots, \beta_{J-1}$ are positive and give attainable switches. Every $j > J$, in particular the even-n singleton $n/2$, is excluded. Canonical attainability and robustness are proved in Theorem 9.3.

Proof. For $1 \le j \le N$, the cell intersects $(0, 1)$ if and only if $\beta_{j-1} > 0$. Necessity follows from its upper endpoint. Sufficiency follows by taking a point strictly between $\max(0, \beta_j)$ and β_{j-1} when $j < N$, with the endpoint case interpreted using V_N. The upper endpoint is less than 1. The cell for 0 intersects $(0, 1)$ since $0 < \beta_0 < 1$. By (84) and $\cos(\pi/n) > 0$,

$$\beta_j > 0 \iff 2j + 1 < n/2 \iff j < (n-2)/4.$$

Indeed the relevant midpoint angles lie in $(0, \pi)$, where cosine changes sign only at $\pi/2$. Hence $\beta_{j-1} > 0$ is equivalent to $j < (n+2)/4$, giving exactly (87). In particular $J < N$ for every $n \ge 5$, $\beta_{J-1} > 0$, and $\beta_J \le 0$. All retained cell intersections have nonempty interiors; there is no additional representative reachable only through an isolated tie. When $T \ge 0$ or $u_* \ge 1$ only 0 is selected, so neither regime enlarges the list. □

The parity information, with $n = 4\ell + r$ in each row, is exact:

r	J	u_J	Nearest representatives at the formal point $u_* = 0$
0	ℓ	0	ℓ
1	ℓ	$\sin(\pi/(2n)) > 0$	ℓ
2	ℓ	$\sin(\pi/n) > 0$	$\ell, \ell+1$
3	$\ell+1$	$-\sin(\pi/(2n)) < 0$	$\ell+1$

To verify the last column independently, the two cosines straddling zero in the odd cases are

$$\begin{aligned} n = 4\ell+1: &\quad u_\ell = \sin(\pi/(2n)), \quad u_{\ell+1} = -\sin(3\pi/(2n)), \\ n = 4\ell+3: &\quad u_\ell = \sin(3\pi/(2n)), \quad u_{\ell+1} = -\sin(\pi/(2n)). \end{aligned}$$

All their arguments are in $(0, \pi/2)$, where sine increases strictly. All remaining cosines are farther from zero by monotonicity of the grid. For $4 \mid n$ there is an exact zero cosine at $j = n/4$, and its cell contains a positive interval. This representative is a conjugate pair, not the real singleton $n/2$. For $n = 4\ell + 2$, the straddling values are $\pm\sin(\pi/n)$, and $\beta_\ell = 0$.

Large restraint at fixed finite dimension. For fixed $C > 0$, as $H - G \to +\infty$, the vertex decreases to zero and J is the unique dominant representative for all sufficiently large finite $H - G$, in every parity class. The formal zero-vertex tie for $n = 4\ell + 2$ is never attained with finite positive $C, H - G$. In fact the two competing real parts have the strictly positive, $H - G$-independent difference

$$r_\ell - r_{\ell+1} = 2C\sin(\pi/n) > 0.$$

Their squared cosines are equal, so this follows directly from (79). Only a leading-order normalization by $H - G$ loses that difference. Thus a formal limiting nearest-grid tie must not be added to the finite-parameter dominant set. No $n \to \infty$ limit is taken. For $n \equiv 3 \pmod 4$, J is the sole reachable representative with a negative cosine; for the other residue classes no negative-cosine representative is reachable.

Complete regions in one dimensionless parameter. Put $\alpha_j = 1/\beta_j$ for $0 \le j < J$. Then $1 < \alpha_0 < \cdots < \alpha_{J-1}$, and for *all* real χ allowed by the canonical coefficients the entire selection rule is

Condition	S
$\chi < \alpha_0$	$\{0\}$
$\alpha_{j-1} < \chi < \alpha_j \quad (1 \le j < J)$	$\{j\}$
$\chi > \alpha_{J-1}$	$\{J\}$
$\chi = \alpha_j \quad (0 \le j < J)$	$\{j, j+1\}$.

(88)

The middle row is empty when $J = 1$. Negative and zero χ are covered by the first row. For positive χ, the inequalities follow by inverting positive midpoint bounds, reversing their order. Thus the continuous vertex boundary $\chi = 1$ is strictly inside the uniform selection region; the first discrete switch is at $\alpha_0 > 1$.

9.6 Canonical parameter realization

The concave condition and vertex in original variables are

$$T < 0 \iff h > \frac{g}{s^2}, \qquad u_* = \frac{c/s}{4x^*(h - g/s^2)} = \frac{cs}{4x^*(hs^2 - g)}, \qquad \chi = \frac{4x^*(hs^2 - g)}{cs}. \tag{89}$$

All denominators in the vertex formula are strictly positive in this regime. For $0 \le j < J$, the switch equations are exactly

$$H - G = \frac{C}{4\beta_j}, \qquad \boxed{h = \frac{g}{(1+x^*)^2} + \frac{c}{4(1+x^*)x^*\beta_j}.} \tag{90}$$

They are subject to P02's equilibrium condition

$$a - (k+h)x^* + \frac{c-g}{1+x^*} = 0. \tag{91}$$

For $\beta_j \le 0$ there is no switch in the $T < 0$ regime: division by zero is not meaningful and negative β_j would require $H - G < 0$. Such negative-midpoint equalities in the convex regime, if realizable as equal lower values, are not dominant-mode switches.

Theorem 9.3 (C: Canonical realizability, open cells and switch geometry)**.** *Every strict cell and every switch in* (88) *is nonempty in the canonical positive-equilibrium parameter space. Each strict cell has a realization persisting on a nonempty open set of positive physical parameters* (a, k, c, g, h) *with a specified smooth positive synchronous branch. Both* $T > 0$ *and* $T < 0$*, and the boundary* $T = 0$*, are realizable. Every positive-midpoint switch is locally a smooth codimension-one hypersurface in physical parameter space along its specified branch; all its positive-equilibrium points are regular branch points. No nonpositive-midpoint dominant switch is realizable. These are local branch statements and assert no global identification of distinct branches.*

Proof of realizability. Choose any $x > 0$, any $c, g > 0$, and any desired vertex $v > 0$. Define

$$s = 1 + x, \qquad h = \frac{g}{s^2} + \frac{c}{4sxv}, \qquad k > \max\left\{0, \frac{c-g}{sx} - h\right\}, \qquad a = (k+h)x - \frac{c-g}{s}. \tag{92}$$

Then h, k, a are strictly positive: the displayed strict lower bound on k is exactly what is required for $a > 0$, together with $k > 0$. There is no upper bound on k, so this choice is always possible. Equation (91) vanishes identically at $x^* = x$; substitution into the original vector field gives, in every component,

$$ax - kx^2 + \frac{cx}{s} - \frac{gx}{s} - hx^2 = x\left[a - (k+h)x + \frac{c-g}{s}\right] = 0.$$

Also $H - G = c/(4sv) > 0$ and $u_* = v$. Hence every retained open Voronoi cell and every positive midpoint can be prescribed exactly. The construction does not treat C, G, H as independent of an equilibrium.

A particularly simple exact family, used below, is

$$x^* = 1, \qquad c = g = 4, \qquad h = 1 + \chi/2, \qquad k > 0, \qquad a = k + 1 + \chi/2, \qquad \chi > -2. \tag{93}$$

Here $C = 2$, $G = 1$, $H = 1 + \chi/2$, so the dimensionless parameter is exactly the displayed χ. Every positive χ is realized. At $k = 1$, choosing $\chi = -1, 0, 1$ realizes respectively $T > 0$, $T = 0$, and $T < 0$ with $u_* = 1$, all selecting 0. Thus the uniform region has examples in both sign regimes. The geometry and openness assertions are proved next. □

9.7 Realizable switch hypersurfaces

Use the equilibrium coordinates (x, k, c, g, h) on the open set

$$\mathcal{U}_{\mathrm{eq}} = \left\{ (x, k, c, g, h) \in (0, \infty)^5 : (k + h)x - \frac{c - g}{1 + x} > 0 \right\}. \tag{94}$$

The map

$$\Phi(x, k, c, g, h) = \left((k + h)x - \frac{c - g}{1 + x},\ k, c, g, h \right)$$

projects this parameterization of the positive equilibrium graph onto physical parameters. With $s = 1 + x$ its Jacobian determinant is

$$\det D\Phi = k + h + \frac{c - g}{s^2}. \tag{95}$$

At every point with $H - G > 0$, this is $k + (h - g/s^2) + c/s^2 > 0$. Equivalently, for the scalar equilibrium function $B(x) = a - (k + h)x + (c - g)/(1 + x)$, $B_x = -\det D\Phi < 0$. The inverse function theorem makes Φ a local diffeomorphism there, and the implicit function theorem gives a specified smooth positive branch $x = x^*(a, k, c, g, h)$. This regularity uses only the scalar relation, not full-system invertibility or a Hopf result.

Strict inequalities in (88) define open subsets of $\mathcal{U}_{\mathrm{eq}}$. Every strict cell has a $\chi > 0$ point from (92), including the uniform cell, so local projection supplies a nonempty open set of physical parameters for each. In fact every concave-regime point of a strict cell has this local property. For a $T > 0$ uniform example, the family at $k = 1, \chi = -1$ has $\det D\Phi = k + h = 3/2 > 0$; strict $T > 0$ persists in a neighborhood. No global openness assertion at an arbitrary scalar fold in the uniform region is required or inferred.

For $\beta_j > 0$ define in equilibrium coordinates

$$\Psi_j(x, k, c, g, h) = h - \frac{g}{(1 + x)^2} - \frac{c}{4(1 + x)x\beta_j}. \tag{96}$$

Its partial derivative with respect to h, holding x, k, c, g fixed, is exactly 1. Hence its zero set is a smooth codimension-one regular level set within $\mathcal{U}_{\mathrm{eq}}$. It is nonempty by the construction, and every one of its points has $H - G > 0$, so the local diffeomorphism Φ transfers this conclusion to physical parameter space on the chosen branch. Moreover, at fixed x, c, g, k,

$$\frac{\partial \chi}{\partial h} = \frac{4(1 + x)x}{c} > 0, \qquad \frac{\partial}{\partial h}(r_j - r_{j+1}) = -4x(u_j - u_{j+1})\beta_j < 0.$$

Here a varies according to (91); this is a legitimate direction in physical parameter space. Distinct midpoint thresholds and the strict gap to all other representatives at the tie show that the two sides locally select j and $j + 1$, respectively. This proves genuine local changes of dominance, and finishes Theorem 9.3. No global manifold structure after forgetting branch labels is claimed.

9.8 Exact examples and small-dimension checks

In (93) take $k = 1$. The tuple $(a, k, c, g, h) = (2, 1, 4, 4, 1)$ selects 0 in every dimension. For $n = 8$, $(3, 1, 4, 4, 2)$ has $\chi = 2$, $u_* = 1/2$, and selects representative 1, since $\sqrt{2}/4 < 1/2 < (2 + \sqrt{2})/4$. For $n = 6$, $(8/3, 1, 4, 4, 5/3)$ has $\chi = 4/3 = 1/\beta_0$ and selects exactly labels $\{0, 1, 5\}$. For $n = 8$, $(2 + \sqrt{2}, 1, 4, 4, 1 + \sqrt{2})$ has $\chi = 2\sqrt{2} = 1/\beta_1$ and selects $\{1, 2, 6, 7\}$. Every example has $x^* = 1$, all five coefficients positive, and original DM residual $a - k - h + (c - g)/2 = 0$.

For exact highest-mode examples across all four residue classes, use $k = 1$ in the same family with $\chi = 2/\beta_{J-1}$. Then $u_* = \beta_{J-1}/2 \in (0, \beta_{J-1})$, while $\beta_J \le 0$, so J is strict. In particular the following are exact specifications:

n	J	χ	u_J
8	2	$4\sqrt{2}$	0
9	2	$4/\cos(\pi/9)$	$\sin(\pi/18)$
10	2	$2/[\cos(\pi/10)\cos(3\pi/10)]$	$\sin(\pi/10)$
11	3	$2/[\cos(\pi/11)\cos(5\pi/11)]$	$-\sin(\pi/22)$

The corresponding tuples are exactly $(2 + \chi/2, 1, 4, 4, 1 + \chi/2)$. No numerical search is used.

The next table records the complete representative grid and positive midpoints for $5 \le n \le 12$. Within each row only, write $\gamma_r = \cos(r\pi/n)$. All listed midpoint values are exact. Their reciprocal values are the switch thresholds in (88); the reachable set is $\{0, \dots, J\}$. The terminal midpoint β_J is negative except at $n = 6, 10$, where it is zero and unreachable. The five-node row is solely a check; the golden-ratio specialization is reserved for P07.

n	$(u_0, \dots, u_N)$	J	$(\beta_0, \dots, \beta_{J-1})$
5	$(1, \gamma_2, -\gamma_1)$	1	(γ_1^2)
6	$(1, 1/2, -1/2, -1)$	1	$(3/4)$
7	$(1, \gamma_2, -\gamma_3, -\gamma_1)$	2	$(\gamma_1^2, \gamma_1\gamma_3)$
8	$(1, \sqrt{2}/2, 0, -\sqrt{2}/2, -1)$	2	$((2+\sqrt{2})/4, \sqrt{2}/4)$
9	$(1, \gamma_2, \gamma_4, -1/2, -\gamma_1)$	2	$(\gamma_1^2, \gamma_1/2)$
10	$(1, \gamma_2, \gamma_4, -\gamma_4, -\gamma_2, -1)$	2	$(\gamma_1^2, \gamma_1\gamma_3)$
11	$(1, \gamma_2, \gamma_4, -\gamma_5, -\gamma_3, -\gamma_1)$	3	$(\gamma_1^2, \gamma_1\gamma_3, \gamma_1\gamma_5)$
12	$(1, \sqrt{3}/2, 1/2, 0, -1/2, -\sqrt{3}/2, -1)$	3	$((2+\sqrt{3})/4, (\sqrt{3}+1)/4, 1/4)$

9.9 Dominant modes versus first instability

The preceding results concern $\arg\max_m r_m$ at a fixed equilibrium. The scalar d cancels from that comparison but affects whether the maximum is negative, zero, or positive. A first loss of stability would require a specified oriented physical path, a positive equilibrium branch, a prior interval with $\max_m r_m < 0$, and a crossing to a positive maximum. Neither a selection switch nor a real-part tie establishes that.

Indeed every reachable switch can occur on a path that stays strictly stable. Fix its threshold α_j, and choose a compact interval $[b_-, b_+] \subset (0, \infty)$ with α_j in its interior and no other threshold in it. In (93) let χ increase across this interval, hold $k > b_+/2$ fixed, and keep $x^* = 1$. The equilibrium is regular, all parameters remain positive, and P04's verified equilibrium diagonal gives $d = -k - 3$. Thus directly from (77),

$$r_m = -k - 3 + 2\cos\theta_m - \frac{\chi}{2}\cos(2\theta_m) \le -k - 1 + \frac{b_+}{2} < -1. \tag{97}$$

The dominant representative changes from j to $j + 1$, with the exact tie at α_j, while every mode remains strictly stable. This also shows that every strict cell has a stable canonical realization, since k can be enlarged without changing χ or $x^* = 1$ in this family. No first-instability subresult is asserted; that separate crossing question is deferred to P11/P12 under their branch/path gates.

Real-part ties are not eigenvalue coincidences. At every switch P05 proves $r_j = r_{j+1}$ only. P04 supplies the signed imaginary parts

$$\omega_m = -C \sin \theta_m + (G + H) \sin(2\theta_m).$$

For particular labels, equality of the complex eigenvalues would additionally require equality of these signed imaginary parts. For conjugate labels the sign changes. None of the selection, realization, or geometry proofs needs an imaginary-part condition. Numerical eigenvalue simplicity, nonzero critical frequency, simultaneous critical pairs and Hopf exclusions remain P11's tasks. Counting two or four selected labels does not certify a simple pair or a multiple-eigenvalue collision.

9.10 Checks and possible failure points

Internal analytic checks. Expansion of (83), its derivative, and the explicit endpoint differences check the quadratic and continuous regimes. The squared-distance difference checks the orientation of every midpoint inequality; strict concavity excludes nonadjacent maximal ties. The midpoint product in (84) and the four straddling cosine calculations independently check J and the negative-cosine case. Equations (89)–(92) check the canonical substitutions, positivity and original vector-field residual. The determinant and $\partial_h \Psi_j = 1$ check the branch regularity and codimension statements. Singleton bookkeeping uses actual Fourier labels; the eigenvalue discussion does not infer simplicity.

Reproducible corroboration. The separately delivered P05 validation support contains a deterministic checker, exact rational symbolic identities, high-precision all-label comparisons for every $5 \le n \le 40$, canonical construction residuals, and build/provenance records. It samples several points in every strict cell, both sides of every switch and the switches themselves, as well as $T = 0$, $u_* = 1$, near-zero vertices and every residue class modulo four. Tie inputs are defined by exact trigonometric expressions; numerical comparisons at those inputs are still finite-precision corroboration. The analytic proofs above do not depend on computed values. Run totals, tolerances and environment are recorded in the support output and manifest. Compilation and these internal checks do not verify or independently audit P05.

Priority checks for the independent P05 auditor.

(1) Authenticate the current P01–P04 closure versions, preserve all their bodies, and keep this P05 version DRAFT until independently audited.

(2) Check the signs in the quadratic, square completion and adjacent gap; retain mode zero, $T = 0$, $u_* = 1$, and the actual discrete first threshold.

(3) Check the descending-grid Voronoi inequalities, both endpoint cells, all maximal ties, and the exclusion of nonadjacent and triple maxima.

(4) Derive $J = \lfloor (n+1)/4 \rfloor$ independently. Inspect the negative cosine for $n \equiv 3 \pmod 4$, zero cosine for $4 \mid n$, and the unattainable zero-midpoint tie for $n \equiv 2 \pmod 4$, including its persistent finite gap.

(5) Verify every constructed coefficient is positive and substitute into the original DM equation, rather than treating spectral coefficients as free.

(6) Distinguish openness in equilibrium coordinates from openness after projection. Check $\det D\Phi > 0$ throughout the concave regime and $\partial_h \Psi_j = 1$ on each switch; assert only the stated local geometry.

(7) Count singleton and paired labels correctly; real-part ties do not establish eigenvalue equality, nonzero frequency, or a simple Hopf pair.

(8) Check the stable switch path and ensure no hidden dependence on P07, P08 or P11–P13, no first-instability assertion, and no novelty claim for the bare quadratic maximization.

10 P06: Large-n selected-wave-number limit

10.1 TODO: Selected-wave-number limit

10.2 TODO: Discretization error and uniformity domain

11 P07: Five-node golden-ratio specialization

Status: DRAFT; independent A07 audit pending. This is a corollary package for the audited P03–P05 demand-modulated results. The direct dependency remains P05 → P07; P03–P04 and the P01–P02 equilibrium foundation are inherited. P08 is present in the current project but supplies no theorem used here.

Fix $a, k, c, g, h > 0$ in the canonical model

$$\dot{x}_i = ax_i - kx_i^2 + \frac{cx_{i-1}}{1+x_i} - \frac{gx_i}{1+x_{i+2}} - hx_ix_{i-2}, \qquad i \in \mathbb{Z}_5. \tag{98}$$

Let $x^*\mathbf{1}$, $x^* > 0$, be a specified synchronous equilibrium. Write

$$s = 1 + x^* > 1, \quad C = \frac{c}{s} > 0, \quad G = \frac{gx^*}{s^2} > 0, \quad H = hx^* > 0, \quad T = G - H, \quad S = G + H. \tag{99}$$

Here s denotes the denominator, not P02's scalar state variable; S is local to P07 and denotes effective DM coefficients, not the physical sum $g+h$ used locally in P08. We use $D_5 := H - G = -T$, leaving P02's discriminant D unchanged. The symbols ω_1, ω_2 below are signed imaginary parts, not a reserved positive Hopf frequency. By P02 and P03,

$$a - (k+h)x^* + \frac{c-g}{s} = 0, \tag{100}$$

$$d = a - 2kx^* - \frac{cx^*}{s^2} - \frac{g}{s} - hx^* = -kx^* - C - \frac{Cx^*}{s}. \tag{101}$$

The last equality uses equilibrium and will only be used when stated.

11.1 Exact fifth-root constants and Fourier spectrum

Lemma 11.1 (Golden-ratio trigonometry)**.** *Set*

$$\phi = \frac{1+\sqrt{5}}{2}, \qquad u_1 = \cos\frac{2\pi}{5}, \qquad u_2 = \cos\frac{4\pi}{5}, \qquad \sigma = \sin\frac{\pi}{5}. \tag{102}$$

Then

$$\phi^2 = \phi + 1, \quad \phi^{-1} = \phi - 1, \quad \phi^{-2} = 2 - \phi = \frac{3-\sqrt{5}}{2}, \quad \phi + \phi^{-1} = \sqrt{5}, \tag{103}$$

$$u_1 = \frac{\sqrt{5}-1}{4} = \frac{1}{2\phi}, \qquad u_2 = -\frac{\sqrt{5}+1}{4} = -\frac{\phi}{2}, \qquad u_1 - u_2 = \frac{\sqrt{5}}{2}, \tag{104}$$

$$\sigma = \frac{\sqrt{10-2\sqrt{5}}}{4} > 0, \qquad \sin\frac{2\pi}{5} = \phi\sigma = \frac{\sqrt{10+2\sqrt{5}}}{4}. \tag{105}$$

Proof. The quadratic identity for ϕ follows by squaring its definition; division by $\phi > 0$ and another multiplication give the remaining identities in (103). Let $z = e^{2\pi i/5}$. Since $1 + z + z^2 + z^3 + z^4 = 0$, real parts give $1 + 2u_1 + 2u_2 = 0$. Also $u_2 = 2u_1^2 - 1$, so $4u_1^2 + 2u_1 - 1 = 0$. As $0 < 2\pi/5 < \pi/2$, the positive root is $u_1 = (\sqrt{5}-1)/4$; substitution gives u_2 and their difference. Moreover $\cos(\pi/5) = -\cos(4\pi/5) = \phi/2$. Thus $\sigma^2 = 1 - \phi^2/4 = (5-\sqrt{5})/8$, with the positive square root required by $0 < \pi/5 < \pi/2$. The double-angle formula then gives $\sin(2\pi/5) = 2\sigma(\phi/2) = \phi\sigma$; squaring and using $\phi^2 = \phi + 1$ gives the second radical. □

With $\theta_m = 2\pi m/5$, all phase signs needed below are displayed explicitly. Each row follows from periodicity and the quadrant signs:

m	$\cos\theta_m$	$\sin\theta_m$	$\cos(2\theta_m)$	$\sin(2\theta_m)$
0	1	0	1	0
1	u_1	$\phi\sigma$	u_2	σ
2	u_2	σ	u_1	$-\phi\sigma$
3	u_2	$-\sigma$	u_1	$\phi\sigma$
4	u_1	$-\phi\sigma$	u_2	$-\sigma$

There are exactly three representative classes: $j = 0$ with labels $\{0\}$, $j = 1$ with labels $\{1, 4\}$, and $j = 2$ with labels $\{2, 3\}$. Odd dimension excludes a Nyquist singleton. The two labels of a conjugate pair are one real-part class, even when their eigenvalues are distinct.

Theorem 11.2 (Exact five-node Fourier spectrum)**.** *The five eigenvalues, with their Fourier labels and algebraic multiplicities retained, are*

$$\begin{aligned}\lambda_0 &= d + C + T, \\ \lambda_1 &= d + \frac{C}{2\phi} - \frac{\phi T}{2} + \mathrm{i}\sigma(S - \phi C), \\ \lambda_2 &= d - \frac{\phi C}{2} + \frac{T}{2\phi} - \mathrm{i}\sigma(C + \phi S), \\ \lambda_4 &= \overline{\lambda_1}, \qquad \lambda_3 = \overline{\lambda_2}.\end{aligned} \tag{106}$$

In particular the three real-part classes are

$$r_0 = d + C + T, \qquad r_1 = d + \frac{C}{2\phi} - \frac{\phi T}{2}, \qquad r_2 = d - \frac{\phi C}{2} + \frac{T}{2\phi}. \tag{107}$$

Here $r_1 = \operatorname{Re}\lambda_1 = \operatorname{Re}\lambda_4$ and $r_2 = \operatorname{Re}\lambda_2 = \operatorname{Re}\lambda_3$.

Proof. Start with the audited P04 spectrum, Theorem 8.1:

$$\lambda_m = d + Ce^{-\mathrm{i}\theta_m} + Ge^{2\mathrm{i}\theta_m} - He^{-2\mathrm{i}\theta_m}.$$

Expansion into sine and cosine, term by term, yields

$$\lambda_m = d + C\cos\theta_m + T\cos(2\theta_m) + \mathrm{i}[-C\sin\theta_m + S\sin(2\theta_m)].$$

In particular $-He^{-2\mathrm{i}\theta_m}$ contributes $-H\cos(2\theta_m)+\mathrm{i}H\sin(2\theta_m)$: the real coefficient is $G-H$, whereas the imaginary coefficient is $G+H$. Insert rows 0, 1 and 2 of the phase table and (104)–(105). Rows 4 and 3 have unchanged cosines and opposite sines, proving the stated conjugacy. As a separate real-part check, P05's quadratic $R(u) = d - T + Cu + 2Tu^2$ gives $R(1) = d + C + T$. The identities $2u_1^2 - 1 = u_2$ and $2u_2^2 - 1 = u_1$ give $R(u_1) = d + Cu_1 + Tu_2$ and $R(u_2) = d + Cu_2 + Tu_1$, exactly (107). □

11.2 Complete ordering and the dominant-mode switch

Theorem 11.3 (Complete ordering of the three real-part classes)**.** *Set $t = T/C$, which is well defined since $C > 0$. The exact differences are*

$$\begin{aligned}r_1 - r_2 &= \frac{\sqrt{5}}{2}(C - T) = \frac{\sqrt{5}C}{2}(1 - t), \\ r_0 - r_1 &= \frac{\sqrt{5}\phi}{2}\left(T + \frac{C}{\phi^2}\right) = \frac{\sqrt{5}\phi C}{2}(t + \phi^{-2}), \\ r_0 - r_2 &= \frac{\sqrt{5}}{2\phi}(T + \phi^2 C) = \frac{\sqrt{5}C}{2\phi}(t + \phi^2).\end{aligned} \tag{108}$$

Consequently the following table is exhaustive, including all equalities:

Condition	*Ordering*	*Tie type*
$t > 1$	$r_0 > r_2 > r_1$	*none*
$t = 1$	$r_0 > r_1 = r_2$	*subdominant*
$-\phi^{-2} < t < 1$	$r_0 > r_1 > r_2$	*none*
$t = -\phi^{-2}$	$r_0 = r_1 > r_2$	*dominant*
$-\phi^2 < t < -\phi^{-2}$	$r_1 > r_0 > r_2$	*none*
$t = -\phi^2$	$r_1 > r_0 = r_2$	*subdominant*
$t < -\phi^2$	$r_1 > r_2 > r_0$	*none*

Ordering is independent of d. No triple tie of real-part classes occurs.

Proof. Subtract (107). The first gap is $(\phi+\phi^{-1})(C-T)/2$. The second is $C(1-1/(2\phi))+T(1+\phi/2)$; the third exchanges these coefficients. Using (103),

$$1+\frac{\phi}{2}=\frac{\sqrt{5}\phi}{2},\qquad 1-\frac{1}{2\phi}=\frac{\sqrt{5}}{2\phi},\qquad \frac{1-1/(2\phi)}{1+\phi/2}=\phi^{-2}.$$

This proves every factor in (108). All prefactors are positive, and $-\phi^2 < -\phi^{-2} < 1$. Starting above 1, the signs of $(r_1 - r_2, r_0 - r_1, r_0 - r_2)$ are $(-,+,+)$. They are $(0,+,+)$ at 1, $(+,+,+)$ down to $-\phi^{-2}$, $(+,0,+)$ there, $(+,-,+)$ down to $-\phi^2$, $(+,-,0)$ there, and $(+,-,-)$ below it. These signs give exactly the seven rows. Distinct gap zeros exclude a triple class tie. The common summand d cancels from each difference; it is not an independent physical parameter by this observation. □

Corollary 11.4 (Exact golden-ratio selection threshold)**.** *The dominant representative set and the full selected Fourier-label set are*

Condition	*Representatives*	*Labels*
$D_5/C < \phi^{-2}$	$\{0\}$	$\{0\}$
$D_5/C = \phi^{-2}$	$\{0,1\}$	$\{0,1,4\}$
$D_5/C > \phi^{-2}$	$\{1\}$	$\{1,4\}$.

(109)

Pair $\{2,3\}$ is never dominant at a canonical positive synchronous equilibrium. The unique dominant-class switch is

$$\boxed{H-G=\frac{C}{\phi^2},\qquad \frac{H-G}{C}=\frac{3-\sqrt{5}}{2}.} \tag{110}$$

Proof. The sign of $r_0 - r_1$ gives the three cases. For a direct exclusion of class 2, if $t < 1$ then $r_1 > r_2$ by the first gap; if $t \geq 1$ then $r_0 > r_2$ by the third gap. Thus class 2 cannot maximize even at either subdominant equality. This argument compares all three classes and does not rely solely on P05's reachability list. □

Check against P05's exact cells. In dimension five,

$$\beta_0=\frac{1+u_1}{2}=\frac{\phi^2}{4},\qquad \beta_1=\frac{u_1+u_2}{2}=-\frac{1}{4},\qquad \alpha_0=\beta_0^{-1}=\frac{4}{\phi^2}=6-2\sqrt{5}. \tag{111}$$

Indeed $2+\phi^{-1}=\phi+1=\phi^2$ and $\phi^{-1}-\phi=-1$. In the concave regime $D_5>0$, P05's vertex is $u_*=C/(4D_5)>0$. It passes β_0 at (110) and never reaches β_1. Theorem 9.1 therefore agrees exactly with the three-class calculation. In P05's variables,

$$\chi=\frac{4(H-G)}{C},\qquad \begin{cases}\chi<4/\phi^2 & :\{0\}\text{ selected},\\ \chi=4/\phi^2 & :\{0,1\}\text{ selected},\\ \chi>4/\phi^2 & :\{1\}\text{ selected}.\end{cases} \tag{112}$$

The current P05 defines χ for all real signs, since $C>0$, so (112) is also valid when $H\le G$: then $\chi\le 0$ and only class 0 is selected. The concave vertex and reciprocal-midpoint argument above are used only when $H>G$.

Canonical switch surface and the equilibrium constraint. Since $D_5=x^*(h-g/s^2)$, solving (110) gives

$$\boxed{h=\frac{g}{(1+x^*)^2}+\frac{c}{\phi^2(1+x^*)x^*}.} \tag{113}$$

The divisions are valid because x^*,s,ϕ,C are positive. The equilibrium relation (100) must hold simultaneously. Thus this equation describes a switch on the positive branch-state manifold; x^* cannot be prescribed independently of a fixed physical tuple. The corresponding strict inequalities in h at that equilibrium give uniform or pair-1 selection, respectively. P05 already supplies regularity and local realizability of this switch.

The other golden-ratio equality surfaces. At $t=1$, equivalently $G-H=C$, one has $r_1=r_2$ but $r_0-r_1=r_0-r_2=5C/2>0$. At $t=-\phi^2$, equivalently $H-G=\phi^2C$, one has $r_0=r_2$ but $r_1-r_0=r_1-r_2=5\phi C/2>0$. These values follow from (108) and $\phi^2=\phi+1$. Both are subdominant real-part ties, not dominant-mode switches. No equality of complex numerical eigenvalues is implied by either tie.

11.3 Frequencies, intersection, and numerical eigenvalue equality

Proposition 11.5 (Signed imaginary parts and frequency degeneracy)**.** *At positive synchrony,*

$$\omega_1:=\operatorname{Im}\lambda_1=\sigma(S-\phi C),\qquad \omega_2:=\operatorname{Im}\lambda_2=-\sigma(C+\phi S)<0. \tag{114}$$

Pair $\{2,3\}$ *is always nonreal. Pair* $\{1,4\}$ *is nonreal if and only if* $S\neq\phi C$*. At the spectral locus*

$$\boxed{G+H=\phi C,} \tag{115}$$

one has $\lambda_1=\lambda_4=r_1$*, a repeated real numerical eigenvalue represented by two Fourier labels. All numerical eigenvalues are semisimple; any further coincident label is counted separately below.*

Proof. The imaginary terms were derived directly in Theorem 11.2. Since $\sigma,C,S,\phi>0$, ω_2 is strictly negative, while $\omega_1=0$ is equivalent to $S=\phi C$. Conjugacy then proves all reality assertions. Semisimplicity is Proposition 8.2, which does not assume distinct numerical eigenvalues. These assertions classify spectra only. □

Proposition 11.6 (Intersection with the dominant-mode switch)**.** *The switch and the frequency-degeneracy locus intersect in the strictly positive coefficient domain exactly at*

$$\boxed{\frac{G}{C}=\frac{1}{\phi},\qquad \frac{H}{C}=1.} \tag{116}$$

They define a nonempty codimension-two spectral locus in $(C, G, H) > 0$, and locally on the canonical positive equilibrium branch. At that locus

$$\lambda_0 = \lambda_1 = \lambda_4 = d + \frac{C}{\phi} = -kx^* - \frac{Cx^*}{s} - \frac{C}{\phi^2} < 0. \tag{117}$$

This numerical eigenvalue is semisimple of multiplicity three, and $r_2 < r_0$. In particular the full linearization is Hurwitz there. This is a spectral codimension statement, not a bifurcation classification.

Proof. Adding and subtracting $H - G = C\phi^{-2}$ and $G + H = \phi C$ gives

$$\frac{G}{C} = \frac{\phi - \phi^{-2}}{2} = \phi - 1 = \phi^{-1}, \qquad \frac{H}{C} = \frac{\phi + \phi^{-2}}{2} = 1.$$

Both are positive. The gradients in (G, H) of $H - G - C/\phi^2$ and $G + H - \phi C$ have determinant -2, so the two linear conditions are independent. For a canonical witness take

$$x^* = 1, \qquad (a, k, c, g, h) = (\sqrt{5}, 1, 2, 4/\phi, 1). \tag{118}$$

The equilibrium bracket equals $\sqrt{5} - 2 + (2 - 4/\phi)/2 = 0$, using $1 + 2/\phi = \sqrt{5}$. The effective coefficients are $C = 1, G = \phi^{-1}, H = 1$.

For the local branch assertion use P05's verified equilibrium chart $(x, k, c, g, h) \mapsto ((k + h)x - (c-g)/(1+x), k, c, g, h)$. Its determinant is positive wherever $H - G > 0$; the intersection lies in that regime. At fixed x, k, c, the two constraint gradients in (g, h) have determinant $-2x^2/(1 + x)^2 \neq 0$. The chart therefore transfers their independent codimension-two conditions locally to physical parameters on the specified branch. There is no claim about a parameter-path crossing or an unfolding.

At the intersection, conjugacy, $\omega_1 = 0$, and $r_0 = r_1$ give the triple value. Substitute (101) and $1 - \phi^{-1} = \phi^{-2}$ for its strict negativity. The ordering table gives $r_2 < r_0$; $\omega_2 < 0$ keeps labels 2 and 3 distinct from the real value and from each other. P04 counts precisely three independent Fourier eigenvectors for the triple value. □

Proposition 11.7 (Complete numerical coincidence classification)**.** *Under the stated positive-equilibrium assumptions, the only numerical eigenvalue coincidences are as follows:*

(i) If $S \neq \phi C$, all five numerical eigenvalues are distinct.

(ii) If $S = \phi C$ and $T \neq -C/\phi^2$, only $\lambda_1 = \lambda_4$ coincide; their real value has multiplicity two.

(iii) If $S = \phi C$ and $T = -C/\phi^2$, exactly $\lambda_0 = \lambda_1 = \lambda_4$ coincide as in (117); their value has multiplicity three.

All multiplicities are algebraic and geometric. Labels 2 and 3 always give distinct, simple numerical eigenvalues, sharing no value with any other label. Every nonreal pair-1 eigenvalue is simple.

Proof. Within pairs, (114) proves that labels 1 and 4 coincide exactly when $S = \phi C$, while labels 2 and 3 never coincide. The real λ_0 cannot equal either nonreal pair-2 eigenvalue. Equality between λ_0 and a pair-1 label requires both $r_0 = r_1$ and $\omega_1 = 0$, exactly the intersection just analyzed.

A cross-pair numerical equality requires $r_1 = r_2$, hence $T = C$. Then $G = H + C$ and $S = C + 2H > C$. At this real-part tie,

$$\omega_1 - \omega_2 = \sigma[(1 + \phi)S - (\phi - 1)C] > 0,$$
$$-\omega_2 - \omega_1 = \sigma[(1 + \phi)C + (\phi - 1)S] > 0.$$

For the first inequality use $S > C$ and $(1+\phi) - (\phi - 1) = 2$; the second has strictly positive summands. Thus $\omega_2 < \omega_1 < -\omega_2$, so ω_1 is neither ω_2 nor $-\omega_2$; conjugating excludes all four possible cross-pair equalities. This exhausts all unordered label pairs. P04 then gives the asserted multiplicities. □

The frequency-degeneracy locus itself has $-\phi < t < \phi$, since $G, H > 0$ imply $|T| < S$. It can intersect the subdominant tie $t = 1$ without a cross-pair coincidence, but cannot intersect $t = -\phi^2$. Equal real parts, repeated real values, and distinct nonreal conjugates are consequently kept separate throughout.

11.4 Exact stability test and critical-mode ledger

Corollary 11.8 (Piecewise five-node linear stability)**.** *The linearization is exponentially asymptotically stable exactly when*

$$\max\{r_0, r_1, r_2\} < 0 \quad \Longleftrightarrow \quad \begin{cases} r_0 < 0, & t > -\phi^{-2}, \\ r_0 = r_1 < 0, & t = -\phi^{-2}, \\ r_1 < 0, & t < -\phi^{-2}. \end{cases} \tag{119}$$

The middle row means that the common value must be negative. At canonical positive equilibria on that tie it is automatically negative:

$$r_0 = r_1 = -kx^* - \frac{Cx^*}{s} - \frac{C}{\phi^2} < 0. \tag{120}$$

Proof. P04 Corollary 8.4 gives the strict all-mode test; Theorem 11.3 gives its piecewise maximum. At the dominant tie, substitute $T = -C/\phi^2$ and (101) into $r_0 = d + C + T$. □

If the maximum is zero, P04 implies that the normal linear system is Lyapunov stable but not asymptotically stable. This does not settle nonlinear stability or establish a bifurcation. A strictly positive maximum means linear instability. The dominant tie cannot itself be a zero-abscissa boundary in the canonical domain, by (120).

Proposition 11.9 (Fixed-state critical-mode ledger)**.** *Only the following classes can attain the spectral abscissa; this is a fixed-state eligibility ledger, not a first-instability theorem.*

Class	*Eligibility for the maximum*	*Signed imaginary part*
0	$t \geq -\phi^{-2}$	0
$\{1,4\}$	$t \leq -\phi^{-2}$	$\pm\sigma(S - \phi C)$
$\{2,3\}$	*never*	$\mp\sigma(C + \phi S)$

At a zero-abscissa state, the inequalities in the first two rows must be strict, because the shared boundary is strictly stable. If $r_2 = 0$, some other class has strictly positive real part, so pair 2 cannot be the sole or tied controlling class of a stability boundary. No oriented parameter path, prior stability, or crossing is asserted.

Proof. The eligibility and exclusions are precisely Corollary 11.4. At every point at least one of r_0, r_1 exceeds r_2 strictly. The zero-abscissa statements follow from that strict comparison and (120). □

11.5 Realizability checks, downstream inputs, and scope

All ordering regions have canonical representatives. For any $C, G, H > 0$, the following single construction gives a canonical positive equilibrium with those exact effective coefficients:

$$x^* = 1, \quad k = C, \quad c = 2C, \quad g = 4G, \quad h = H, \quad a = H + 2G > 0, \quad d = -\frac{5C}{2}. \tag{121}$$

Indeed (100) becomes $H + 2G - (C + H) + (2C - 4G)/2 = 0$. Every original component equals x^* times that zero bracket. For any real t, take $C = 1$, $G = 1 + \max(t, 0)$ and $H = 1 + \max(-t, 0)$. Then $G - H = t$ and both coefficients are positive. Thus no row of the ordering table is merely an unattached-coefficient possibility. This elementary five-node check uses no MB comparison or new general selection theorem.

Exact inputs available to P11. The unique nonuniform dominant class, when present, is $\{1, 4\}$. Its signed imaginary part is $\sigma(S - \phi C)$, and its exact nonzero-frequency condition is $S \neq \phi C$. Whenever pair 1 is maximal, $r_2 < r_1$ strictly, including the uniform/pair-1 dominant tie. When pair 1 is uniquely dominant, $r_0 < r_1$ as well. Every nonreal pair-1 eigenvalue is simple; pair 2 never coincides numerically with another label. The only uniform/pair-1 numerical coincidence is the strictly stable triple value (117); more generally the entire dominant tie is strictly stable. These are exact spectral inputs, without a Hopf existence or first-crossing conclusion. No transversality, crossing derivative, Lyapunov coefficient, normal form, periodic orbit, or rotating wave is analyzed here; P09–P18 remain unbegun by this module.

Remark 11.10 (Golden ratio is network geometry). The constants ϕ and σ arise solely from the fifth roots of unity and the five-node Fourier grid. They establish no physiological golden-ratio law, biological optimum, medical significance, or empirical TCM validation. P07 is a supporting specialization of the audited general program, not standalone golden-ratio novelty. The frozen scope and novelty documents need no amendment.

Supplementary corroboration. The separate `P07_validation_support.zip` contains exact algebraic checks, direct 5×5 matrix comparisons at positive canonical equilibria, current-input hashes, and build/preservation records. Checks cover all seven ordering rows, both sides of the dominant switch, the other two ties, and the frequency locus and its neighborhood. The analytic proofs above do not depend on a finite check, floating-point equality, compilation, or numerical continuation. P07 remains DRAFT.

11.6 Priority checks for the independent P07 auditor

1. Reconstruct the fifth-root cosine quadratic, positive sine radicals, and every use of $\phi^2 = \phi + 1$; check all five phase-table rows.

2. Start from P04's oriented shift phases and check the minus sign in $-He^{-2i\theta_m}$. The real coefficient is $T = G - H$; the imaginary coefficient is $S = G + H$. Verify both signed frequencies.

3. Recompute all three factorized gaps and reconstruct all seven ordering rows. Distinguish the dominant threshold $-\phi^{-2}$ from the subdominant threshold $-\phi^2$ and the other subdominant tie $t = 1$.

4. Check all three classes directly, the impossibility of pair-2 dominance, the midpoint constants, the all-sign P05 convention for χ, and the canonical switch together with the equilibrium relation.

5. Check $S = \phi C$, the positive intersection $G/C = \phi^{-1}, H/C = 1$, the codimension gradients and branch regularity, and the strictly negative triple value. Count numerical multiplicity rather than conjugate labels or tied real parts.

6. Exhaust all label-pair coincidences, particularly the cross-pair real-part tie $T = C$, where $S > C$ is essential. Verify the independent strict negativity of the whole dominant tie, not only its intersection.

7. Check the piecewise Hurwitz test and the critical-mode ledger; retain the distinction between a fixed-state maximizer and a first instability along a path. No Hopf or other bifurcation claim is licensed.

8. Authenticate the current P03–P05 and audit closures, preserve P05 $\rightarrow$ P07 without a P08 edge, and keep P07 DRAFT until a separate exact-version audit is completed. Computational corroboration and a successful build are not that audit.

12 P08: Matched bilinear analysis and exact comparison

Status: DRAFT; independent A08 audit required. All coefficients are strictly positive, $n \geq 5$, $1 \leq q < n$, and indices are modulo n. The comparator is exactly the five-parameter model ($\mathrm{MB}_{n,q}$). The same tuple (a, k, c, g, h), n, q and time variable are used in DM and MB; each equilibrium amplitude is derived for its own equation. Only current audited P01–P05 are imported. Their historical DRAFT body sentences are superseded by their current headers and audit-closure records.

The substantive results are open same-tuple stability contrasts in both directions at $n = 7$, a universal forward contrast, and realization of every ordered pair of reachable dominant representatives with both models stable. None is a first-instability, Hopf, criticality, or literature-priority theorem. Here "strictly stable" means Hurwitz; "unstable" means a strictly positive spectral abscissa. Zero abscissa is treated separately. Selection maximizes over all labels, including zero, retaining every maximal tie.

12.1 MB local flow, synchrony, and synchronous growth

Put $S = g + h > 0$ and $M = k + S - c$; the subspace remains denoted by $\mathcal{S}$.

Theorem 12.1 (MB foundation)**.** *For arbitrary canonical (n, q) the following hold.*

(i) MB has a unique maximal classical solution through every point of $\mathbb{R}^n$. Every coordinate face is invariant. Nonnegative initial data stay nonnegative, and initially positive coordinates remain strictly positive on the maximal interval. No uniform asymptotic lower bound is meant by this finite-time positivity statement.

(ii) $\mathcal{S}$ is invariant, with exact restriction

$$\dot{s} = as - Ms^2. \tag{122}$$

The real synchronous equilibria are $0, a/M$ if $M \neq 0$, and only 0 if $M = 0$. A positive synchronous equilibrium exists if and only if $M > 0$, and is uniquely $y^* = a/M$. For $M < 0$ the nonzero root is negative.

(iii) For $s_0 > 0, M > 0$, synchronous dynamics are global, $0 < s(t) \leq \max(s_0, a/M)$ and $s(t) \to a/M$. For $M = 0$ they are $s(t) = s_0 e^{at}$, global and unbounded. For $M < 0$ they blow up at

$$t_b = \frac{1}{a} \log \left(1 + \frac{a}{|M| s_0} \right). \tag{123}$$

These classify nonnegative synchronous dynamics, not all MB trajectories.

Proof. The polynomial field is smooth and locally Lipschitz on $\mathbb{R}^n$, so local existence, uniqueness, continuous dependence and maximal continuation apply. On the coordinate face $y_i = 0$, direct substitution in the original five terms gives $F_i^{\mathrm{MB}}(y) = 0$, even when other coordinates are negative. Along any solution the integrating-factor identity is

$$y_i(t) = y_i(0) \exp \int_0^t \left[a - k y_i(r) + c y_{i-1}(r) - g y_{i+q}(r) - h y_{i-q}(r) \right] dr. \tag{124}$$

The integrand is continuous on each compact subinterval of the maximal interval. The identity holds also for zero initial coordinates and proves sign preservation and face invariance without assuming global existence.

At $y = s\mathbf{1}$, each component of the original field is $as - ks^2 + cs^2 - gs^2 - hs^2 = as - Ms^2$. The scalar solution lifts to a vector solution, uniqueness identifies the two, and continuation makes their maximal intervals coincide. Zeros of $s(a - Ms)$ give the complete synchronous equilibrium list, retaining zero before division. For $s_0 > 0$, solve $(1/s)' = -a/s + M$ to obtain

$$s(t) = \frac{s_0 e^{at}}{1 + (Ms_0/a)(e^{at} - 1)}. \tag{125}$$

For $M > 0$ the denominator stays positive and the bound and limit follow; for $M = 0$ it is one; for $M < 0$ its first zero is exactly (123), approached from above with positive numerator. The zero initial state stays zero in every case. □

12.2 Sufficient full-system bounds and their scope

Proposition 12.2 (Maximum comparison)**.** *For nonnegative MB solutions and $Y = \max_i y_i$,*

$$D^+ Y \le aY - (k - c)Y^2. \tag{126}$$

Thus $k \ge c$ suffices for global nonnegative existence. If $k > c$, put $R = a/(k - c)$. Then $Y(t) \le \max\{Y(0), R\}$, $[0, R]^n$ is invariant and uniformly attracts bounded subsets, and every $[0, R+\varepsilon]^n$, $\varepsilon > 0$, absorbs bounded subsets in finite time. MB is therefore dissipative in this sufficient regime. The comparison alone asserts neither finite-time absorption into the exact cube $[0, R]^n$ nor boundedness when $k = c$.

Proof. Apply the generic finite-maximum identity of Lemma 5.1. At any maximizing coordinate $y_i = Y$, all neighbors lie in $[0, Y]$, so $cy_i y_{i-1} \le cY^2$ and both inhibitory terms are nonpositive. This includes tied maxima and coincident offsets. For $k \ge c$, the inequality gives $Y(t) \le Y(0)e^{at}$, excluding finite-time escape; polynomial continuation yields global existence. For $k > c$, logistic comparison bounds Y and proves invariance. A logistic solution starting from the supremum of any bounded set of initial maxima converges to R and enters $[0, R + \varepsilon]$ in finite time, proving the uniform attraction and absorption assertions. □

Consequently $M > 0$ is necessary for boundedness of every nonnegative MB trajectory, and $M \ge 0$ is necessary for global existence of every such trajectory, by the synchronous counterexamples. No converse is inferred. In particular existence of positive synchrony is not used to assert full-system boundedness. The sufficient condition $k > c$ is not necessary, as an explicit example below will prove.

12.3 Homogeneous MB Jacobian and arbitrary-offset spectrum

At $y = s\mathbf{1}$, differentiate the original field before using equilibrium. None of $i - 1, i + q, i - q$ equals i in the canonical range, so the diagonal is

$$d_{\mathrm{MB}}(s) = a - 2ks + cs - gs - hs = a - (k + M)s.$$

The three off-diagonal contributions are $cs, -gs, -hs$. Therefore

$$DF^{\mathrm{MB}}(s\mathbf{1}) = [a - (k + M)s]I + s(cP - gP^{-q} - hP^q). \tag{127}$$

In full, with row i and column j, the general derivative entry is

$$[a - 2ky_i + cy_{i-1} - gy_{i+q} - hy_{i-q}]\delta^{(n)}_{ji} + cy_i\delta^{(n)}_{j,i-1} - gy_i\delta^{(n)}_{j,i+q} - hy_i\delta^{(n)}_{j,i-q}.$$

Coincident columns add; none is overwritten. At $y^* = a/M > 0$ use $a = My^*$ only now, giving $d_{\mathrm{MB}} = -ky^*$.

Theorem 12.3 (Full MB spectrum, normality, and uniform direction)**.** *For $M > 0$, positive MB synchrony has*

$$J_{\mathrm{MB}} = y^*[-kI + cP - gP^{-q} - hP^q], \tag{128}$$

$$\lambda_m^{\mathrm{MB}} = y^*[-k + ce^{-\mathrm{i}\theta_m} - ge^{\mathrm{i}q\theta_m} - he^{-\mathrm{i}q\theta_m}], \tag{129}$$

$$\operatorname{Re}\lambda_m^{\mathrm{MB}} = y^*[-k + c\cos\theta_m - S\cos(q\theta_m)], \tag{130}$$

$$\operatorname{Im}\lambda_m^{\mathrm{MB}} = y^*[-c\sin\theta_m + (h-g)\sin(q\theta_m)]. \tag{131}$$

The n Fourier labels give the full spectrum with algebraic multiplicity; J_{MB} is normal and unitarily diagonalized by the project's DFT. In particular

$$\boxed{\lambda_0^{\mathrm{MB}} = -a < 0.} \tag{132}$$

Any linear instability at positive MB synchrony involves a nonuniform mode. This does not classify a bifurcation as Hopf or steady.

Proof. The preceding differentiation proves the matrix identity. In the verified convention $P^\ell v^{(m)} = e^{-\mathrm{i}\ell\theta_m} v^{(m)}$. Applying these powers yields the stated spectrum, including the negative coefficient on the positive-phase g term. Euler's formula gives the real and imaginary parts; the imaginary contributions are $-c\sin\theta_m, -g\sin(q\theta_m), +h\sin(q\theta_m)$. The DFT is unitary by the generic geometric-sum proof in P04, whence $J_{\mathrm{MB}} = \mathsf{Q}\,\mathrm{diag}(\lambda_m^{\mathrm{MB}})\mathsf{Q}^*$, proving the full multiplicity statement and normality. Independently, J_{MB} and its adjoint are sums of mutually commuting powers of the unitary shift, so they commute.

For the first check of the uniform value, Fourier substitution at $m = 0$ gives $y^*(-k+c-g-h) = -My^* = -a$. For a second independent check, the derivative of the scalar restriction is $a - 2Ms$, equal to $-a$ at $s = a/M$. Its tangent vector is $\mathbf{1}$, so it is precisely the uniform eigenvalue. Its strict negativity proves the instability claim. □

The complete offset-collision checks, using P03/P04 only for their index geometry, are

$q = 1$	$J_{\mathrm{MB}}/y^* = -kI + (c-h)P - gP^{-1}$
$q = n-1$	$J_{\mathrm{MB}}/y^* = -kI + (c-g)P - hP^{-1}$
$q = n/2$ (n even)	$J_{\mathrm{MB}}/y^* = -kI + cP - SP^{n/2}$.

Their imaginary parts are respectively $y^*(-c+h-g)\sin\theta_m$, $y^*(-c+g-h)\sin\theta_m$, and $-y^*c\sin\theta_m$. The last real restraint term is $-y^*S(-1)^m$. Each also follows from the master spectrum, with cancellation retained. For even n, the singleton is $\lambda_{n/2}^{\mathrm{MB}} = y^*[-k-c-S(-1)^q]$; the uniform value is unchanged in every collision case. Label conjugacy and semisimplicity follow from the real circulant matrix and its unitary diagonalization; a pair may have zero imaginary part or share an eigenvalue with other labels. No numerical-eigenvalue simplicity is inferred from label counts.

Corollary 12.4 (Exact stability and another sufficient global bound)**.** *Define*

$$K_{n,q}(c,S) = \max_{0\le m<n} [c\cos\theta_m - S\cos(q\theta_m)]. \tag{133}$$

At positive MB synchrony, strict stability is equivalent to $k > K_{n,q}$, and instability to $k < K_{n,q}$. At equality the normal linear system is Lyapunov stable but not asymptotically stable; nonlinear stability is undecided here. Hyperbolicity requires every real part to be nonzero, not merely a nonzero maximum.

Moreover $k > K_{n,q}$ itself implies $M > 0$, global boundedness and dissipativity for nonnegative MB data. This is another sufficient bound, not a necessary condition or an identification with $k > c$.

Proof. The abscissa is $y^*(K_{n,q}-k)$ and unitary diagonalization gives $\|e^{tJ_{\mathrm{MB}}}\|_2 = e^{t\,\mathrm{sa}(J_{\mathrm{MB}})}$, proving the linear claims. For the bound put $\delta = k - K_{n,q} > 0$, $V = \sum_i y_i$. Summing the original equations yields

$$V' = aV + y^{\mathsf{T}}[-kI + cP - gP^{-q} - hP^q]y \le aV - \delta\|y\|_2^2 \le aV - \frac{\delta}{n}V^2. \tag{134}$$

The symmetric part of the displayed matrix has eigenvalues $-k + c\cos\theta_m - S\cos(q\theta_m) \le -\delta$, and its skew part contributes zero to the real quadratic form. Logistic comparison bounds V before any maximal endpoint and proves global continuation. The simplex $V \le na/\delta$ attracts bounded subsets, and every strictly larger one absorbs them. Finally $K_{n,q} \ge c - S$ by its zero label, so $M > 0$. □

For example, for $q = 2$ the tuple $(9/2, 7, 8, 1, 1)$ has $K_{n,2} = 6$ for every $n \ge 5$: $2 + 8u - 4u^2$ uniquely maximizes at $u = 1$ on the grid. MB is dissipative although $k = 7 < c = 8$. Its own root is $y^* = 9/2$, whereas DM's unique positive root is $x^* = 1$. Direct original residuals are $9/2 - 8 + 7/2 = 0$ for DM after division by x^*, and $9/2 - (9/2) = 0$ for MB after division by y^*. This refutes necessity of $k > c$. Conversely, $k > c$ need not imply spectral stability when S is large, so the sufficient bounds have different scopes.

12.4 Exact step-two MB selected-mode rule

Set $q = 2$ and $u_m = \cos\theta_m$. Since $\cos(2\theta) = 2u^2 - 1$,

$$p(u_m) := \mathrm{Re}\,\lambda_m^{\mathrm{MB}}/y^* = -k + S + cu_m - 2Su_m^2, \tag{135}$$

$$p(u) = -k + S + \frac{c^2}{8S} - 2S(u - v_{\mathrm{MB}})^2,$$

$$v_{\mathrm{MB}} = \frac{c}{4S} = \frac{1}{\chi_{\mathrm{MB}}}, \qquad \chi_{\mathrm{MB}} = \frac{4S}{c} > 0. \tag{136}$$

This is always strictly concave, $p'' = -4S < 0$. The unconstrained vertex is v_{MB}; on $[-1, 1]$ the continuous maximizer is $\min(1, v_{\mathrm{MB}})$. Its departure from 1 need not switch a discrete mode.

Use P05's notation

$$N = \lfloor n/2 \rfloor, \quad u_j = \cos(2\pi j/n), \quad \beta_j = (u_j + u_{j+1})/2, \quad J = \lfloor (n+1)/4 \rfloor, \quad \alpha_j = 1/\beta_j \ (0 \le j < J).$$

P05 proves $\beta_j = \cos(\pi/n)\cos((2j+1)\pi/n)$ and exactly $\beta_0, \ldots, \beta_{J-1} > 0$. Its nearest-grid-cell argument, midpoint ordering, parity and label bookkeeping are generic finite-grid facts. Its DM realization theorem is not an MB realization proof.

Theorem 12.5 (MB cells, all ties, reachability, and cell stability)**.** *On $M > 0$, the complete MB representative selection is*

$$\begin{array}{c|c} 0 < \chi_{\mathrm{MB}} < \alpha_0 & \{0\} \\ \alpha_{j-1} < \chi_{\mathrm{MB}} < \alpha_j \ (1 \le j < J) & \{j\} \\ \chi_{\mathrm{MB}} > \alpha_{J-1} & \{J\} \\ \chi_{\mathrm{MB}} = \alpha_j \ (0 \le j < J) & \{j, j+1\}. \end{array} \tag{137}$$

The middle row is empty for $J = 1$. Equivalently, closed vertex cells are $[\beta_0, \infty)$, $[\beta_j, \beta_{j-1}]$ $(1 \le j < N)$, and $(-\infty, \beta_{N-1}]$, intersected with $v_{\mathrm{MB}} > 0$. Every listed strict cell and tie is realizable with $M > 0$; the exact reachable representatives are $0, \ldots, J$.

A nonzero selected representative gives $\{j, n-j\}$, while zero is a singleton. A tie involving zero gives three labels, other reachable ties four. No nonadjacent or triple representative maximum

occurs. The even Nyquist singleton is excluded. These are label counts, not numerical-eigenvalue multiplicities or frequency assertions. On the cell for j strict stability is exactly

$$\boxed{k > S + cu_j - 2Su_j^2.} \tag{138}$$

Either selected representative gives the same bound at a tie. Reverse strict inequality gives instability and equality gives zero abscissa. In the uniform cell, including a maximal tie with zero, $M > 0$ already makes the maximum $-a < 0$.

Proof. Square completion reduces selection to minimizing $|u_j - v_{\mathrm{MB}}|$. Apply the generic parts of Theorems 9.1 and 9.2. An independent sign check is

$$p(u_j) - p(u_{j+1}) = c(u_j - u_{j+1})(1 - \chi_{\mathrm{MB}}\beta_j),$$

confirming the switch orientation. For MB-specific realizability, prescribe any $v > 0$, take $g = h = 1/2$, $c = 4v$, $k > \max(0, c-1)$ and any $a > 0$. Then $M > 0$ and $v_{\mathrm{MB}} = v$. Each required strict cell and positive midpoint can be attained; strict cells are open by continuity. The grid theorem gives the labels, exclusions and ties. Taking the maximum of the derived quadratic proves the exact cell-stability test. □

The precise parity effects follow from the same generic grid calculation:

n	J	u_J	nearest representatives at formal vertex 0
4ℓ	ℓ	0	ℓ
$4\ell+1$	ℓ	$\sin(\pi/(2n))$	ℓ
$4\ell+2$	ℓ	$\sin(\pi/n)$	$\ell, \ell+1$
$4\ell+3$	$\ell+1$	$-\sin(\pi/(2n))$	$\ell+1$.

Every finite positive tuple has $v_{\mathrm{MB}} > 0$; the formal tie for $n = 4\ell+2$ is unattainable. Its two real parts differ by $2y^* c \sin(\pi/n) > 0$. Only the gap divided by y^* is independent of S; the unnormalized gap need not stay away from zero in parameter limits. At fixed n, sufficiently large finite χ_{MB} uniquely selects J in every parity class. No large-n theorem is asserted.

12.5 Common domain, named branches, and structural comparison

For DM set $A = k + h$, $\mu = a + c - g$, $b = a - A$, and $D = b^2 + 4A\mu$. By audited Theorem 6.4, the full common existence set is

$$\begin{aligned}\mathcal{C} = \{(a,k,c,g,h) > 0 : M > 0, \quad [\mu > 0 \text{ or } (\mu = 0, b > 0) \\ \text{or } (\mu < 0, b > 0, D \geq 0)]\}.\end{aligned} \tag{139}$$

It is not entirely an open set. The branch-labelled comparison pairs a tuple with one of its allowed positive DM roots $x_\pm = (b \pm \sqrt{D})/(2A)$ and the sole MB root a/M. Most constructions use the open domain

$$\mathcal{C}_+ = \{(a,k,c,g,h) > 0 : M > 0,\ \mu > 0\}, \qquad x^* = x_+. \tag{140}$$

DM then has exactly one positive synchronous equilibrium; its scalar uniform derivative is $-x^*\sqrt{D}/(1+x^*) < 0$ by P04. A separate construction below explicitly follows x_- on a two-positive-root domain.

At a specified positive DM equilibrium put $s = 1 + x^*$. Quote P03–P05:

$$\begin{gathered} d = -kx^* - c(1+2x^*)/s^2, \quad C = c/s, \quad G = gx^*/s^2, \quad H = hx^*, \\ r_m^{\mathrm{DM}} = d + Cu_m + (G-H)(2u_m^2 - 1), \\ \chi_{\mathrm{DM}} = \frac{4(H-G)}{C} = \frac{4x^*[h(1+x^*)^2 - g]}{c(1+x^*)}, \qquad \chi_{\mathrm{MB}} = \frac{4(g+h)}{c}. \end{gathered} \tag{141}$$

DM's parameter can be negative, zero or positive; its positive-vertex interpretation applies only if $h > g/s^2$. P05's full selection rule uses $\chi_{\mathrm{DM}} < \alpha_0$ for every uniform case. At fixed k, c, g, h, MB selection is independent of a, whereas DM selection can vary with a through the equilibrium branch.

Proposition 12.6 (Structural functional-form distinctions)**.** *MB real dispersion for $q = 2$ is always strictly concave in u. DM dispersion is strictly convex, linear, or strictly concave according as $h < g/(1+x^*)^2$, $h = g/(1+x^*)^2$, or $h > g/(1+x^*)^2$. When $h \le g/(1+x^*)^2$, DM uniquely selects zero. All three curvature regimes occur in $\mathcal{C}_+$.*

MB has invariant coordinate faces. DM has $F_i^{\mathrm{DM}}|_{x_i=0} = cx_{i-1}$ and propagates positivity from any nonzero nonnegative initial state by P01. MB's uniform eigenvalue at positive synchrony is always $-a$. DM's is $-x^*\sqrt{D}/s$ on x_+, $+x^*\sqrt{D}/s$ on a distinct positive x_-, and zero at a positive double root.

DM is globally nonnegative and dissipative for the entire canonical positive class by P01. MB has the sufficient dissipative regimes above, but global unbounded synchronous growth at $M = 0$ and finite-time synchronous blow-up at $M < 0$. Thus the DM functional form is structurally dissipative over the entire canonical class, whereas MB is not. The growth/blow-up examples are outside $\mathcal{C}$.

Proof. The second derivatives are $-4y^*S < 0$ and $4(G-H)$, respectively; use P05 for uniform selection. To realize the three curvature signs, set $x^* = 1$, $c = g = 4$, $k = 1$, $a = 1 + h$, with $h = 1/2, 1, 2$. Then $\mu = a > 0$, $M = 1 + h > 0$, $y^* = 1$, $G - H = 1 - h$, and both original scalar residuals vanish. The face and global statements quote P01 and the MB proofs above. The uniform DM values result from inserting P02's roots in (73). No full-system boundedness is inferred from MB equilibrium existence. □

Remark 12.7 (Meaning and limits of matching). Matching fixes n, q, the five positive coefficients, directed offsets and coefficient roles, and the original time variable. It does not calibrate equilibrium amplitude, local derivatives, empirical behavior, or nonlinear response. The successor derivative is positive for DM and negative for MB at positive synchrony, despite the negative term in both fields. No biological or quantitative-superiority claim follows. For a fixed source $z > 0$, as receiver $r \downarrow 0$, the predecessor inputs satisfy $cz/(1+r) \to cz$ for DM and $crz \to 0$ for MB. This last assertion is only a limit under the explicitly fixed-source convention.

12.6 Every ordered dominant-mode pair with both models stable

Theorem 12.8 (Independent realization of the two selection parameters)**.** *For each fixed $n \ge 5, q = 2$, every ordered pair $(j_{\mathrm{DM}}, j_{\mathrm{MB}}) \in \{0, \dots, J\}^2$ occurs with unique selected representatives and both equilibria strictly stable on a nonempty open subset of $\mathcal{C}_+$. In particular reverse uniform/nonuniform selection is possible; distinct nonuniform/nonuniform pairs are possible when $J \ge 2$, beginning at $n = 7$. This is not a full stability partition.*

Proof. Choose positive parameters in the desired strict cells; explicit choices for representative j are

$$z_0 = \alpha_0/2, \quad z_j = (\alpha_{j-1} + \alpha_j)/2 \ (1 \le j < J), \quad z_J = 2\alpha_{J-1}.$$

When $J = 1$ use only the first and last expressions. Put $\eta = z_{j_{\mathrm{DM}}}$, $\zeta = z_{j_{\mathrm{MB}}}$, and define

$$\begin{gathered} c = 4, \quad S = \zeta, \quad x = 1 + \eta/\zeta, \quad s = 1 + x, \\ h = \frac{\zeta + s\eta/x}{s^2 + 1}, \quad g = \zeta - h, \\ k > \max\{c + S,\ (c - g)/(xs) - h\}, \qquad a = (k + h)x - (c - g)/s. \end{gathered} \tag{142}$$

Here $xs > \eta/\zeta$, so $0 < h < S$ and $g, h > 0$. The bound on k gives $a > 0$ and $M > 0$. Further, $\mu = x[k + h + (c - g)/s] > 0$, since $k > S > g$. Substitution into the *original* vector fields gives

$$F_i^{\mathrm{DM}}(x\mathbf{1}) = x[a - (k + h)x + (c - g)/s] = 0, \qquad F_i^{\mathrm{MB}}(y^*\mathbf{1}) = y^*(a - My^*) = 0. \tag{143}$$

Algebra gives $\chi_{\mathrm{MB}} = \zeta$ and $x(hs^2 - g)/s = \eta$, hence $\chi_{\mathrm{DM}} = \eta$. For all MB labels $\operatorname{Re}\lambda_m^{\mathrm{MB}}/y^* \le -k + c + S < 0$. For DM the audited formula gives

$$\frac{r_m^{\mathrm{DM}}}{x} \le -k - \frac{c}{s^2} + \frac{g}{s^2} + h < -k + S < 0. \tag{144}$$

Indeed the two c contributions after division by x are at most $-c/s^2$, and $|g/s^2 - h| \le g/s^2 + h$. This proves all claimed strict inequalities. Openness in all physical parameters, not just this displayed family, is proved below. □

A rational reverse-selection witness at $n = 8$ is

$$(a, k, c, g, h) = (113/4, 9, 8, 1, 1), \quad x^* = 3, \quad y^* = 113/12. \tag{145}$$

Here $M = 3$, $\mu > 0$, $\chi_{\mathrm{MB}} = 1$ selects 0, while $\chi_{\mathrm{DM}} = 45/8 > 2\sqrt{2} = \alpha_1$ selects 2. The abscissae are $-443/16$ for DM and $-113/4$ for MB. Original residuals after division by the respective amplitudes are $113/4 - 10 \cdot 3 + 7/4 = 0$ and $113/4 - 3(113/12) = 0$.

12.7 DM stable and MB unstable for every size

Theorem 12.9 (Universal same-tuple opposite stability)**.** *For each fixed $n \ge 5, q = 2$, a nonempty open subset of $\mathcal{C}_+$ has DM strictly stable with unique dominant representative 0, while MB is unstable with a unique nonzero dominant representative. A single rational tuple works for every dimension:*

$$\boxed{(a, k, c, g, h) = (1/4, 1/8, 1, 1, 1/8).} \tag{146}$$

At this tuple the independently computed amplitudes happen to be $x^* = y^* = 1$; amplitude equality is not a comparison assumption.

Proof. All coefficients are positive, $M = 1/4$, $\mu = 1/4$. At amplitude one the original DM residual is $1/4 - 1/8 + 1/2 - 1/2 - 1/8 = 0$ and the MB residual is $1/4 - 1/8 + 1 - 1 - 1/8 = 0$. P02 ensures the unique positive DM root. Its coefficients are $d = -7/8, C = 1/2, G = 1/4, H = 1/8$. Thus $G - H = 1/8 > 0$, zero is strictly dominant and $r_0^{\mathrm{DM}} = -1/4$; every DM real part is negative. For MB,

$$S = 9/8, \qquad p(u) = 1 + u - \frac{9}{4}u^2, \qquad v_{\mathrm{MB}} = 2/9, \qquad \chi_{\mathrm{MB}} = 9/2.$$

The parity formulas give $-1/4 < u_J \le 1/2$. In the only negative case $n \equiv 3 \pmod 4$, $n \ge 7$ and $|u_J| = \sin(\pi/(2n)) < \pi/14 < 1/4$. The other cases give the upper bound, attained at $n = 6$. Hence the actual finite label J satisfies

$$p(u_J) \ge 1 - \frac{1}{4} - \frac{9}{4}\left(\frac{1}{2}\right)^2 = \frac{3}{16} > 0.$$

MB is unstable, and its negative uniform eigenvalue excludes zero from the selected set. There is no representative tie: a tie would force $\beta_j = 2/9$, but $4\beta_j = 2\cos(2\pi j/n) + 2\cos(2\pi(j+1)/n)$ is an algebraic integer, being a sum of roots of unity and their inverses. The rational number $8/9$ is not an integer, whereas every rational algebraic integer is an integer (apply the rational-root argument to a monic integer polynomial). This excludes a tie. Strict gaps and branch regularity give openness by the argument below. □

For $n = 5, 6, 7$ this witness selects MB representative 1; for $n = 8$ it selects representative 2. Other sizes follow the exact finite-grid rule and need not retain either label. The instability proof uses a finite Fourier label rather than only a positive continuous maximum.

12.8 Reverse stability: parity obstruction and constructive regions

For a positive DM equilibrium write $x = x^*$, $s = 1 + x$, $u = \cos\theta_m$, $w = -\cos(2\theta_m)$. The audited DM formula and the derived MB formula give

$$\frac{r_m^{\mathrm{DM}}}{x} = -k + (h - g/s^2)w - \frac{c(1 + 2x - su)}{xs^2}, \qquad \frac{r_m^{\mathrm{MB}}}{y^*} = -k + Sw + cu. \tag{147}$$

Notice $1 + 2x - su \geq x > 0$ for $u \leq 1$.

Theorem 12.10 (Reverse stability on the unique positive branch)**.** *On $\mathcal{C}_+$ with $q = 2$, a tuple with DM unstable and MB strictly stable exists if and only if $n \equiv 3 \pmod 4$. For every such fixed size there is a nonempty open region realizing this contrast, with both models uniquely selecting J. For all other sizes, MB strict stability implies DM strict stability on $\mathcal{C}_+$.*

Proof of the obstruction. On $\mathcal{C}_+$ the uniform DM eigenvalue is strictly negative. If $G - H \geq 0$, P05 makes it largest, so DM is strictly stable. Otherwise $h - g/s^2 > 0$. A nonnegative DM real part in (147) forces $w > 0$. If its cosine $u \geq 0$,

$$\frac{r_m^{\mathrm{MB}}}{y^*} - \frac{r_m^{\mathrm{DM}}}{x} = g(1 + s^{-2})w + cu + \frac{c(1 + 2x - su)}{xs^2} > 0. \tag{148}$$

That MB mode is therefore positive. Unless $n \equiv 3 \pmod 4$, all reachable DM representatives have nonnegative cosines by P05. Any nonnegative DM maximum would contradict MB strict stability. This also excludes a zero DM abscissa and proves the strict implication. □

Proof of the remaining-size realization. Let $n \equiv 3 \pmod 4$, $\tau = \sin(\pi/(2n))$, so $0 < \tau < 1/4$, $u_J = -\tau$, and $W = 1 - 2\tau^2 > 7/8$. Also $u_{J-1} = \sin(3\pi/(2n)) = 3\tau - 4\tau^3$, giving $\beta_{J-1} = \tau W$. Set

$$\begin{gathered} h = 1, \quad c = \tau, \quad g = \tau^2/4, \quad k = 1 - 5\tau^2/2, \quad x = 2/\tau, \quad s = 1 + x, \\ a = (k+1)x - (\tau - \tau^2/4)/s. \end{gathered} \tag{149}$$

Every coefficient is positive: $k > 27/32$, $x > 8$, the subtracted term in a is less than $1/4$, and $(k+1)x > 8$. Further $M = 2 - \tau - 9\tau^2/4 > 0$, and $\mu > 0$ since $c > g$. The original residuals vanish by (143).

For MB, $v_{\mathrm{MB}} = \tau/(4 + \tau^2) < \tau/4 < \beta_{J-1}$ and $\beta_J < 0$, so it selects J strictly. Its maximum divided by y^* is

$$-k + (1 + \tau^2/4)W - \tau^2 = -\tau^2(1/4 + \tau^2/2) < 0.$$

For DM the J-mode divided by x is

$$\frac{r_J^{\mathrm{DM}}}{x} = \frac{\tau^2}{2} - \frac{gW}{s^2} - \frac{\tau(1+2x+s\tau)}{xs^2}.$$

Since $1+2x+s\tau \le (2+\tau)s$, $s > x$, and $x = 2/\tau$, the two subtracted terms are less than

$$\frac{g+\tau(2+\tau)}{x^2} < \frac{37}{256}\tau^2.$$

Here $g < 1/64$ and $\tau(2+\tau) < 9/16$. Thus $r_J^{\mathrm{DM}}/x > 91\tau^2/256 > 0$. DM is concave, and its vertex obeys

$$v_{\mathrm{DM}} = \frac{\tau}{4x(s-g/s)} < \frac{\tau}{256} < \tau W = \beta_{J-1},$$

since $x > 8$ and $s - g/s > 8$. So DM also strictly selects J. These are strict canonical common-domain inequalities; openness is established below. □

Rational reverse-stability witness. At $n = 7$ a compact exact witness is

$$(a,k,c,g,h) = (20301/1100, 17/20, 1/2, 1/100, 1), \quad x^* = 10, \quad M = 34/25, \quad y^* = 20301/1496. \tag{150}$$

Original residuals divided by their amplitudes are $20301/1100-(37/20)10+(49/100)/11 = 0$ for DM and $20301/1100-(34/25)(20301/1496) = 0$ for MB. All coefficients and μ are positive. The ordered seven-node cosines u_1, u_2, u_3 are the three roots of $8u^3+4u^2-4u-1$: use $1+2(\cos\theta+\cos 2\theta+\cos 3\theta) = 0$ at nonzero seventh roots and the Chebyshev identities. Rational isolating intervals are

$$31/50 < u_1 < 63/100, \quad -23/100 < u_2 < -11/50, \quad -91/100 < u_3 < -9/10. \tag{151}$$

The endpoint signs of the cubic are respectively $(-,+),(+,-),(-,+)$. Three disjoint sign changes exhaust its degree; cosine ordering assigns the labels. Thus no decimal trigonometric value is a proof premise. MB representatives $1,2,3$ have normalized real parts $-17/20 + u_1/2 - 101u_2/100$, $-17/20 + u_2/2 - 101u_3/100$, and $-17/20 + u_3/2 - 101u_1/100$. The interval upper bounds are respectively $-3027/10000$, $-409/10000$, $-19262/10000$, all negative; mode zero is $-34/25$. DM instead satisfies

$$\frac{r_2^{\mathrm{DM}}}{10} > -\frac{17}{20} - \frac{21}{2420} - \frac{23}{22000} + \frac{12099}{12100}\frac{9}{10} = \frac{9729}{242000} > \frac{1}{25} > 0.$$

Both select 2 strictly: $v_{\mathrm{MB}} = 25/202 < 39/200 < \beta_1$, $v_{\mathrm{DM}} = 55/48396 < 39/200 < \beta_1$, and $\beta_2 < 0$.

The full domain requires explicit branch labels. The preceding obstruction cannot be extended to all of $\mathcal{C}$ without qualification. A lower-branch construction gives the exact existence answer on that larger branch-labelled domain.

Proposition 12.11 (Reverse contrast when both DM branches are allowed)**.** *If $4 \mid n$ and MB is strictly stable, every positive DM synchronous equilibrium is strictly stable. If $4 \nmid n$, a nonempty open common region has MB strictly stable and the specified lower positive DM branch unstable with unique dominant representative zero.*

Proof. When $4 \mid n$, MB's label $n/4$ has normalized real part $-k+S$. Its strict stability forces $k > S$. Bound (144), which did not require $\mu > 0$, makes every DM real part negative.

Otherwise put $\rho_n = \max_m[-\cos(2\theta_m)]$. One has $0 < \rho_n < 1$: equality to one requires $4m = n$ modulo $2n$, impossible when $4 \nmid n$, and a label nearest $n/4$ has negative double-angle cosine. Define

$$\begin{gathered} \varepsilon = (1-\rho_n)/24, \quad k = 1, \quad c = h = \varepsilon, \quad g = (1+\varepsilon)^3 + 2\varepsilon, \\ a = (1+\varepsilon)(1+2\varepsilon) + \frac{\varepsilon}{1+\varepsilon}. \end{gathered} \tag{152}$$

All coefficients are positive and $M = 1 + g > 0$. With $A = 1 + \varepsilon$, the two DM roots are exactly

$$x_- = \varepsilon, \qquad x_+ = \varepsilon + \frac{\varepsilon}{(1+\varepsilon)^2}.$$

Direct substitution of x_- in $a - Ax + (c-g)/(1+x)$ gives zero. The second is checked by the quadratic factorization $A(x-x_-)(x-x_+)$ of P02's equilibrium polynomial, or by substitution. In particular $\mu = -A\varepsilon^2 - \varepsilon^2/(1+\varepsilon) < 0$, $b > 0$, $D > 0$, and both roots are positive. On the named lower branch,

$$\lambda_0^{\mathrm{DM}} = \frac{\varepsilon^2}{(1+\varepsilon)^2} > 0, \qquad G - H = \varepsilon\left(1 + \frac{2\varepsilon}{(1+\varepsilon)^2}\right) > 0.$$

Thus zero is uniquely dominant and unstable. For MB, $S = (1+\varepsilon)^3 + 3\varepsilon < 1 + 10\varepsilon$, since $0 < \varepsilon < 1$, and every normalized real part obeys

$$-1 + \varepsilon\cos\theta_m - S\cos(2\theta_m) \le -1 + \varepsilon + S\rho_n < -\frac{13}{24}(1-\rho_n) < 0.$$

The MB original residual vanishes at $y^* = a/M$. Distinct simple positive DM roots and all strict inequalities persist locally on the specified lower branch. At $n = 6$, $\rho_6 = 1/2$, so $\varepsilon = 1/48$ gives an entirely rational witness. □

12.9 Mode/stability combinations and physical-parameter openness

All five requested types of combination have realizations, without a claim that each occurs in every dimension. Both stable with identical or different modes follows from Theorem 12.8; DM stable / MB unstable is universal by Theorem 12.9; the reverse has the branch and size restrictions just proved. For both unstable with different dominant modes take $n = 8$ and

$$(a, k, c, g, h) = (16351/1100, 1/2, 3/2, 1/100, 1), \quad x^* = 10, \quad M = 1/100, \quad y^* = 16351/11. \tag{153}$$

The original residuals divided by their amplitudes are $16351/1100 - 15 + (149/100)/11 = 0$ and $16351/1100 - (1/100)(16351/11) = 0$. Coefficients and μ are positive. MB's vertex $75/202$ lies strictly between $\sqrt{2}/4$ and $(2+\sqrt{2})/4$, so MB selects 1 with normalized maximum $-1/2+3\sqrt{2}/4 > 0$. DM's vertex $55/16132 < \sqrt{2}/4$ selects 2, whose normalized maximum is

$$-\frac{1}{2} - \frac{63}{2420} + \frac{12099}{12100} = \frac{2867}{6050} > 0.$$

The small value of M is strictly positive. MB's large amplitude is its own computed equilibrium, not an amplitude imported from DM.

Proposition 12.12 (Openness in all five physical coefficients)**.** *Every strict contrast and strict mode pair constructed here persists on a nonempty open set in $(a, k, c, g, h) \in (0, \infty)^5$ for its stated fixed size, offset, and DM branch. No openness assertion rests merely on an interval inside a displayed lower-dimensional family.*

Proof. On $\mathcal{C}_+$, $D > 0$ and the unique positive root x_+ is smooth in all five coefficients. Equivalently, the scalar equilibrium function $B(x) = a-(k+h)x+(c-g)/(1+x)$ has $B_x = -\sqrt{D}/(1+x_+) < 0$, so the implicit function theorem applies. MB's branch a/M is smooth for $M > 0$. Coefficient positivity, $M > 0$, and $\mu > 0$ are strict open conditions. At fixed n there are finitely many spectral inequalities; each is strict and continuous. A unique selected representative has a strictly positive gap to every other representative, while its conjugate labels remain tied. A sufficiently small intersection of these neighborhoods preserves all branch, sign, stability and selection assertions.

For the lower-branch family use instead $\mu < 0, b > 0, D > 0$. Both positive roots are simple and the specified x_- is smooth, with $B_x = +\sqrt{D}/(1+x_-) > 0$. The same continuity argument preserves that branch and its strict conclusions. No full-system IFT, later module, uniform neighborhood radius in n, or global branch identification is used. □

12.10 Interpretation, deferred gates, and internal checks

The strongest completed DRAFT results are a universal forward stability contrast, both stability directions at $n = 7$ on the unique-positive-root domain, every ordered pair of reachable modes with both models stable, and exact parity/branch limitations on reverse stability. Identical abstract reachable labels do not imply identical behavior at the same physical tuple. The unrestricted reverse-stability hope fails on $\mathcal{C}_+$ unless $n \equiv 3 \pmod 4$; even allowing both DM branches cannot produce it when $4 \mid n$. The rejected phrase "MB is generically unstable" is not reinstated: MB has explicit open stable regions and an exact stability criterion.

For structural growth examples, $(a, k, c, g, h) = (1, 1, 3, 1, 1)$ gives $M = 0$, and $(1, 1, 4, 1, 1)$ gives $M = -1$. At $s_0 = 1$ the latter blows up at $\log 2$. DM has its P01 bounded flow for both. Its own positive root is 1 or $(\sqrt{33} - 1)/4$, respectively, each satisfying the original residual $x[a - (k + h)x + (c - g)/(1 + x)] = 0$. MB has no positive synchronous root in either case; its origin has zero residual, and the stated scalar trajectories have the exact original MB derivative. These are not common-equilibrium stability comparisons.

The six-term model is explicitly deferred. Existing source-provenance records are retained, but no archived six-term equation is imported. Full MB Hopf, first Lyapunov coefficient, crossing transversality, and criticality comparisons are deferred as well. The spectra and open regions are inputs for later P11–P13, not proofs of their hypotheses. A selected nonuniform pair, positive real part, or candidate imaginary crossing is insufficient for Hopf. P07, P09, P10 and all later proof bodies are untouched. The frozen package already calls for this comparison, so no scope amendment or novelty-priority promotion is needed.

Internal analytic and computational checks. The proofs independently check the original scalar reduction, coordinate faces, reciprocal scalar solution, maximum inequality, pre-equilibrium diagonal, Fourier phases, both uniform-eigenvalue derivations, quadratic completion, midpoint orientation, and parameter constructions. Each displayed exact equilibrium has an original-field residual check; the general constructions have symbolic residual identities. The separate validation-support package supplies an independent term-by-term field/Jacobian implementation, direct matrix eigenvalues, Fourier comparisons for $n = 5, 6, 7, 8$ and collision offsets, exact rational residual and isolating-interval checks, and deterministic finite-grid checks. Tolerances and per-case data are recorded. Floating-point corroboration is not interval certification, proof of openness, or an independent audit; the module remains DRAFT.

12.11 Priority checks for the independent P08 auditor

(1) Authenticate the P05 audited snapshot, P01–P05 body hashes, A01–A05, and A05 E01 closure. Separate historical computational provenance notes from the current audited proof inputs.

(2) Check the full common domain, unique x_+ on $\mathcal{C}_+$, and the explicitly fixed lower branch in the two-root construction. Do not identify amplitudes by coefficient names.

(3) Check positivity only on the maximal interval; separate sufficient full-system bounds, scalar growth and blow-up, and necessary conditions. Do not infer global boundedness merely from $M > 0$.

(4) Recompute the MB pre-equilibrium diagonal, all collision sums, complex phases and imaginary signs, normality and both proofs of $\lambda_0^{\mathrm{MB}} = -a$.

(5) Check closed/strict selection cells, all maximal ties, parity, finite-parameter endpoints, MB realizability, and each cell's exact stability inequality. Generic P05 grid facts and DM realization are distinct.

(6) Reconstruct the all-pair family, positivity, original residuals, $\mu > 0$, and both strict all-mode stability bounds.

(7) Check the universal rational witness, finite-label bound $p(u_J) \geq 3/16$, and algebraic-integer exclusion of a tie at $2/9$.

(8) Check the modal-gap obstruction and its branch restriction, the all-$n \equiv 3$ reverse family and $37/256$ estimate, and the rational seventh-root isolating intervals and stability inequalities. Independently check the $n = 8$ both-unstable example.

(9) Check the lower-root family and full-domain $4 \mid n$ obstruction; verify both distinct roots and that the compared branch was fixed in advance.

(10) Verify physical five-dimensional openness using simple scalar roots and finitely many strict spectral and representative gaps.

(11) Confirm no use of P06, P07, P09–P18, no later bifurcation assertion, no six-term substitution, and no claim that compilation verifies mathematics.

13 P09: Transcritical bifurcation at the origin

Status: DRAFT; independent audit pending. Fix canonical $a, k, c, h > 0$, $n \geq 5$, and $1 \leq q < n$, and vary g. Throughout this module put

$$A = k + h > 0, \qquad b = a - A, \qquad g_c = a + c > 0, \qquad \mu = g_c - g. \tag{154}$$

Here b is a local abbreviation, not an archived comparator coefficient. Write $F(x, \mu)$ for the original canonical DM field with $g = g_c - \mu$:

$$F_i(x, \mu) = ax_i - kx_i^2 + \frac{cx_{i-1}}{1 + x_i} - \frac{(g_c - \mu)x_i}{1 + x_{i+q}} - hx_i x_{i-q}. \tag{155}$$

For $|\mu| < g_c/2$ all coefficients remain positive. The field is jointly C^∞ on $\mathcal{U} \times (-g_c/2, g_c/2)$, $\mathcal{U} = (-1, \infty)^n$, by its nonvanishing denominators and P01's local foundation (Proposition 5.2). No global invariance of $\mathcal{U}$ is needed. Neighborhoods may depend on all fixed parameters and on n; no uniform-in-n assertion is made.

The direct proof inputs are P02, P03 and P04. P02's Proposition 6.1 and (25) supply

$$F(r\mathbf{1}, \mu) = f(r, \mu)\mathbf{1}, \qquad f(r, \mu) = \frac{r}{1 + r}(\mu + br - Ar^2), \qquad r > -1. \tag{156}$$

This is the imported exact invariant restriction, not a new global classification.

13.1 Critical spectrum and full local equilibrium reduction

Lemma 13.1 (Simple synchronous kernel and transverse gap)**.** *At $(x, \mu) = (0, 0)$,*

$$J_c = D_x F(0, 0) = c(P - I), \qquad \lambda_m^c = c(e^{-\mathrm{i}\theta_m} - 1), \quad \theta_m = \frac{2\pi m}{n}. \tag{157}$$

Zero is algebraically simple, with real eigenspace $\operatorname{span}\{\mathbf{1}\}$*. Every transverse eigenvalue satisfies*

$$\operatorname{Re} \lambda_m^c \leq -\delta_n \quad (1 \leq m < n), \qquad \delta_n = c\left(1 - \cos\frac{2\pi}{n}\right) > 0. \tag{158}$$

No condition on q or $\gcd(n, q)$ *is imposed beyond the canonical range.*

Proof. At zero the only linear terms are $(a - g)x_i + cx_{i-1}$; every remaining Jacobian contribution contains a state coordinate. Thus, independently checking P03's (46), $D_x F(0, \mu) = \mu I + c(P - I)$. The canonical Fourier basis satisfies $Pv^{(m)} = e^{-\mathrm{i}\theta_m} v^{(m)}$, which gives the spectrum. The n roots of unity are distinct, so only $m = 0$ gives zero, with algebraic multiplicity one. Equivalently, $(\lambda + c)^n - c^n$ has nonzero derivative nc^{n-1} at zero. The equation $Pz = z$ forces $z_i = z_{i-1}$ at every index, giving the kernel. For $m \neq 0$, let $j = \min(m, n - m)$. Then $2\pi j/n \in [2\pi/n, \pi]$, and monotonicity of cosine gives $\cos\theta_m \leq \cos(2\pi/n) < 1$. This proves the gap, including all singleton and conjugate-pair labels. □

Theorem 13.2 (Every sufficiently small equilibrium is synchronous)**.** *For every fixed canonical tuple, including $b = 0$, there are state and parameter neighborhoods of zero in which the entire full-system equilibrium equation is equivalent to*

$$x = r\mathbf{1} + y, \qquad y = 0, \qquad r(\mu + br - Ar^2) = 0. \tag{159}$$

In particular, no nearby nonsynchronous equilibrium branch is omitted.

Proof. Use the unnormalized amplitude, to agree exactly with (156). Set

$$w^{\mathsf{T}} = \frac{1}{n}\mathbf{1}^{\mathsf{T}}, \quad E = \{y : w^{\mathsf{T}} y = 0\}, \quad \Pi_{\perp} = I - \mathbf{1}w^{\mathsf{T}}, \quad r = w^{\mathsf{T}} x, \quad y = \Pi_{\perp} x.$$

This is a real linear isomorphism $x \leftrightarrow (r, y)$, and E is the real nonuniform Fourier subspace. The smooth map

$$T(r, y, \mu) = \Pi_{\perp} F(r\mathbf{1} + y, \mu) \in E$$

has $D_y T(0, 0, 0) = J_c|_E$, since J_c preserves E. The complexification of this restriction has exactly the nonzero-mode eigenvalues of Lemma 13.1. None vanishes, so it is an invertible real linear map $E \to E$.

The finite-dimensional implicit-function theorem gives a product neighborhood and a unique smooth function $y = \psi(r, \mu)$ such that, within that product, $T = 0$ if and only if $y = \psi(r, \mu)$. Synchrony gives $T(r, 0, \mu) = 0$ for every nearby r, μ. After shrinking the product, zero belongs to the IFT's y neighborhood for all these parameters. Uniqueness therefore proves

$$\boxed{\psi(r, \mu) \equiv 0.} \tag{160}$$

Every full equilibrium solves $T = 0$, hence has $y = 0$. Conversely, on $y = 0$ the full field vanishes precisely when $f = 0$; $1 + r > 0$ makes that equivalent to (159). A smaller state neighborhood times a parameter interval lies in the IFT product. No part of this argument requires $b \neq 0$. □

13.2 Local derivatives and full-vector nondegeneracy

Proposition 13.3 (Transcritical coefficients and parameter orientation)**.** *The local scalar derivatives are*

$$f(0, \mu) = 0, \quad f_r(0, 0) = 0, \quad f_{r\mu}(0, 0) = 1, \quad f_{rr}(0, 0) = 2b. \tag{161}$$

Choose $v = \mathbf{1}$, $w^{\mathsf{T}} = n^{-1}\mathbf{1}^{\mathsf{T}}$, so $J_c v = 0$, $w^{\mathsf{T}} J_c = 0$, and $w^{\mathsf{T}} v = 1$. The full vector field satisfies

$$F(0, \mu) = 0, \quad w^{\mathsf{T}} F_{x\mu}(0, 0) v = 1, \quad w^{\mathsf{T}} F_{xx}(0, 0)[v, v] = 2b. \tag{162}$$

The quadratic Taylor coefficient includes the factor $1/2$ and is b. With g as parameter the mixed coefficient is instead $w^{\mathsf{T}} F_{xg}(0, g_c) v = -1$. Ordinary transcritical nondegeneracy holds exactly when $b \neq 0$.

Proof. Equation (156) gives $f_r(0, \mu) = \mu$ and $f_\mu(r, \mu) = r/(1 + r)$, whose r derivative at zero is one. At $\mu = 0$, $f(r, 0) = br^2 - (A + b)r^3 + O(r^4)$, so $f_{rr} = 2b$.

Independently, every term of the original field vanishes at $x = 0$. Its parameter derivative is $(F_\mu)_i = x_i/(1 + x_{i+q})$, whose state derivative at zero is the identity. Using g instead gives $-x_i/(1 + x_{i+q})$, hence minus the identity. The second directional derivatives of the five original terms along v are respectively $0, -2k, -2c, 2g_c, -2h$. Their sum is $2(g_c - c - k - h) = 2b$ in each component. Averaging by w proves the full-vector identities without a missing factor of n. Also $F_\mu(0, 0) = 0$, as required by persistence of the origin branch. □

13.3 Generic full-system transcritical theorem

Theorem 13.4 (Nondegenerate transcritical bifurcation at $g_c = a + c$)**.** *Suppose $b = a - k - h \neq 0$. For sufficiently small $|\mu|$, all full-system equilibria sufficiently near the origin are exactly*

$$x = 0 \quad \textit{and} \quad x = r_*(\mu)\mathbf{1}, \qquad r_*(0) = 0, \quad r_*(\mu) \neq 0 \ (\mu \neq 0), \tag{163}$$

where r_* is smooth and

$$r_*'(0) = -\frac{1}{b}, \qquad r_*(\mu) = -\frac{\mu}{b} + \frac{A\mu^2}{b^3} + O(\mu^3), \qquad \frac{dr_*}{dg}(g_c) = \frac{1}{b}. \tag{164}$$

The two curves have distinct tangent directions in the synchronous state–parameter plane. The critical eigenspace is one-dimensional and synchronous; all transverse modes are strictly stable. In $\mathcal{U}$, the full nonlinear stability exchange is

Side	*Origin*	*Nearby nonzero branch*
$\mu < 0$ $(g > g_c)$	*Locally exponentially stable*	*Unstable*
$\mu > 0$ $(g < g_c)$	*Unstable*	*Locally exponentially stable*

Together with the full-vector nondegeneracy coefficients, these facts give a nondegenerate full-system local transcritical bifurcation.

Proof. Theorem 13.2 first makes the equilibrium problem complete in the full state space. Apart from the persistent origin, solve $Q(r,\mu) = Ar^2 - br - \mu = 0$. Since $Q_r(0,0) = -b \neq 0$, the implicit-function theorem gives a unique smooth root $r_*(\mu)$ in a product neighborhood. The identity $Q(0,\mu) = -\mu$ shows that this root is nonzero exactly for $\mu \neq 0$. This proves the full local list. For fixed $b \neq 0$, the other root b/A at $\mu = 0$ is excluded by shrinking the state neighborhood.

Differentiating $Q(r_*(\mu),\mu) = 0$ gives $(2Ar_* - b)r_*' = 1$, hence $r_*'(0) = -1/b$. Differentiating again gives $2A(r_*')^2 + (2Ar_* - b)r_*'' = 0$, hence $r_*''(0) = 2A/b^3$. Smoothness proves the expansion. In (r,μ) coordinates the tangents are $(0,1)$ and $(-1/b,1)$, which are independent. This is transverse crossing in that plane, not a claim that two curves span the full $(n+1)$-dimensional ambient space. Since $d\mu/dg = -1$, the physical g slope has the opposite sign. Increasing g decreases μ.

The exact center dynamics also identifies the transcritical bifurcation. At $\mu = 0$ the synchronous line is smooth, invariant, and tangent to the entire center eigenspace, so is a local one-dimensional center manifold. For the extended system $(\dot{x},\dot{\mu}) = (F(x,\mu),0)$, its product with the parameter interval is an exact two-dimensional center manifold: $F_\mu(0,0) = 0$, the extended linearization is $\operatorname{diag}(J_c,0)$, and all remaining eigenvalues are strictly stable. On this product use

$$u = Ar^2 - br, \qquad \frac{du}{dr}(0) = -b \neq 0.$$

This is a local smooth coordinate. Direct substitution yields

$$\dot{u} = \alpha(r)u(\mu - u), \qquad \alpha(r) = \frac{2Ar - b}{(Ar - b)(1 + r)}, \qquad \alpha(0) = 1. \tag{165}$$

Near zero the denominators do not vanish and $\alpha > 0$. Thus $d\tau/dt = \alpha(r)$ gives exactly $du/d\tau = u(\mu - u)$, the nondegenerate transcritical center dynamics. This is orbital equivalence of the exact center restriction, not exact decoupling of the full field.

Full stability still requires all Fourier modes. Differentiating first,

$$f_r(r,\mu) = \frac{\mu + br - Ar^2}{(1+r)^2} + \frac{r}{1+r}(b - 2Ar).$$

At the nonzero equilibrium the first numerator vanishes. By P04's (72), its uniform eigenvalue is

$$\lambda_0^{\mathrm{nz}} = \frac{r_*}{1+r_*}(b - 2Ar_*) = -\mu\frac{b - 2Ar_*}{(1+r_*)(b - Ar_*)}. \tag{166}$$

The last equality uses $\mu = -r_*(b - Ar_*)$. Choose $|r_*| < 1/2$ and $2A|r_*| < |b|/2$. Both $b - Ar_*$ and $b - 2Ar_*$ have the sign of b, so the final fraction is positive. Therefore, exactly,

$$\operatorname{sgn} \lambda_0^{\mathrm{nz}} = -\operatorname{sgn} \mu, \qquad \lambda_0^{\mathrm{nz}} = -\mu + O(\mu^2). \tag{167}$$

The sign argument, rather than the expansion, gives stability.

For the transverse modes use the general P04 coefficients, valid also at zero or negative $r > -1$:

$$\begin{aligned} d(r,\mu) &= a - 2kr - \frac{cr}{(1+r)^2} - \frac{g_c - \mu}{1+r} - hr, \\ C(r) &= \frac{c}{1+r}, \quad G(r,\mu) = \frac{(g_c - \mu)r}{(1+r)^2}, \quad H(r) = hr, \\ \lambda_m(r,\mu) &= d + Ce^{-\mathrm{i}\theta_m} + Ge^{\mathrm{i}q\theta_m} - He^{-\mathrm{i}q\theta_m}. \end{aligned} \tag{168}$$

Every expression is continuous at $(r,\mu) = (0,0)$, where it equals λ_m^c. Finitely many labels allow one common neighborhood with $|\lambda_m(r,\mu) - \lambda_m^c| < \delta_n/2$ for every $m \neq 0$. Thus for some $\eta > 0$,

$$|r| + |\mu| < \eta \quad \Longrightarrow \quad \operatorname{Re} \lambda_m(r,\mu) < -\delta_n/2 \quad (m \neq 0). \tag{169}$$

No mode-selection theorem is used.

At the origin $\lambda_0 = \mu$, and every other real part is $\mu - c(1 - \cos\theta_m) < -\delta_n/2$ if $|\mu| < \delta_n/2$. On the nonzero branch use (167) and (169). A Hurwitz Jacobian gives local exponential stability of the smooth nonlinear equilibrium; a positive eigenvalue gives nonlinear instability, as in P04's nonlinear implications. Every unstable equilibrium here has exactly one unstable real direction. This proves the table and the full-system theorem, without using P05, P07 or P08. □

Corollary 13.5 (Physical positive-branch orientation)**.** *For sufficiently small nonzero μ, $r_*(\mu) > 0$ exactly when $\operatorname{sgn}\mu = -\operatorname{sgn} b$. If $a < k + h$, this branch is positive and locally exponentially stable for $\mu > 0$, equivalently $g < g_c$, while the origin is unstable. If $a > k + h$, it is positive and unstable for $\mu < 0$, equivalently $g > g_c$, while the origin is locally exponentially stable. Origin instability for $\mu > 0$ is also visible relative to $\mathbb{R}^n_{\geq 0}$; for $\mu < 0$ exponential stability restricts to that state space.*

Proof. Use the exact relation $\mu = -r_*(b - Ar_*)$ and the sign of $b - Ar_*$, then the full stability table. The unstable origin eigenvector $v = \mathbf{1}$ is strictly positive. Explicitly, at any fixed small $\mu > 0$, $f(r,\mu)/r \to \mu$ as $r \downarrow 0$. Choose $\varepsilon > 0$ such that $f(r,\mu) \geq (\mu/2)r$ on $0 < r \leq \varepsilon$. Arbitrarily small positive synchronous states reach $r = \varepsilon$ in finite time. Their full-system lifts remain nonnegative, proving relative instability. For $\mu < 0$, full local exponential stability and P01's nonnegative invariance give the restriction. An unstable positive branch point is interior to $\mathbb{R}^n_{\geq 0}$; small positive synchronous perturbations also show its relative instability. P09's local branch is not any additional positive root away from zero allowed by P02 when $b > 0$. □

Proposition 13.6 (Generic critical origin in the open domain)**.** *For $\mu = 0$, $b \neq 0$, the origin is nonlinearly unstable in $\mathcal{U}$. For $b > 0$ this is visible from nonnegative synchronous states. No full relative-orthant stability conclusion is asserted at $\mu = 0$, $b < 0$.*

Proof. The exact invariant-line dynamics is

$$\dot{r} = \frac{r^2(b - Ar)}{1 + r}. \tag{170}$$

Choose $0 < \varepsilon < 1/2$ with $A\varepsilon < |b|/2$. For $0 < |r| \leq \varepsilon$ the sign is that of b, and the magnitude is at least $|b|r^2/3$. For $b > 0$ take $r(0) > 0$; for $b < 0$ take $r(0) < 0$. On that side $s = |r|$ satisfies $\dot{s} \geq (|b|/3)s^2$ until $s = \varepsilon$, which must occur in finite time. Arbitrarily small data therefore exit a fixed neighborhood. The exact synchronous lift stays in $\mathcal{U}$ up to that exit. When $b < 0$ these initial states are negative, so this proof does not establish relative-orthant instability. □

13.4 The cubic-degenerate case and its precise scope

Proposition 13.7 (Degenerate geometry and full nonlinear stability)**.** *Let $b = 0$, that is $a = A = k + h$. Ordinary transcritical quadratic nondegeneracy fails. All sufficiently small full-system equilibria are synchronous, with the following exact local geometry and stability.*

(i) For $\mu < 0$, only the origin exists locally; it is locally exponentially stable.

(ii) At $\mu = 0$, only the origin exists locally. It is a triple zero of the scalar field and is locally asymptotically stable in the full open domain $\mathcal{U}$, but not locally exponentially stable.

(iii) For $\mu > 0$, there are precisely the origin and

$$x = r_{\pm}\mathbf{1}, \qquad r_{\pm} = \pm\sqrt{\mu/A}. \tag{171}$$

For sufficiently small $\mu > 0$, both nonzero equilibria belong to $\mathcal{U}$ and are locally exponentially stable there. The origin is unstable, also relative to $\mathbb{R}^n_{\geq 0}$. Only $r_+\mathbf{1}$ is physically nonnegative.

The two nonzero local roots occur for $g < g_c$; for $g > g_c$ only the stable origin remains locally. This is a cubic-degenerate origin with square-root nonzero branches. No pitchfork classification or two-parameter unfolding is asserted.

Proof. Theorem 13.2 does not require $b \neq 0$. Now $f(r, \mu) = r(\mu - Ar^2)/(1 + r)$, and the nonzero equation $Ar^2 = \mu$ gives the complete list. For $0 < \mu < A$, the negative root exceeds -1; shrink the interval further to put both roots within the reduction and spectral-gap neighborhoods. At $\mu = 0$,

$$f(r, 0) = -\frac{Ar^3}{1 + r}, \qquad f_{rrr}(0, 0) = -6A \neq 0. \tag{172}$$

The scalar zero is therefore triple. At either nonzero root,

$$\lambda_0 = f_r(r_{\pm}, \mu) = -\frac{2Ar_{\pm}^2}{1 + r_{\pm}} < 0. \tag{173}$$

The proof of (169) never used $b \neq 0$; it supplies strict stability of all other Fourier modes. Thus both full equilibria are locally exponentially stable. At the origin $\lambda_0 = \mu$ gives the off-critical stability; the positive synchronous escape argument above proves relative instability for $\mu > 0$.

At the fixed value $\mu = 0$, the smooth invariant synchronous line passes through zero and is tangent to the whole one-dimensional center subspace. All complementary eigenvalues have real part at most $-\delta_n$, and there is no unstable subspace. On this exact center manifold, $r^2/2$ has derivative $-Ar^4/(1 + r) < 0$ for small $r \neq 0$, so its origin is locally asymptotically stable. These verify the hypotheses of the finite-dimensional center-manifold stability reduction principle: a smooth local system with center and strictly stable spectra inherits local asymptotic stability from its center restriction. No parameter direction is included in this fixed-parameter stability statement.

A direct full-system Lyapunov proof independently checks this application. Use $x = r\mathbf{1} + y$, $y \in E$, at the fixed degenerate tuple. By P04, $J(r) = D_xF(r\mathbf{1}, 0)$ is circulant even at nonstationary synchronous states and preserves E, so $w^{\mathsf{T}}J(r)y = 0$. Taylor expansion with uniformly bounded second derivatives on a compact neighborhood gives, for some $K \geq 0$,

$$\dot{r} = -\frac{Ar^3}{1 + r} + R(r, y), \qquad |R(r, y)| \leq K \|y\|_2^2 . \tag{174}$$

Synchrony makes the transverse field vanish at $y = 0$, hence

$$\dot{y} = J_c y + B(r, y)y, \qquad B(r, y) = \Pi_\perp \int_0^1 [D_x F(r\mathbf{1} + ty, 0) - J_c]dt\Big|_E .$$

Continuity gives $\|B(r, y)\|_2 \le \delta_n/2$ in a smaller neighborhood. Fourier diagonalization gives $y^\mathsf{T} J_c y \le -\delta_n \|y\|_2^2$ for real $y \in E$; therefore $y^\mathsf{T}\dot{y} \le -(\delta_n/2)\|y\|_2^2$. Choose $0 < \rho < 1/2$ inside these neighborhoods so that $K\rho \le \delta_n/4$. The positive definite function $V(r, y) = (r^2 + \|y\|_2^2)/2$ then satisfies

$$\dot{V} \le -\frac{A}{1+\rho} r^4 - \frac{\delta_n}{4}\|y\|_2^2 < 0 \quad \text{for } (r, y) \ne (0, 0), \quad |r|, \|y\|_2 < \rho. \tag{175}$$

Small closed sublevel sets lie compactly in this neighborhood and in $\mathcal{U}$, since $|x_i| \le |r| + \|y\|_2 < 1$. They are positively invariant. Local continuation and the strict Lyapunov theorem give full local asymptotic stability; equivalently, zero is the only invariant set in $\dot{V} = 0$. This also checks the conclusion for arbitrary q.

Convergence is not exponential. Each sufficiently small nonzero synchronous solution keeps its sign, tends to zero, and obeys

$$\frac{d}{dt}(r^{-2}) = \frac{2A}{1 + r(t)} \longrightarrow 2A.$$

It follows that $|r(t)| \sim (2At)^{-1/2}$ as $t \to \infty$. These exact full-system solutions exclude a local exponential stability estimate. No common asymptotic rate for every nonsynchronous trajectory is asserted. □

Proposition 13.8 (Codimension and the meeting of the algebraic loci)**.** *For fixed admissible (n, q), the degenerate origin locus is a smooth codimension-two submanifold of $(0, \infty)^5$, given by*

$$a - k - h = 0, \qquad g - a - c = 0. \tag{176}$$

The branch-intersection and P02 repeated-root loci meet there. This is an algebraic/spectral codimension statement only.

Proof. In coordinates (a, k, c, g, h) the gradients are

$$(1, -1, 0, 0, -1), \qquad (-1, 0, -1, 1, 0).$$

They are independent: the g coordinate forces the coefficient of the second vector to be zero, and the k coordinate then forces the first to be zero. The regular level-set theorem proves codimension two. The locus is nonempty: any $k, c, h > 0$ give $a = k + h$, $g = a + c > 0$. The center eigenvalue remains simple, not a double-zero spectral degeneracy.

P02's Proposition 6.5 gives

$$g_f = a + c + \frac{(a - A)^2}{4A}, \qquad r_f = \frac{a - A}{2A}.$$

Hence $g_f = g_c$ exactly at $a = A$, where $r_f = 0$. The repeated root is positive only for $a > A$; the meeting point is the origin, not a positive nonzero equilibrium. Only this algebraic overlap is recorded. It is not a saddle-node proof; that remains P10. □

13.5 Scope, corroboration and independent audit priorities

B02 is resolved for P09 only: g varies with a, k, c, h, n, q fixed. NB07 is used only for the cubic case, its one-parameter geometry and codimension; two-parameter unfolding remains excluded. P09 retains its frozen supporting role. No frozen-document amendment is needed.

The exact checks are in the proofs: critical Jacobian, simple kernel and gap; transverse invertibility and uniqueness; both branch slopes; full mixed and quadratic coefficients; uniform eigenvalue and exact signs; all-mode stability; physical side; generic critical escape; degenerate roots, spectra and full critical Lyapunov estimate; independent constraints and P02 locus overlap. A separate reproducible checker evaluates the original field and independently formed Jacobians for several dimensions and offsets, including collisions and noncoprime cases. Exact arithmetic and floating-point checks are corroboration, not proof dependencies or independent audit. Counts and diagnostics are recorded in the manifest and separate validation companion.

Priority checks for the independent P09 auditor.

1. Check the real transverse IFT and product neighborhoods, and the uniqueness step $\psi \equiv 0$ for both generic and degenerate cases.

2. Check $r_*'(0) = -1/b$, $dr_*/dg = 1/b$, $d\mu/dg = -1$, and the local exclusion of the other quadratic root for fixed $b \neq 0$.

3. Recompute $w^{\mathsf{T}} F_{x\mu} v = 1$, $w^{\mathsf{T}} F_{xx}[v, v] = 2b$, the Taylor factor $1/2$, and the opposite g mixed coefficient from the original field.

4. Check the exact center coordinate and positive time factor in (165); scalar branch crossing alone does not prove full-system equilibrium completeness.

5. Check the exact uniform sign and finite-mode transverse continuity, including negative synchronous states, collisions and every Fourier label.

6. Check the physical branch side for both signs of b and separate open-domain instability from relative-orthant claims at criticality.

7. At $b = 0$, check both square-root roots and their full spectra, the center-manifold hypotheses, the independent Lyapunov estimate (175), and the nonexponential synchronous decay.

8. Check the rank-two parameter constraints and the zero repeated-root meeting point. No pitchfork, cusp or other named codimension-two bifurcation is claimed without further proof.

9. Preserve the boundary to P10: no positive-fold saddle-node, true bistability, basin description or unfolding is claimed. No Hopf, transversality, Lyapunov coefficient, rotating wave, large-n result or DM–MB comparison is part of P09.

14 P10: Saddle-node bifurcation and true bistable region

Status: DRAFT; independent A10 audit pending. Fix $a, k, c, h > 0$, $n \geq 5$, $1 \leq q < n$, and vary $g > 0$ in the canonical DM field ($\mathrm{DM}_{n,q}$). Put

$$A = k + h, \quad b = a - A > 0, \quad g_c = a + c, \quad g_f = g_c + \frac{b^2}{4A}, \quad r_f = \frac{b}{2A}, \quad s_f = 1 + r_f. \tag{177}$$

Here r is the synchronous amplitude and b is a local abbreviation. *Notation reconciliation:* P02's roots s_-, s_+ are the amplitudes called r_-, r_+ here. P02's repeated-root amplitude s_f in (35) is our r_f; P10's s_f denotes its denominator $1 + r_f$, following P03/P04's denominator convention. No earlier notation or file is changed. All neighborhoods below may depend on the fixed parameters and on (n, q); no uniform spectral gap as $n \to \infty$ is claimed.

The direct inputs are P02's invariant restriction and root formulas, P03's original-field Jacobian and origin stability, P04's complete Fourier spectrum, and P09's origin branch organization. P01 supplies smoothness, uniqueness and global nonnegative existence. We also use the real implicit-function theorem, the usual smooth local stable/unstable manifold theorem, and linearized stability/instability. These standard local results introduce no new project-module dependency or literature work.

14.1 Exact fold geometry

P02 gives, for $r > -1$,

$$F(r\mathbf{1}, g) = f(r, g)\mathbf{1}, \qquad f(r, g) = ar - Ar^2 + \frac{(c - g)r}{1 + r} = \frac{r}{1 + r}(\mu + br - Ar^2), \quad \mu = g_c - g. \tag{178}$$

The nonzero equilibrium equation is $Ar^2 - br - \mu = 0$. Completing the square, or using its discriminant, verifies independently

$$g_f = c + \frac{(a + A)^2}{4A}, \qquad s_f = 1 + \frac{a - A}{2A} = \frac{a + A}{2A}, \qquad g_f - g_c = \frac{b^2}{4A} > 0. \tag{179}$$

Define the state and parameter coordinates

$$u = r - r_f, \qquad \nu = g_f - g. \tag{180}$$

Then $\mu = \nu - b^2/(4A)$, $br_f - Ar_f^2 = b^2/(4A)$, and $b - 2Ar_f = 0$. Expanding in u therefore proves, with no approximation,

$$\boxed{\mu + br - Ar^2 = \nu - Au^2, \qquad \dot{u} = \frac{r_f + u}{s_f + u}(\nu - Au^2).} \tag{181}$$

Choose $|u| < r_f/2$. Both amplitude and denominator are positive. For $\nu < 0$ there is no nonzero synchronous equilibrium anywhere on $r > -1$; in particular none near r_f. For $\nu = 0$ there is one double positive scalar equilibrium r_f. For small $\nu > 0$ there are two distinct nearby positive roots

$$\boxed{r_\pm = r_f \pm \sqrt{\nu/A} = s_\pm \text{ of P02.}} \tag{182}$$

Both are positive exactly when $0 < \nu < b^2/(4A)$ within this real root regime. The origin always exists, but is excluded from the chosen fold state neighborhood.

14.2 Complete fold spectrum and the real transverse reduction

For a synchronous state let $s = 1 + r$ and import P03/P04:

$$d = a - 2kr - \frac{cr}{s^2} - \frac{g}{s} - hr, \qquad C = \frac{c}{s}, \quad G = \frac{gr}{s^2}, \quad H = hr.$$

$$\begin{aligned} J(r, g) &= dI + CP + GP^{-q} - HP^q, \\ \lambda_m &= d + Ce^{-\mathrm{i}\theta_m} + Ge^{\mathrm{i}q\theta_m} - He^{-\mathrm{i}q\theta_m}, \end{aligned} \tag{183}$$

where $\theta_m = 2\pi m/n$ and $0 \le m < n$. Colliding offsets are added, as in P03; none is removed. At a positive equilibrium P04 gives

$$\lambda_0 = f_r(r, g) = \frac{r}{1 + r}(b - 2Ar). \tag{184}$$

Thus $\lambda_0^f = 0$, also obtained by differentiating (181) at $(u, \nu) = (0, 0)$.

Lemma 14.1 (Automatic transverse stability at every positive fold)**.** *At the fold, with subscript f denoting evaluation there,*

$$\boxed{G_f - H_f = r_f \left(k + \frac{4A^2 c}{(a + A)^2} \right) > 0.} \tag{185}$$

The exact complex fold spectrum is (183)*, with $d_f = -C_f - G_f + H_f$. In particular,*

$$\operatorname{Re} \lambda_m^f = C_f(\cos \theta_m - 1) + (G_f - H_f)(\cos(q\theta_m) - 1), \tag{186}$$

$$\operatorname{Im} \lambda_m^f = -C_f \sin \theta_m + (G_f + H_f) \sin(q\theta_m). \tag{187}$$

For every $m \neq 0$,

$$\boxed{\operatorname{Re} \lambda_m^f \le -\delta_f < 0, \qquad \delta_f = \frac{c}{s_f} \left(1 - \cos \frac{2\pi}{n} \right) > 0.} \tag{188}$$

The only eigenvalue on the imaginary axis is zero. It has algebraic and geometric multiplicity one and real eigenspace $\operatorname{span}\{\mathbf{1}\}$.

Proof. Using (179), directly

$$\frac{g_f}{s_f^2} - h = \frac{c + As_f^2}{s_f^2} - h = k + \frac{c}{s_f^2} = k + \frac{4A^2 c}{(a + A)^2} > 0. \tag{189}$$

Multiplication by $r_f > 0$ proves (185). Since $0 = \lambda_0^f = d_f + C_f + G_f - H_f$, taking real and imaginary parts of the complete P04 formula proves the two identities. For $1 \le m < n$, set $j = \min(m, n - m)$. The angle $2\pi j/n$ belongs to $[2\pi/n, \pi]$, so $\cos \theta_m \le \cos(2\pi/n) < 1$. The first cosine difference is strictly negative, while the second is nonpositive. This proves the margin. For even n the real singleton $m = n/2$ is included; for odd n the same argument ends before π. Collisions $q = 1, n - 1, n/2$ and $\gcd(n, q) > 1$ change neither argument. Even if $q\theta_m$ is a multiple of 2π, the first term is strictly negative. P04's unitary diagonalization lists exactly n eigenvalues with algebraic multiplicity. Only label zero can vanish, so its multiplicities are one and its eigenvector is $\mathbf{1}$. No simplicity of the other numerical eigenvalues is asserted. □

Proposition 14.2 (Full-system local equilibrium completeness)**.** *In a state neighborhood of $r_f\mathbf{1}$ and parameter interval about g_f, every full-system equilibrium is synchronous. The entire local equilibrium equation reduces exactly to $\nu - Au^2 = 0$.*

Proof. Let $v = \mathbf{1}$, $w^\mathsf{T} = n^{-1}\mathbf{1}^\mathsf{T}$, $E = \{y : w^\mathsf{T} y = 0\}$, and $\Pi_\perp = I - vw^\mathsf{T}$. Write uniquely $x = r\mathbf{1} + y$, $y \in E$. The circulant J_f preserves E, and its restriction to the complexification of E has exactly the nonuniform eigenvalues. Lemma 14.1 therefore makes $J_f|_E : E \to E$ an invertible real map.

For the smooth E-valued map $T(r, y, g) = \Pi_\perp F(r\mathbf{1} + y, g)$ one has $D_y T(r_f, 0, g_f) = J_f|_E$. The real implicit-function theorem gives a product neighborhood and a unique smooth transverse solution $y = \psi(r, g)$. Invariance of synchrony gives $T(r, 0, g) = 0$ for every nearby r, g. After shrinking the product, uniqueness forces $\psi \equiv 0$. Every full equilibrium solves $T = 0$ and hence has $y = 0$. Conversely, at $y = 0$ its scalar equation is $f = 0$. Here $r > 0$ and $1 + r > 0$, so (181) makes it precisely $\nu - Au^2 = 0$. A smaller state ball times a parameter interval lies in the IFT product. This is a full equilibrium reduction, not an inference from the presence of synchronous solutions alone. □

14.3 Full-system nondegeneracy and exact center dynamics

Proposition 14.3 (Original-vector-field saddle-node coefficients)**.** *With the above v, w, one has $J_f v = 0$, $w^\mathsf{T} J_f = 0$, $w^\mathsf{T} v = 1$, and*

$$\alpha_{\mathrm{SN}} := w^\mathsf{T} F_g(r_f\mathbf{1}, g_f) = -\frac{r_f}{s_f} < 0, \tag{190}$$

$$w^\mathsf{T} F_{xx}(r_f\mathbf{1}, g_f)[v, v] = -\frac{2Ar_f}{s_f} \neq 0, \tag{191}$$

$$\beta_{\mathrm{SN}} := \tfrac{1}{2} w^\mathsf{T} F_{xx}[v, v] = -\frac{Ar_f}{s_f} < 0. \tag{192}$$

Derivatives hold all other physical coefficients fixed. With ν as parameter the parameter coefficient has the opposite sign, $+r_f/s_f$.

Proof. The row and column sums of J_f are both $\lambda_0^f = 0$, giving the right and left nullvectors and their normalization. In the original vector field, $(F_g)_i = -x_i/(1 + x_{i+q})$, hence its average at the fold is $-r_f/s_f$.

Independently differentiate each original summand twice along $x = (r_f + t)\mathbf{1}$ at fixed $g = g_f$. The five componentwise second derivatives, retaining all terms even at coincident offsets, are

$$0, \qquad -2k, \qquad -\frac{2c}{s_f^3}, \qquad \frac{2g_f}{s_f^3}, \qquad -2h.$$

Their sum is $-2A + 2(g_f - c)/s_f^3$. Since $g_f - c = As_f^2$, this equals $-2A + 2A/s_f = -2Ar_f/s_f$ in every component. Averaging produces no extra factor of n. The Taylor coefficient of u^2 is half this full contraction; the derivatives themselves contain no Taylor factorial. □

Theorem 14.4 (Nondegenerate full-system positive saddle-node)**.** *For every fixed canonical $a > k + h$, $c, k, h > 0$ and admissible (n, q), the equilibrium $r_f\mathbf{1}$ at*

$$\boxed{g = g_f = a + c + \frac{(a - k - h)^2}{4(k + h)}}$$

is a nondegenerate full-system saddle-node. It has a simple uniform zero eigenvalue and strictly stable transverse spectrum. Locally there are no equilibria for $g > g_f$, one double equilibrium for $g = g_f$, and precisely the two positive equilibria (182) *for $g < g_f$ sufficiently close. The exact center restriction is orbitally equivalent, by a positive change of time, to*

$$\boxed{\frac{du}{d\tau} = \nu - Au^2, \qquad \frac{d\nu}{d\tau} = 0.} \tag{193}$$

Proof. Smoothness holds in an open neighborhood contained in $\mathbb{R}^n_{>0}$ and with $g > 0$. Lemma 14.1 supplies the simple kernel and stable complement. Proposition 14.3 verifies both nonzero coefficients of the standard saddle-node nondegeneracy test. Proposition 14.2 proves local equilibrium completeness, so no nonsynchronous branch is omitted.

For completeness the normal form is exact on a known center manifold. At fixed g_f the synchronous line is invariant and tangent to the entire one-dimensional center eigenspace. For the parameter extension $(\dot{x}, \dot{\nu}) = (F(x, g_f - \nu), 0)$, the linearization at the fold, in coordinates (u, y, ν), is

$$\dot{u} = (r_f/s_f)\nu, \qquad \dot{y} = (J_f|_E)y, \qquad \dot{\nu} = 0. \tag{194}$$

Its center block is a nilpotent two-by-two block: unlike P09 at the origin, the parameter derivative here is nonzero. The generalized center subspace is precisely $y = 0$. Synchrony times the parameter interval is smooth, invariant and tangent to this subspace, hence an exact local two-dimensional center manifold. Its exact restriction is (181) with $\dot{\nu} = 0$. For $|u| < r_f/2$ the time factor satisfies

$$\frac{r_f + u}{s_f + u} \geq \frac{r_f/2}{1 + 3r_f/2} > 0.$$

Consequently $d\tau/dt = (r_f + u)/(s_f + u)$ proves (193). This is orbital equivalence on the center restriction, not a global decoupling of the full field. The quadratic equation gives the stated branches and sides. At the fold itself the center flow is negative for $u \neq 0$; it attracts from $u > 0$ and escapes from $u < 0$, so the fold is not a locally asymptotically stable equilibrium in the full space. □

14.4 Both branches and all Fourier modes

Proposition 14.5 (Exact uniform signs throughout the two-root interval)**.** *For every $g_c < g < g_f$, define D as in P02 and $\Delta = \sqrt{D}$. Then*

$$0 < \Delta = \sqrt{b^2 + 4A(a + c - g)} = 2\sqrt{A(g_f - g)} < b,$$

$$r_\pm = \frac{b \pm \Delta}{2A}, \qquad b - 2Ar_\pm = \mp\Delta, \tag{195}$$

$$\boxed{\lambda_0(r_\pm) = \mp\frac{r_\pm}{1 + r_\pm}\Delta.} \tag{196}$$

Thus the lower branch is always unstable in the uniform direction, whereas the upper uniform eigenvalue is always strictly negative. Any upper-branch loss of stability within this open interval must involve a nonuniform mode.

Proof. P02's discriminant factorization is also obtained directly by inserting $g_f = g_c + b^2/(4A)$. The strict parameter inequalities give $0 < \Delta < b$, so both roots are positive. Substitution into $b - 2Ar_\pm$ cancels b and leaves $\mp\Delta$. Equation (184) gives the exact signs, and the positive lower eigenvalue implies nonlinear instability. This last implication needs no hyperbolicity of the other lower-branch eigenvalues. □

Corollary 14.6 (Near-fold full branch stability)**.** *There exists $\varepsilon_{\mathrm{sp}} > 0$ such that both branches for $0 \le \nu < \varepsilon_{\mathrm{sp}}$ satisfy*

$$\operatorname{Re} \lambda_m(r_\pm, g_f - \nu) < -\delta_f/2 \quad (m \neq 0).$$

For $0 < \nu < \min(\varepsilon_{\mathrm{sp}}, b^2/(4A))$, the upper branch is Hurwitz and locally exponentially stable. The lower branch is a hyperbolic saddle with exactly one simple positive real eigenvalue and $n-1$ stable real dimensions. Its local stable manifold has dimension $n-1$ and codimension one, and its unstable manifold has dimension one.

Proof. Every labeled eigenvalue in (183) is continuous in (r, g) near (r_f, g_f). There are finitely many labels. A common neighborhood makes each transverse real part differ from its fold value by less than $\delta_f/2$. Both continuous roots $r_\pm(\nu)$ tend to r_f as $\nu \downarrow 0$; smoothness in ν at zero is not needed. Shrink $\varepsilon_{\mathrm{sp}}$ so both branches lie in that neighborhood and remain positive. Combine the margin with (196). Only label zero has positive real part on the lower branch, so its eigenvalue is simple. Real conjugate-pair blocks and real singleton modes together give $n-1$ stable dimensions, even if stable numerical eigenvalues repeat. P04's nonlinear stability implications and the smooth local stable/unstable manifold theorem apply. □

14.5 The nonempty open bistability set

Theorem 14.7 (Stable coexistence and openness in physical parameters)**.** *For each fixed $a, k, c, h > 0$ with $a > k + h$ and each admissible (n, q), there is $\varepsilon > 0$ with*

$$(g_f - \varepsilon, g_f) \subset (g_c, g_f) \tag{197}$$

such that throughout this interval the full system has the locally exponentially stable origin, a positive index-one saddle $r_-\mathbf{1}$, and the locally exponentially stable positive equilibrium $r_+\mathbf{1}$. At every strict point of a sufficiently small such interval this configuration persists on an open neighborhood in $(0, \infty)^5$ with coordinates (a, k, c, g, h) and fixed (n, q). Here "bistability" means coexistence of at least two locally stable equilibria; it does not mean that these are the only attractors.

Proof. At the origin P03 gives

$$\operatorname{Re} \lambda_m(0) = g_c - g - c(1 - \cos\theta_m) \le g_c - g < 0 \quad (g_c < g < g_f).$$

Thus the origin is Hurwitz and locally exponentially stable in $\mathcal{U}$, and also relative to $\mathbb{R}^n_{\ge 0}$ using P01 invariance. Take $0 < \varepsilon < \min(\varepsilon_{\mathrm{sp}}, b^2/(4A))$ and apply Corollary 14.6. The interval is nonempty and lies strictly above g_c.

At a strict witness all three Jacobians are invertible and hyperbolic. The implicit-function theorem continues each equilibrium smoothly under changes of all five physical coefficients. The origin stays identically zero. Alternatively the positive roots are explicitly smooth because $D > 0$; uniqueness identifies them with the IFT continuations. Their separation and strict positivity persist after shrinking the parameter neighborhood. Spectral continuity preserves both Hurwitz properties and the lower saddle's unstable dimension. The inequalities $a > A$, $g_c < g < g_f$ and positivity of every coefficient also persist. Such neighborhoods can depend on (n, q); no common all-dimension persistence neighborhood for the index-one configuration is asserted. □

Proposition 14.8 (Exact coexistence set and a simple sufficient criterion)**.** *For fixed (n, q) define the open two-positive-root domain*

$$\mathcal{D} = \{\boldsymbol{\pi} \in (0, \infty)^5 : a > k + h,\ g_c < g < g_f\}.$$

For $\boldsymbol{\pi} \in \mathcal{D}$, evaluate (183) *at its upper root and define*

$$\boxed{\mathcal{B}_{n,q} = \left\{ \boldsymbol{\pi} \in \mathcal{D} : \max_{0 \le m < n} \operatorname{Re} \lambda_m(r_+, g) < 0 \right\}.} \tag{198}$$

This set is open and nonempty. It is exactly the set in $\mathcal{D}$ where the origin and upper synchronous equilibrium are both locally exponentially stable. Every point has an unstable lower positive branch, although its unstable dimension is asserted to be one only in the near-fold neighborhood above. A sufficient condition for membership in $\mathcal{B}_{n,q}$ is

$$\boxed{G \ge H \quad \Longleftrightarrow \quad \frac{g}{(1+r_+)^2} \ge h.} \tag{199}$$

This condition allows equality.

Proof. The positive upper root and all its labeled eigenvalues are continuous (and smooth before taking the finite maximum) on $\mathcal{D}$. The strict finite set of inequalities defines an open subset of $(0,\infty)^5$, hence of $\mathbb{R}^5$. Nonemptiness follows from Theorem 14.7. The origin is automatically Hurwitz on $\mathcal{D}$. A Hurwitz Jacobian gives nonlinear local exponential stability. Conversely, a smooth equilibrium satisfying a local exponential estimate has a Jacobian semigroup satisfying that estimate by differentiation of the flow at the equilibrium, so all eigenvalues have strictly negative real part. Thus the definition is exact for exponential stable coexistence, not a classification of possible nonhyperbolic asymptotically stable boundary points or other attractors.

Subtracting the uniform row sum in (183) gives, at either branch and in fact at any synchronous state,

$$\operatorname{Re} \lambda_m = \lambda_0 + C(\cos\theta_m - 1) + (G - H)(\cos(q\theta_m) - 1). \tag{200}$$

On the upper branch $\lambda_0 < 0$ and $C > 0$. If $G - H \ge 0$, every mode has negative real part, including $m = 0$; for $m \ne 0$ the C contribution is strictly negative. Since $r_+ > 0$, dividing $G - H$ by it gives the physical-coefficient condition. The lower uniform sign follows from Proposition 14.5. □

One can give an explicit, dimension-independent sufficient interval for the two stable equilibria (not necessarily for the saddle's index). Set

$$M = g_f - h s_f^2 = c + k s_f^2 > 0, \qquad t_* := \min\left\{ \frac{r_f}{2}, \frac{M}{4 h s_f}, \sqrt{\frac{M}{2(A+h)}} \right\} > 0. \tag{201}$$

For $0 < \nu < A t_*^2$, put $t = \sqrt{\nu/A} < t_*$. Then $g = g_f - A t^2$, $r_+ = r_f + t$, and directly

$$g - h(1+r_+)^2 = M - 2 h s_f t - (A+h) t^2 > 0. \tag{202}$$

Indeed each of the last two terms is strictly less than $M/2$. Also $A t_*^2 \le b^2/(16A) < g_f - g_c$, so this whole interval lies in $\mathcal{D}$ and in $\mathcal{B}_{n,q}$ for every admissible (n,q). This supplies a quantitative alternative to the continuity argument for stable coexistence. Equation (189) also proves directly that $G > H$ persists on the upper branch near the fold. Intersecting this interval with that of Corollary 14.6 retains the index-one conclusion.

Remark 14.9 (An exact obstruction to whole-interval stability). Take

$$n = 6, \quad q = 1, \qquad (a, k, c, g, h) = \left(6, \frac{1}{10}, \frac{1}{10}, \frac{101}{10}, \frac{9}{10}\right). \tag{203}$$

Here $A = 1$, $b = 5$, $g_c = 61/10$, $g_f = 247/20$, $\mu = -4$, and the equilibrium polynomial is $(r-1)(r-4)$. At $r_+ = 4$ the original-field equilibrium residual is exactly zero, and

$$d = -\frac{109}{250}, \quad C = \frac{1}{50}, \quad G = \frac{202}{125}, \quad H = \frac{18}{5}, \quad G - H = -\frac{248}{125}.$$

Consequently $\lambda_0 = -12/5 < 0$, but the real mode $m = 3$ has

$$\lambda_3 = d - C - G + H = \frac{191}{125} > 0. \tag{204}$$

The origin is stable and the upper equilibrium unstable at this strict interior point of the two-root interval. This refutes universal upper-branch stability, hence universal origin/upper stable coexistence on that interval. At the lower root $r_- = 1$, one has $\lambda_0 = 3/2$ and $\operatorname{Re}\lambda_1 = \operatorname{Re}\lambda_5 = 53/80 > 0$; the other three real parts are negative. Thus its unstable dimension is three here, also precluding a whole-interval index-one claim. This makes no assertion about other attractors there. It is a static full-spectrum counterexample, not a theorem about a crossing, a Hopf bifurcation or a periodic branch.

14.6 Synchronous threshold and the precise local basin statement

Proposition 14.10 (Exact basin threshold on the nonnegative synchronous ray)**.** *For every $g_c < g < g_f$, the scalar flow on the ray satisfies*

r	$(0, r_-)$	(r_-, r_+)	(r_+, ∞)
$f(r, g)$	$-$	$+$	$-$

If $0 < r(0) < r_-$, then $r(t) \to 0$. If $r(0) > r_-$, then $r(t) \to r_+$; this includes the constant solution $r(0) = r_+$. The data $r(0) = r_-$ and $r(0) = 0$ give their respective constant solutions. These are exact full-system solutions by synchrony, even when the upper equilibrium is transversely unstable.

Proof. Factor (178) as

$$f(r, g) = -\frac{Ar}{1 + r}(r - r_-)(r - r_+). \tag{205}$$

The prefactor is negative for $r > 0$, giving the three signs. Uniqueness prevents crossing an equilibrium. P01 and P02 give global nonnegative existence and boundedness. More specifically solutions below r_- decrease and are bounded below by zero; solutions between the roots increase and are bounded above by r_+; those above r_+ decrease and are bounded below by it. Each therefore has a limit. If that limit were not a zero of f, continuity would bound $|f|$ away from zero near the limit, contradicting convergence. The monotone interval and the initial strict inequality select the claimed zero. No nonsynchronous basin conclusion follows from this argument alone. □

Proposition 14.11 (Unstable synchrony and a local separating stable disk)**.** *At a near-fold parameter tuple covered by Corollary 14.6 and Theorem 14.7, let $p = r_-\mathbf{1}$. The local stable manifold is a smooth embedded $(n-1)$-disk tangent to E. Its germ locally separates a sufficiently small neighborhood into two sides. The local unstable manifold is exactly a segment of the synchronous line through p. Its two forward branches tend, respectively, to the origin and to $r_+\mathbf{1}$.*

In fact one may choose a neighborhood $V \subset \mathbb{R}^n_{>0}$ of p such that its local stable disk S has

$$V \setminus S = V_- \dot\cup V_+, \qquad V_- \subset \mathcal{A}(0), \quad V_+ \subset \mathcal{A}(r_+\mathbf{1}), \tag{206}$$

where the two sets are connected and $\mathcal{A}(z)$ denotes the basin of the equilibrium z relative to $\mathbb{R}^n_{\geq 0}$. Every point of S tends to p. Thus S is the common basin boundary relative to this neighborhood. *No assertion identifies the full global basin boundary or globally continued stable manifold.*

Proof. The lower saddle has stable space E and unstable space $\mathrm{span}\{\mathbf{1}\}$. The stable/unstable manifold theorem gives the dimensions and tangencies. Tangency and invariance alone would not establish an arbitrary invariant curve's equality with an unstable manifold, so check the backward-orbit characterization. On a sufficiently small synchronous interval around r_- the scalar field has a simple repelling zero, $f_r(r_-, g) > 0$. Every point of that interval has a negative-time orbit staying in the interval and converging exponentially to r_- as $t \to -\infty$. Its vector lift has the same property. Uniqueness of the local unstable manifold, as the set of such backward trajectories, identifies its germ with this interval. Both are one-dimensional, so after restriction it is exactly the stated segment. Proposition 14.10 gives the two forward limits. The open branches lie on $0 < r < r_-$ and $r_- < r < r_+$ and furnish synchronous heteroclinic trajectories.

Here is a local argument also identifying the two basin sides, rather than extrapolating a global boundary from codimension. Write $x = p + s\mathbf{1} + y$, $y \in E$. The stable disk is a graph $s = \phi(y)$ with $\phi(0) = 0$, $D\phi(0) = 0$. Use the smooth local coordinates $\xi = s - \phi(y)$, $z = y$, so the stable disk is $\xi = 0$ and the unstable segment is $z = 0$. Local invariance of these two manifolds and smoothness imply

$$\dot{\xi} = \xi\big(\lambda + \alpha(\xi, z)\big), \qquad \dot{z} = Bz + R(\xi, z), \quad \lambda = \lambda_0(p) > 0, \quad B = J(p)|_E, \tag{207}$$

where $\alpha(0,0) = 0$, $R(\xi, 0) = 0$ and $D_z R(0,0) = 0$. P04 normality and the transverse gap give $\langle z, Bz\rangle \le -\gamma\|z\|^2$ for real $z \in E$, with some $\gamma > 0$. Shrink the coordinate neighborhood so $|\alpha| < \lambda/2$ and $\|R(\xi, z)\| \le (\gamma/2)\|z\|$. Hence while in it, $\|z(t)\| \le e^{-\gamma t/2}\|z(0)\|$ and $|\xi(t)|$ grows at least at rate $\lambda/2$ if nonzero.

Choose $\rho > 0$ small so $(\pm\rho, 0)$ lie on the respective synchronous unstable branches and the closed coordinate box is in this neighborhood and in $\mathbb{R}^n_{>0}$. These two exit points belong to the two sinks' basins. Those basins are relatively open: each sink has a local attracting neighborhood and continuous dependence pulls it back along any finite segment of a converging orbit. Thus choose $\sigma > 0$ small enough that both exit caps $\{(\pm\rho, z) : \|z\| \le \sigma\}$ lie in the respective basins. The image V of $(-\rho, \rho) \times \{\|z\| < \sigma\}$ is an open neighborhood of p. A solution starting with $\xi \neq 0$ cannot exit through a z boundary, because its norm decreases, and reaches the cap with the sign of ξ in finite time. It then converges to the corresponding sink. If $\xi = 0$, it stays on the stable graph and converges to p. The two half-boxes are connected and map to V_-, V_+. Each disk point is approached from both halves, so the disk is exactly the boundary of each basin relative to V. This proves (206) with the claimed local scope. The disk is locally invariant (and forward invariant in this chosen box), not necessarily invariant under arbitrary negative-time continuation within a fixed small neighborhood. □

Global scope limit. The ray threshold is global only within the invariant synchronous ray. The preceding full-space statement concerns one sufficiently small neighborhood of the lower saddle at a fixed near-fold tuple. It neither identifies the complete global boundary between the sinks nor gives a partition of the whole orthant. Other equilibria, periodic attractors, or other invariant attracting sets elsewhere are not excluded. In particular "bistability" here does not assert exactly two global attractors.

14.7 Connection to P09, degenerate limit, and the P11 boundary

For $b > 0$, as $g \downarrow g_c$ from above,

$$r_- \longrightarrow 0, \qquad r_+ \longrightarrow b/A, \qquad \frac{dr_-}{dg} = \frac{1}{\Delta} \longrightarrow \frac{1}{b}. \tag{208}$$

Thus P09's small positive branch on $g > g_c$ is precisely P02/P10's lower branch, unstable in the uniform direction throughout (g_c, g_f). It meets the upper branch at g_f. No upper stability all the way to g_c follows; Remark 14.9 shows why this would be false in general.

For fixed $k, c, h > 0$, as $b = a - A \downarrow 0$,

$$r_f = \frac{b}{2A} \to 0, \qquad g_f - g_c = \frac{b^2}{4A} \to 0, \qquad \alpha_{\mathrm{SN}}, \beta_{\mathrm{SN}} \to 0. \tag{209}$$

The limiting locus is P09's $a = A$, $g = a+c$, where the center field is $-Ar^3/(1+r)$, its cubic derivative is $-6A$, and the center zero remains simple. P09 supplies the codimension-two algebraic/spectral statement. The P10 nondegeneracy theorem excludes this limit. There is no two-parameter unfolding or cusp classification here.

The exact facts available for later P11 work are the uniform signs on the entire open two-root interval, the complete fold spectrum, and automatic all-mode upper stability sufficiently near the fold. A subsequent loss of upper stability inside this interval must be nonuniform. P10 establishes no Hopf existence or frequency, critical pair simplicity, crossing transversality, Lyapunov coefficient, or periodic branch. The static counterexample does not begin those tasks. The frozen direct edge into P10 remains P02/P03/P04/P09, with P01 inherited; recording potential later inputs does not alter the frozen DAG.

14.8 Checks and independent audit priorities

B03 is resolved for this DRAFT as stable coexistence, the exact synchronous threshold, and the explicitly proved local saddle basin statement; global exclusivity and complete basin geometry remain open. This fits the frozen S4 supporting role, so no scope amendment is needed. A separate checker evaluates exact rational identities, original-field residuals and direct Jacobians, tests the full spectrum at and near the fold, and searches wider two-root parameters for counterexamples. It has no theorem status and cannot certify equilibrium completeness or manifold claims by numerical sampling. All analytical claims above remain DRAFT pending independent A10 audit.

Priority checks for the independent P10 auditor.

1. Recheck the denominator/amplitude notation, square completion, $g_f - g_c > 0$, and the exact branch labels inherited from P02.

2. Derive $G_f - H_f = r_f(k + c/s_f^2) > 0$ from the physical fold data; check the transverse margin for all labels, parity, offset collisions and noncoprime cases. Distinguish a simple zero from other repetitions.

3. Check the real transverse IFT, its product neighborhood, and uniqueness forcing $y = 0$ for all nearby full equilibria.

4. Recompute F_g and the second directional derivative from the original field; check $w^{\mathsf{T}} v = 1$ and the half-Taylor convention.

5. Check the parameter-extended nilpotent center block and generalized center tangent space, the exact invariant center manifold, and the strictly positive time factor. No full-space decoupling is used.

6. Check the exact uniform signs on the entire two-root interval and finite-mode transverse persistence on both roots. The index-one claim is local to the fold, not extended without proof.

7. Check the stable origin, nonempty coexistence interval, smooth hyperbolic persistence in all five coefficients, openness of $\mathcal{B}_{n,q}$, and the sufficient criterion including equality.

8. Check the explicit sufficient interval and the exact unstable upper-branch counterexample, independently of numerical searches.

9. Check all scalar sign intervals and limiting equilibria; justify unstable-manifold equality by backward convergence, not tangency alone.

10. Check local stable-disk separation and, especially, the cap-exit argument for local basin membership. The statement is relative to V and does not identify the complete global basin boundary.

11. Preserve the P09 degenerate-limit qualification and strict P11 boundary: no Hopf, critical-pair, transversality, nonlinear periodic or two-parameter unfolding theorem is supplied.

15 P11: Hopf spectral conditions and spectral simplicity

Status: DRAFT; independent A11 audit pending. We prove spectral assertions for the canonical demand-modulated model at positive synchrony. Direct inputs are P02, P04 and P05; P07 is used only for the five-node specialization. P03's coefficients are inherited through P04. P08–P10 and P12 or later results supply no mathematical premise. In particular, an imaginary pair here is only a *spectral Hopf candidate*. No parameter crossing, nonlinear nondegeneracy, periodic orbit or propagation theorem is asserted. The physical path is frozen for P12 without differentiating along it. The optional realizability result below concerns spectral boundaries, not first instability.

15.1 Setting, representative classes and frequency

Fix $q = 2$, $n \geq 5$, $a, k, c, g, h > 0$, and a specified positive synchronous equilibrium $x^*\mathbf{1}$. Set

$$s = 1 + x^*, \qquad C = \frac{c}{s} > 0, \qquad G = \frac{gx^*}{s^2} > 0, \qquad H = hx^* > 0. \tag{210}$$

Use P04's verified spectrum and unitary diagonalization:

$$\begin{aligned} \lambda_m &= d + Ce^{-\mathrm{i}\theta_m} + Ge^{2\mathrm{i}\theta_m} - He^{-2\mathrm{i}\theta_m}, \quad \theta_m = 2\pi m/n, \\ r_m &= d + C\cos\theta_m + (G - H)\cos 2\theta_m, \\ \omega_m &= -C\sin\theta_m + (G + H)\sin 2\theta_m, \\ J &= \mathsf{Q}\,\mathrm{diag}(\lambda_m)\mathsf{Q}^*, \qquad \lambda_{n-m} = \overline{\lambda_m} \quad (1 \leq m < n). \end{aligned} \tag{211}$$

Here $J = DF^{\mathrm{DM}}(x^*\mathbf{1})$; ω_m is a signed imaginary part. The reserved positive frequency is chosen separately when needed.

Let $N = \lfloor n/2 \rfloor$, $u_j = \cos(2\pi j/n)$ and L_j be P05's representative label classes: $L_0 = \{0\}$, $L_N = \{n/2\}$ if n is even, and otherwise $L_j = \{j, n - j\}$. Write

$$\mathcal{S} = \operatorname*{arg\,max}_{0 \leq j \leq N} r_j$$

for the *complete* selected representative set, including every tie. The selected Fourier labels are $\bigcup_{j \in \mathcal{S}} L_j$. This notation distinguishes $\mathcal{S}$ from the scalar $G + H$.

Lemma 15.1 (Reachable nonuniform representatives are genuine pairs)**.** *Every reachable nonuniform representative satisfies* $1 \leq j \leq j_{\max} := \lfloor (n + 1)/4 \rfloor < n/2$. *Consequently* $0 < \theta_j < \pi$, $\sin\theta_j > 0$, *and* $L_j = \{j, n - j\}$ *has two different labels. No even-n Nyquist singleton is included.*

Proof. P05 Theorem 9.2 supplies the reachable list. Since $(n + 1)/4 < n/2$ for $n > 1$, its nonzero members satisfy the claimed strict bounds. The angle and sine statements follow. This counts Fourier labels only, not numerical eigenvalue multiplicities. □

Proposition 15.2 (Exact signed frequency and its zero locus)**.** *For any genuine nonuniform pair representative* $0 < j < n/2$,

$$\boxed{\omega_j = \sin\theta_j\,[2(G + H)u_j - C].} \tag{212}$$

Thus $\omega_j = 0$ *if and only if* $2(G + H)u_j = C$. *If* $u_j \leq 0$, *then* $\omega_j < 0$ *automatically. If* $u_j > 0$, *the zero-frequency locus is exactly*

$$G + H = \frac{C}{2u_j}. \tag{213}$$

For positive u_j *the sign of* ω_j *is the sign of* $G + H - C/(2u_j)$. *These are spectral statements, not bifurcation classifications.*

Proof. Insert $\sin 2\theta_j = 2\sin\theta_j\cos\theta_j$ in (211). The factor $\sin\theta_j$ is strictly positive. If $u_j \le 0$, the bracket is at most $-C < 0$; otherwise divide by $2u_j > 0$. The Nyquist singleton is outside the proposition's scope. □

15.2 Strict candidates, simplicity and the stable complement

Definition 15.3 (Strict nonuniform spectral Hopf candidate)**.** A positive synchronous equilibrium of the canonical $q = 2$ model is a *strict nonuniform spectral Hopf candidate in representative j* if $j \neq 0$, $\mathcal{S} = \{j\}$, $r_j = 0$, and $\omega_j \neq 0$. Uniqueness refers to representatives and includes the exclusion of all maximal ties. This definition does not assert a Hopf bifurcation.

Theorem 15.4 (General strict spectral criterion)**.** *At a strict candidate, the entire imaginary-axis spectrum is exactly*

$$\lambda_j = \mathrm{i}\omega_j, \qquad \lambda_{n-j} = -\mathrm{i}\omega_j, \qquad \omega_j \neq 0. \tag{214}$$

Each value is algebraically and geometrically simple over $\mathbb{C}$ and the two values are distinct. Every other Fourier label has $r_\ell < 0$. The number

$$\eta = -\max_{\ell \notin L_j} r_\ell \tag{215}$$

is strictly positive, and all noncritical eigenvalues satisfy $\operatorname{Re}\lambda_\ell \le -\eta$. The real critical space has dimension two and its real invariant complement has dimension $n-2$. Moreover

$$0,\ 2\mathrm{i}\omega_j \notin \operatorname{spec}(J), \qquad J^{-1} \text{ and } (2\mathrm{i}\omega_j I - J)^{-1} \text{ exist.} \tag{216}$$

Proof. P05 reachability and Lemma 15.1 give two distinct labels. P04 conjugacy proves (214). For each other representative t, complete strict selection gives $r_t < r_j = 0$: equality would place t in $\mathcal{S}$. Every remaining Fourier label belongs to one of those other classes, so every remaining real part is negative, including the uniform label and any Nyquist label.

If another label outside L_j produced λ_j, it would have real part zero, contradicting that strict inequality. The other label $n-j$ cannot produce λ_j either, since $\omega_j \neq 0$. Thus precisely one label produces λ_j, and the same argument applies to λ_{n-j}. P04 Proposition 8.2 equates both complex multiplicities with the number of coincident labels. This proves simplicity, rather than inferring it merely from the Fourier basis. The real plane and complement follow from the conjugate Fourier spaces; an explicit orthonormal real basis is given below.

There are $n - 2 \ge 3$ noncritical labels, a nonempty finite set with strictly negative real parts. Its maximum is negative, proving $\eta > 0$ and the claimed bound. Neither zero nor $2\mathrm{i}\omega_j$ can equal a noncritical eigenvalue because their real parts are zero. Neither equals $\pm\mathrm{i}\omega_j$: the first exclusion uses $\omega_j \neq 0$, and the second would require $2 = 1$ or $2 = -1$ after division by $\mathrm{i}\omega_j$. This proves both inverse gates. □

Proposition 15.5 (Exact real and adjoint normalization)**.** *At the strict candidate, choose*

$$v_H = v^{(j)}, \qquad p_H = v_H, \qquad e_c = \sqrt{2}\,\operatorname{Re} v_H, \quad e_s = \sqrt{2}\,\operatorname{Im} v_H. \tag{217}$$

Under the project's Hermitian pairing $\langle p, v\rangle = p^* v$,

$$Jv_H = \mathrm{i}\omega_j v_H, \qquad J^* p_H = -\mathrm{i}\omega_j p_H, \qquad \langle p_H, v_H\rangle = 1. \tag{218}$$

The real vectors e_c, e_s are orthonormal and span the real critical plane. In the ordered basis (e_c, e_s) the matrix is

$$J|_{\mathrm{crit}} = \begin{pmatrix} 0 & \omega_j \\ -\omega_j & 0 \end{pmatrix}. \tag{219}$$

In its coefficient plane the signed angular velocity is $-\omega_j$.

Proof. P04 gives $J = \mathsf{Q}\Lambda\mathsf{Q}^*$, hence $J^* = \mathsf{Q}\overline{\Lambda}\mathsf{Q}^*$. The same normalized Fourier column therefore solves the opposite-sign adjoint equation. This agrees with Section 2, including conjugate linearity in the first argument.

Write $v_H = A + \mathrm{i}B$ with real vectors A, B. The columns v_H and $\overline{v_H} = v^{(n-j)}$ are orthogonal and each has norm one. Thus $v_H^{\mathsf{T}} v_H = 0$ and $v_H^* v_H = 1$, giving $\|A\|^2 = \|B\|^2 = 1/2$ and $A^{\mathsf{T}} B = 0$. Every real vector in the complex critical span is a real linear combination of A, B, proving the plane assertion. Taking real and imaginary parts of $J(A + \mathrm{i}B) = \mathrm{i}\omega_j(A + \mathrm{i}B)$ gives $Je_c = -\omega_j e_s$ and $Je_s = \omega_j e_c$, which are the columns of (219). For coefficients (z_c, z_s), $\dot{z}_c = \omega_j z_s$ and $\dot{z}_s = -\omega_j z_c$. This is a linear orientation statement only. If $\omega_j < 0$, one may instead choose label $n - j$ to obtain the positive frequency $|\omega_j|$; here the signed representative convention is retained. □

Remark 15.6 (Ties and zero frequency are excluded separately). If two different representative classes have $r_j = r_t = 0$, their labels give additional imaginary-axis values or larger multiplicities. Either possibility fails the single-simple-pair-with-stable-complement hypothesis. A tie of real parts alone is not equality of numerical eigenvalues. In particular choosing one member of a maximal tie does not satisfy Definition 15.3.

If a genuine pair has $r_j = \omega_j = 0$, both labels produce the same real zero eigenvalue. P04 makes it semisimple, with complex algebraic and geometric multiplicity exactly $|\{m : \lambda_m = 0\}| \geq 2$. Under strict selection of that representative its multiplicity is exactly two and its real kernel has dimension two. Without strict selection other zero labels must also be counted. This is ineligible for the simple Hopf package. No named multiple-criticality or steady-state bifurcation is classified. The two-dimensional real plane in Theorem 15.4, by contrast, corresponds to two distinct simple complex eigenvalues.

15.3 Regular spectral surfaces and physical admissibility

Fix a reachable nonuniform representative $1 \leq j \leq j_{\max}$ throughout this subsection, so $0 < \theta_j < \pi$. Use P02's nonzero equilibrium equation in branch coordinates:

$$a = (k + h)x - \frac{c - g}{1 + x} > 0, \qquad x, k, c, g, h > 0. \tag{220}$$

This is exactly P05's open chart domain $\mathcal{U}_{\mathrm{eq}}$. Combining P04's diagonal coefficient with the equilibrium equation gives

$$d = a - 2kx - \frac{cx}{(1+x)^2} - \frac{g}{1+x} - hx = -kx - \frac{c(1+2x)}{(1+x)^2}.$$

For fixed representative j, put $v_j = \cos 2\theta_j = 2u_j^2 - 1$; this scalar v_j is not a Fourier vector $v^{(j)}$. Define

$$\begin{aligned} R_j(x, k, c, g, h) = & -kx - \frac{c(1+2x)}{(1+x)^2} + \frac{c}{1+x} u_j \\ & + x\left(\frac{g}{(1+x)^2} - h\right) v_j. \end{aligned} \tag{221}$$

Substitution in P04's real dispersion shows exactly $R_j = r_j$.

Proposition 15.7 (Regular spectral boundary and excluded frequency locus)**.** *At fixed x, c, g, h,*

$$\frac{\partial R_j}{\partial k} = -x < 0, \qquad k_H = \frac{1}{x}\left[-\frac{c(1+2x)}{(1+x)^2} + \frac{c}{1+x} u_j + x\left(\frac{g}{(1+x)^2} - h\right) v_j\right]. \tag{222}$$

Thus $R_j = 0$ is a regular codimension-one level set wherever it meets $\mathcal{U}_{\mathrm{eq}}$, explicitly the graph $k = k_H$. A physical point requires $k_H > 0$ and $a_H = (k_H + h)x - (c - g)/(1 + x) > 0$. Within a strict nonuniform P05 cell, the portions with $\omega_j \neq 0$ are strict candidates, and this regular surface transfers locally to physical parameter space along the specified positive branch.

For $u_j > 0$, any nonempty intersection $R_j = \omega_j = 0$ is locally a regular codimension-two level set in branch coordinates. It is codimension one inside $R_j = 0$. The same statement transfers to physical parameters wherever P05's concave-chart hypotheses hold. For $u_j \leq 0$ this intersection is empty.

Proof. In (221) only $-kx$ depends on k, proving the partial derivative and every sign in the solved formula. Strict selection and $\omega_j \neq 0$ are open conditions. P05 Theorem 9.1 implies $H - G > 0$ at any nonuniform selected representative. On precisely this concave part of the positive chart, P05 supplies

$$\Phi(x, k, c, g, h) = (a, k, c, g, h), \qquad \det D\Phi = k + h + \frac{c - g}{(1 + x)^2} = k + \left(h - \frac{g}{(1 + x)^2}\right) + \frac{c}{(1 + x)^2} > 0.$$

Its local diffeomorphism therefore transfers the regular level set. No global injectivity or topology after forgetting the branch is claimed. The derivative $\partial_k R_j$ holds x, c, g, h fixed and lets a follow (220); it is not a derivative along the later fixed-a continuation path.

For the additional frequency equation, at fixed branch coordinates

$$\partial_k \omega_j = 0, \qquad \partial_g \omega_j = \frac{2x u_j \sin\theta_j}{(1 + x)^2} > 0 \quad (u_j > 0).$$

The Jacobian of (R_j, ω_j) in columns (k, g) has determinant $-2x^2 u_j \sin\theta_j/(1 + x)^2 \neq 0$. The regular level set theorem proves the local codimension assertion, conditional on existence and physical positivity; the local diffeomorphism applies under the same concavity hypotheses just stated. Proposition 15.2 excludes all $u_j \leq 0$ intersections. No unfolding is inferred. □

Proposition 15.8 (Exact obstruction and cell realizability criterion)**.** *Fix a reachable nonuniform representative $1 \leq j \leq j_{\max}$. Let $\beta_j = (u_j + u_{j+1})/2$ be P05's midpoint, and put*

$$w_j = 1 - 2u_j^2, \quad L_j^{\mathrm{vert}} = \max(0, \beta_j), \qquad B_j = \frac{w_j}{4(1 - u_j)}. \tag{223}$$

Its strict P05 cell contains a physical strict spectral Hopf candidate if and only if

$$\boxed{w_j > 0 \quad \textit{and} \quad L_j^{\mathrm{vert}} < B_j.} \tag{224}$$

More precisely, for a prescribed positive vertex $z = C/[4(H - G)]$ in that strict cell and prescribed $x > 0$, such a candidate can be realized by some positive physical coefficients if and only if

$$\frac{w_j}{4z} > 1 - u_j + \frac{x}{1 + x}. \tag{225}$$

Consequently not every reachable nonuniform cell supports a candidate. For example $n = 8, j = 1$ has $w_j = 0$ and never reaches $r_j = 0$. For $n = 12, j = 2$, $u_j = 1/2$, $L_j^{\mathrm{vert}} = B_j = 1/4$; this cell also has no physical candidate despite $w_j > 0$.

Proof. Write $D_c = H - G > 0$ and $C = c/(1+x)$ within the selected cell. At a candidate, (221) reads

$$kx = D_c w_j - C\left(1 - u_j + \frac{x}{1+x}\right), \qquad \frac{kx}{C} = \frac{w_j}{4z} - \left(1 - u_j + \frac{x}{1+x}\right). \tag{226}$$

Because $k, x, C > 0$ and $u_j < 1$, this forces $w_j > 0$ and (225). It further forces $z < B_j$. P05's strict cell is $L_j^{\mathrm{vert}} < z < \beta_{j-1}$, where $\beta_{j-1} > 0$; β_j exists since $j_{\max} < N$. Thus necessity of (224) follows.

Conversely, if (224) holds, choose

$$L_j^{\mathrm{vert}} < z < \min(\beta_{j-1}, B_j).$$

Both upper bounds exceed the lower bound. Choose $x > 0$ sufficiently small that (225) holds. For fixed x, z satisfying that inequality, the following construction proves the more precise sufficiency as well. Choose any $C > 0$, set $D_c = C/(4z)$, and define $k > 0$ by (226). For a free $G > 0$, set

$$c = C(1+x), \qquad g = \frac{G(1+x)^2}{x}, \qquad h = \frac{G + D_c}{x}, \qquad a = kx + D_c - C + G\left(1 + \frac{1+x}{x}\right). \tag{227}$$

These equations give $H - G = D_c$ and exactly $a = (k+h)x - (c-g)/(1+x)$. Increasing G makes $a > 0$ without changing z, k or $r_j = 0$. The signed frequency bracket is $2(D_c + 2G)u_j - C$. For $u_j \le 0$ it is automatically negative; for $u_j > 0$ it vanishes for at most one G, which can be avoided while retaining positivity. Hence a strict candidate is obtained. The two examples follow by exact substitution of their cosines and midpoints. They also show why P05's realization of dominance alone cannot imply realization of a spectral boundary. □

15.4 An exact five-node physical candidate

Use the verified P07 constants and spectra, without a new P07 derivation:

$$\phi = \frac{1+\sqrt{5}}{2}, \quad \sigma = \frac{\sqrt{10 - 2\sqrt{5}}}{4} > 0, \quad u_1 = \frac{1}{2\phi}, \quad u_2 = -\frac{\phi}{2}.$$

P07 shows that $\{1, 4\}$ is the only potentially dominant nonuniform pair and that $\omega_1 = \sigma[(G+H) - \phi C]$.

Theorem 15.9 (Exact five-node spectral Hopf candidate)**.** *For $n = 5, q = 2$ set*

$$\begin{gathered} x_H = 1, \quad c = 2, \quad g_H = 16, \quad h = 6, \\ k_H = \frac{3\phi - 4}{2} = \frac{3\sqrt{5} - 5}{4}, \qquad a_H = \frac{3\phi + 22}{2} = \frac{47 + 3\sqrt{5}}{4}. \end{gathered} \tag{228}$$

This is a positive physical tuple with a positive synchronous equilibrium $x_H \mathbf{1}$. It is a strict spectral candidate in representative 1*. Its full labeled spectrum is*

$$\begin{aligned} &\lambda_0 = -\frac{3\phi + 1}{2}, && && \\ &\lambda_1 = \mathrm{i}\omega_H, \qquad \lambda_4 = -\mathrm{i}\omega_H, && \omega_H = \sigma(10 - \phi) > 0, \\ &\lambda_2 = -\frac{3\sqrt{5}}{2} - \mathrm{i}\sigma(1 + 10\phi), && \lambda_3 = \overline{\lambda_2}. \end{aligned} \tag{229}$$

The critical values are each algebraically and geometrically simple. The ordering is $r_1 = 0 > r_0 > r_2$, with exact noncritical gap

$$\eta_H = \frac{3\phi + 1}{2} > 0. \tag{230}$$

The equilibrium is the upper of two distinct positive synchronous roots. The other root is

$$x_- = \frac{3(2-\phi)}{3\phi+8} \in (0,1). \tag{231}$$

Proof. Since $5/3 < \sqrt{5} < 3$, $k_H > 0$ and $a_H > 0$; the remaining three coefficients are positive. In the original canonical field, each component at $\mathbf{1}$ is exactly

$$F_i^{\mathrm{DM}}(\mathbf{1}) = a_H - k_H + \frac{2}{2} - \frac{16}{2} - 6 = a_H - k_H - 13 = 0.$$

This verifies the original model residual before using any spectral data. Direct substitution gives

$$\begin{gathered} s = 2, \quad C = 1, \quad G = 4, \quad H = 6, \quad T = G - H = -2, \quad S = G + H = 10, \\ d = -k_H - \frac{3}{2} = \frac{1-3\phi}{2}. \end{gathered} \tag{232}$$

The raw P04 expression independently gives $d = a_H - 2k_H - 1/2 - 8 - 6 = -k_H - 3/2$. P07's exact real-part formulas now yield

$$\begin{aligned} r_1 &= \frac{1-3\phi}{2} + \frac{1}{2\phi} + \phi = 0, \\ r_0 &= \frac{1-3\phi}{2} + 1 - 2 = -\frac{3\phi+1}{2}, \\ r_2 &= \frac{1-3\phi}{2} - \frac{\phi}{2} - \frac{1}{\phi} = \frac{3-6\phi}{2} = -\frac{3\sqrt{5}}{2}. \end{aligned}$$

Here $\phi^{-1} = \phi - 1$ and $2\phi - 1 = \sqrt{5}$. Their noncritical difference is $r_0 - r_2 = (3\phi - 4)/2 = k_H > 0$. Equivalently, $-\phi^2 < -2 < -\phi^{-2}$, so P07's full ordering table gives the same strict ordering. In particular pair $\{2,3\}$ is excluded.

For the imaginary parts, P07 gives $\omega_1 = \sigma(10 - \phi)$ and $\omega_2 = -\sigma(1 + 10\phi)$. The exact bounds $1 < \phi < 2$ imply $10 - \phi > 8 > 0$, so the critical frequency is nonzero without decimal evidence. P04 conjugacy fixes the signs at labels $4, 3$. This proves every entry of (229). The general theorem proves simplicity and the inverse gates, and the strict noncritical ordering gives (230).

For the branch identification, use only P02's abbreviations and polynomial:

$$A = k_H + h = \frac{3\phi+8}{2}, \quad b = a_H - A = 7, \quad \mu = a_H + 2 - 16 = \frac{3\phi-6}{2} < 0. \tag{233}$$

Because 1 is a root and $A - b = \mu$, that polynomial factors as

$$Q(X) = AX^2 - bX - \mu = A(X-1)(X-x_-), \qquad x_- = -\mu/A.$$

The displayed x_- is positive because $\phi < 2$, and is less than one since $3(2-\phi) < 3\phi + 8$. Thus the discriminant is $A^2(1-x_-)^2 > 0$. Together with $b > 0$ and $\mu < 0$ this checks P02's two-positive-root regime fully; those two signs alone would not suffice without the positive discriminant. The larger root is exactly $x_H = 1$. No P10 theorem is used. □

15.5 Frozen continuation path and exact spectral data

Proposition 15.10 (Local branch, path, vectors and resolvents for P12/P13)**.** *Hold $a = a_H, k = k_H, c = 2, h = 6$ fixed and freeze the physical path*

$$\boxed{\pi(g) = (a_H, k_H, 2, g, 6), \qquad g_H = 16.} \tag{234}$$

There is a neighborhood of 16 *with a unique smooth positive synchronous root near* 1*, denoted* $x_+(g)$*, with* $x_+(16) = 1$*. It continues the upper root. With the project's node ordering* $0, \dots, 4$*,*

$$\begin{gathered} v_H = \frac{1}{\sqrt{5}}(1, e^{2\pi \mathrm{i}/5}, e^{4\pi \mathrm{i}/5}, e^{6\pi \mathrm{i}/5}, e^{8\pi \mathrm{i}/5})^{\mathsf{T}}, \\ p_H = v_H, \qquad \overline{v_H} = v^{(4)}, \qquad J_H v_H = \mathrm{i}\omega_H v_H, \quad J_H^* p_H = -\mathrm{i}\omega_H p_H, \quad p_H^* v_H = 1. \end{gathered} \tag{235}$$

The real orthonormal basis $e_c = \sqrt{2}\,\mathrm{Re}\, v_H$*,* $e_s = \sqrt{2}\,\mathrm{Im}\, v_H$ *has block* $\left(\begin{smallmatrix} 0 & \omega_H \\ -\omega_H & 0 \end{smallmatrix}\right)$*. Both required inverse operators exist and are exactly*

$$\begin{aligned} J_H^{-1} &= \sum_{m=0}^{4} \lambda_m^{-1} v^{(m)} v^{(m)*}, \\ (2\mathrm{i}\omega_H I - J_H)^{-1} &= \sum_{m=0}^{4} (2\mathrm{i}\omega_H - \lambda_m)^{-1} v^{(m)} v^{(m)*}. \end{aligned} \tag{236}$$

In particular the mode-0 and mode-2 denominators useful for later work are

$$\lambda_0 = -\eta_H \neq 0, \qquad 2\mathrm{i}\omega_H - \lambda_2 = \frac{3\sqrt{5}}{2} + \mathrm{i}\sigma(21 + 8\phi) \neq 0. \tag{237}$$

Proof. Let $B(X, g) = a_H - (k_H + 6)X + (2 - g)/(1 + X)$. P02's scalar field is $f(X, g) = XB(X, g)$. P04's uniform identity $\lambda_0 = \partial_X f$ at synchrony, together with $B(1, 16) = 0$, gives $B_X(1, 16) = \lambda_0 = -\eta_H < 0$. The scalar implicit-function theorem supplies the unique smooth root near 1. Positivity persists in a sufficiently small neighborhood, and P02's distinct roots and their strict ordering persist there, so this is the locally continued upper root, not uniqueness of all positive roots. The parameter tuple stays positive for g near 16. No derivative of $x_+(g)$ is calculated.

Proposition 15.5 gives all vector identities and the real block with its stated sign. The spectral theorem and Theorem 15.4 give the two sums, all of whose denominators are nonzero. Finally, $2(10 - \phi) + (1 + 10\phi) = 21 + 8\phi$ verifies the second denominator in (237); its real part is positive. No tensor, nonlinear Fourier selection rule or Taylor-factor convention is required for these spectral identities. □

Proposition 15.11 (Spectral hypothesis ledger at the five-node point)**.** *The following is the precise P12/P13 handoff, subject to independent A11 audit of this DRAFT.*

H1. All five physical coefficients in (228) *are positive.*

H2. $x_H = 1$ *is an exact positive synchronous equilibrium of the original DM field.*

H3. The path varying g *in* (234) *has the smooth local upper branch* $x_+(g)$ *through* $x_+(16) = 1$*.*

H4. Labels $1, 4$ *give* $\pm\mathrm{i}\omega_H$ *with* $\omega_H = \sigma(10 - \phi) > 0$*.*

H5. Each critical complex eigenvalue is algebraically and geometrically simple.

H6. Every other eigenvalue has real part at most $-\eta_H < 0$*, where* $\eta_H = (3\phi + 1)/2$*.*

H7. $0 \notin \mathrm{spec}(J_H)$*.*

H8. $2\mathrm{i}\omega_H \notin \mathrm{spec}(J_H)$*.*

H9. Exact normalized right and adjoint vectors are (235)*; the pairing and real-plane orientation are fixed.*

H10. Spectral transversality is ***not established****; no state or eigenvalue derivative along* (234) *has been computed.*

H11. The first Lyapunov coefficient and nonlinear nondegeneracy are ***not established****; no tensor or normal-form coefficient is computed.*

H12. No periodic orbit, periodic-branch stability or rotating wave is established.

Proof. Items H1–H2 and H4–H6 are Theorem 15.9; H3 and H9 are Proposition 15.10; H7–H8 follow from Theorem 15.4. H10–H12 explicitly delimit unproved later conclusions, rather than adding assumptions already certified here. □

Scope, evidence and frozen planning language. The exact five-node candidate and the optional cell criterion supply analytic nonemptiness results, with no numerical premise. Separate computational checks are corroboration only. The archived decimal candidate C073 is not used or reproduced. B02 is fixed for this DRAFT to the upper branch and path (234); B06 is met for these spectral data by exact algebra. B07's pairing and adjoint sign agree with the current convention, but its remaining Taylor and nonlinear normalization issues remain for P13. Neither a first loss along the path nor an open transversality/nonlinear-nondegeneracy regime has been proved. The accompanying `PROPOSED_P11_scope_clarification.md` records the proposed refinement of older first-crossing wording; all frozen planning and novelty documents remain unchanged. P12 is not begun.

16 P12: Hopf transversality

16.1 TODO: Equilibrium derivative along the chosen path

16.2 TODO: Total spectral crossing derivative

16.3 TODO: Nonzero transversality on the stated parameter set

17 P13: First Lyapunov coefficient and nondegenerate Hopf theorem

17.1 TODO: Bilinear and trilinear derivative tensors

17.2 TODO: Fourier selection rules and resolvent domains

17.3 TODO: First Lyapunov coefficient and rigorous sign or nonvanishing

17.4 TODO: Assembled nondegenerate Hopf theorem and branch stability conditions

18 P14: Equivariant rotating waves

18.1 TODO: Exact cyclic spatiotemporal symmetry of the periodic branch

18.2 TODO: Isotropy, phase clusters and propagation direction

18.3 TODO: Relation of the exact wave symmetry to its leading-order expression

19 P15: Persistence under parameter heterogeneity

19.1 TODO: Nearby equilibrium existence, uniqueness and smooth dependence

19.2 TODO: Positivity and stability persistence with distinct hypotheses

19.3 TODO: Optional periodic-orbit persistence gate

20 P16: Static sensitivity and localized Green function

20.1 TODO: First-order parameter forcing and equilibrium correction

20.2 TODO: Fourier sensitivity and static norm bound

20.3 TODO: Localized Green function and directional response

20.4 TODO: Remainder estimate and its domain

21 P17: Transient and harmonic response

21.1 TODO: Transient Fourier kernel and Euclidean norm behavior

21.2 TODO: Conditional late-time modal dominance

21.3 TODO: Harmonic transfer function and frequency response

21.4 TODO: Scope of comparison with nonlinear dynamics

22 P18: Stability radius and Hopf-threshold sensitivity

22.1 TODO: Explicit Jacobian perturbation bound including equilibrium motion

22.2 TODO: Sufficient stability radius with controlled remainder

22.3 TODO: First-order eigenvalue and Hopf-threshold shift

22.4 TODO: Persistence of the relevant Hopf hypotheses under the perturbation

A N01: Independent validation protocol

Every check below is a future task. Before execution, record the model version, claim IDs, formula version, verified proof dependencies and acceptance criteria. Choose scale-aware error measures and tolerances in advance; do not invent a pass threshold after seeing a result. Numerical agreement does not prove a theorem.

V1. **Analytic Jacobian versus finite differences.** Implement the vector field independently of the analytic Jacobian. Use several perturbation steps and scales, interiors and allowed boundaries, and test coincident offsets at $q = 1$, $q = n - 1$ and even n with $q = n/2$. Gate: P03; P08 for MB.

V2. **Equilibrium residuals.** Record both absolute and scaled residuals, branch labels and admissibility. Test origin, all synchronous branches and near-fold cases; later test heterogeneous equilibria. Gates: P02, P08, P15.

V3. **Direct eigendecomposition versus Fourier eigenvalues.** Match unordered spectra with multiplicity; check real and imaginary parts, conjugacy, even-n singleton modes, ties, and $\gcd(n, q) > 1$. Gate: P04; P05–P08 for selection and comparison.

V4. **Symmetry/equivariance tests.** Test the homogeneous vector field and Jacobian against the fixed group action and the synchronous restriction; later test exact wave phase relations. Gates: P02–P04 and P14.

V5. **Continuation versus direct simulation.** Compare stable equilibrium and periodic branches, periods and amplitudes. For unstable branches use residuals and variational/Floquet checks; lack of convergence in forward simulation is not a branch validation. Gates: P09–P14 as applicable.

V6. **First Lyapunov coefficient from two independent implementations.** Use independently assembled tensors or derivative methods and a separate normal-form implementation. Record all inner-product, scaling, frequency, factorial and software conventions. Shared code paths do not count as independence. Reconcile magnitudes as well as signs. Gate: P13.

V7. **Convergence under tighter tolerances.** Refine integration steps, relative/absolute tolerances, continuation mesh, arithmetic precision, finite difference increments and sampling horizons as appropriate. Record convergence of each reported observable; a default tolerance is not evidence.

V8. **Reproducibility and software-version recording.** Archive exact versions, dependency lockfiles, operating system, precision, solver choices, seeds, parameters, initial data, time horizons, continuation controls, raw outputs, figure scripts and code/version hashes. A fresh environment must reproduce outputs within the predeclared tolerance.

V9. **Additional theorem-specific checks.** Boundary-face and box tests after P01; tensor derivatives and transversality after P12–P13; Floquet multipliers and phase shifts after P14; first-order remainder rates and Green kernels after P15–P16; transient and harmonic responses after P17; stability bounds and threshold shifts after P18. Record tests near failure modes.

A.1 Per-check result form to fill later

Claim and theorem version; independent implementation identifiers; inputs; observable and norm; predicted result; predeclared tolerance; measured error; refinement table; pass/fail/unresolved; failure diagnosis; reviewer and date. No field contains a computed result at initialization.

B N02: Continuation plan

1. Resolve B02, B06 and NB11–NB12: choose branch, parameter path, admissible parameter range, software, precision, event definitions and stopping criteria. The master baseline and Hopf coordinates are unverified seed suggestions; retrieve their exact archival values from the registry.

2. After P02–P04, continue all admissible synchronous equilibrium branches and retain the origin. Tag residuals, full spectra, mode labels and every inadmissible segment. Use P08 for a separately implemented MB comparison.

3. After P09–P10, identify transcritical and fold candidates and compare the continuation with proved statements. Separate root multiplicity, full stability and any basin claim. A sampled basin picture cannot establish a global separatrix or exclude additional attractors.

4. After P11–P13 and N01 validation, continue a periodic branch from the verified Hopf setting. Record amplitude convention, period, mesh convergence, Floquet multipliers, neutral phase multiplier and transverse stability.

5. After P14, check the exact cyclic time-shift relation with the declared orientation, phase convention and possible phase clusters.

6. For two-parameter work, choose the plane before plotting. Continue only curves justified by the completed analysis. Label fold–Hopf, zero–Hopf, Hopf–Hopf, generalized-Hopf and mode-tie points as candidates until separately validated; no existence of such points is assumed in this plan.

7. After P15–P18, continue selected heterogeneous equilibria and Hopf thresholds for explicitly named profiles. Compare first-order predictions and remainders across decreasing amplitudes, keeping coefficients positive.

B.1 Required continuation record

Model and proof versions; full parameters; network size and step; branch and seed provenance; free parameters; step limits; mesh and collocation settings; solver tolerances and precision; event rules; raw branch files; failed runs; independent checks and exact scripts to regenerate the diagrams.

C N03: Figure plan

Every caption must identify what is theorem, computation or interpretation. Use a figure only after its stated dependencies and N01 checks are satisfied.

ID	Planned figure	Evidence gate and required disclosure
F1	Network family	Canonical offsets and indexing; mark the connected-step condition, offset coincidences and $(5, 2)$ specialization. No dynamical claim.
F2	Interaction laws	Canonical definitions and B05; declare fixed source/receiver values and calibration. Do not label the input standard Holling type II.
F3	Synchronous branches	P02, P03, P09, P10; distinguish existence, stability, two-root region and any proved bistability.
F4	Dispersion and selection	P04–P07; all mode classes, ties, even-n cases, large-n family and exact error quantity.
F5	DM–MB comparison	P01–P05, P08; matched parameter plane, eligibility of equilibria and full-spectrum stability.
F6	Two-parameter organization	P09–P13; separate proved curves, numerical continuation and unresolved special points.
F7	Hopf and rotating waves	P11–P14; amplitude, period, Floquet multipliers, trajectories, phase-shift errors and tolerance refinement.
F8	Heterogeneity and response	P15–P18; named perturbation profiles, static Green function, transient/harmonic response, first-order errors and stability/threshold bounds.
F9	Five-node synthesis	P07, P08 and the applicable verified bifurcation/response modules; no physiological inference from algebraic constants.

C.1 Figure record to fill later

Claim IDs; verified theorem versions; model, parameters and calibration; data provenance; initial data; solver and continuation settings; tolerances; random seed if any; code version; reproduction command; error bars or error metrics; accessible labels and caption; independent checker. Optional random-prior maps and extra network sizes remain supplementary choices. No favorable-regime-only selection is prescribed.

D Adversarial mathematical audit template

Audit ID, reviewer, date and independence:
Record whether the reviewer is independent of the derivation and which version or hash is reviewed.

Statement being audited:
Exact theorem text, claim registry IDs, proof-file path and statement label. Do not audit a paraphrase.

Exact assumptions:
Every quantified domain, parameter restriction, regularity hypothesis, branch, norm, symmetry, exclusion and nondegeneracy.

Dependency verification:
List each input theorem, exact version, VERIFIED/AUDITED status and applicability; identify circular reasoning.

Line-by-line proof checks:
For each step record justification, algebra, signs, quantifiers, logical implication, and missing hypotheses. Require independent reconstruction of critical calculations.

Edge cases:
Zero and repeated roots; parameter boundaries; even/odd n; q=1, q=n-1 and q=n/2 when applicable; gcd cases; coincident shifts; self-conjugate modes; ties and resonances; singular denominators; state boundary.

Counterexample search:
Challenge the strongest quantifier; try excluded and limiting cases; distinguish a successful proof from failure to find a counterexample. Record whether the search was analytic or computational.

Symbolic checks:
Independent derivatives, identities, normalization, adjoints and remainders; source of any computer-algebra expression.

Numerical checks:
Independent implementation, residuals, conditioning, precision and convergence; whether ordinary or rigorously validated numerics; which analytic assertion the check can and cannot support.

Literature overlap:
Exact source and location, hypotheses, model dictionary, novelty limitations and any result imported by citation.

Verdict:
Select exactly one: **verified** / **revision needed** / **false** / **unresolved**. Explain the decisive evidence and remaining obligations.

Disposition and sign-off:
Corrections, owner, downstream invalidations, recheck version and date; do not mark AUDITED until the verdict is verified.